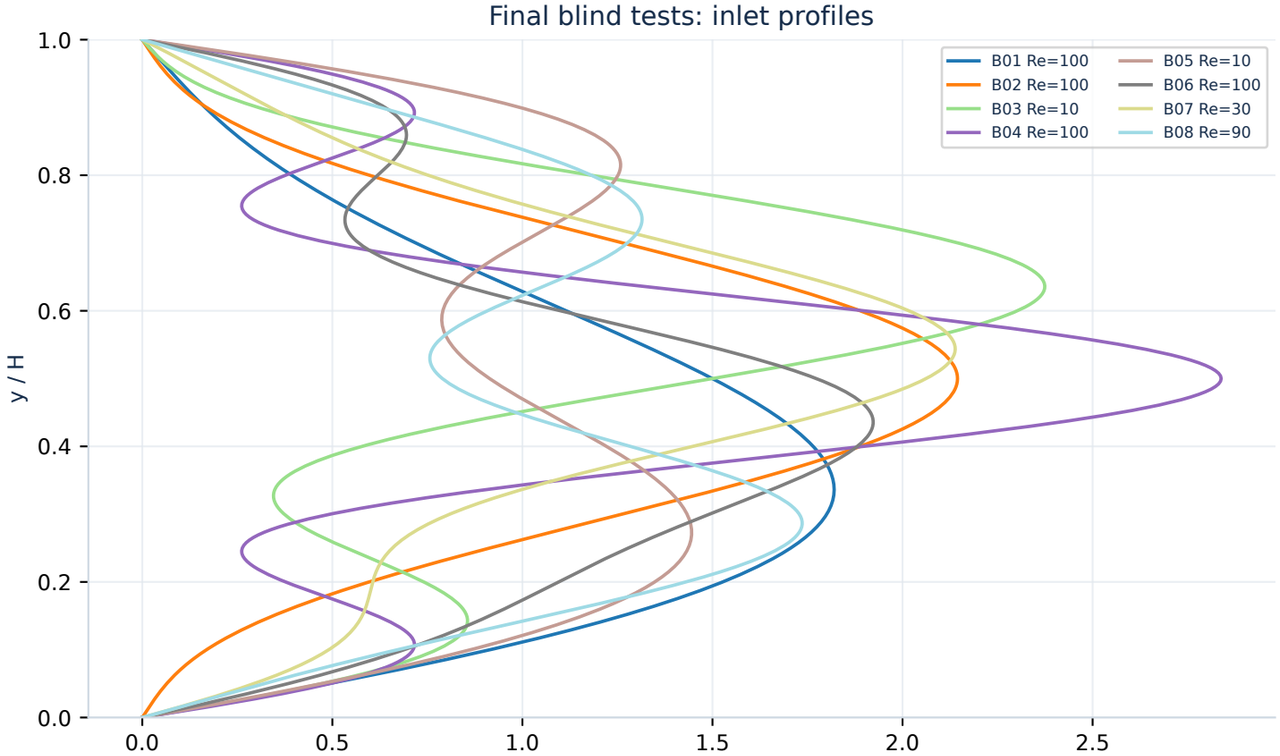
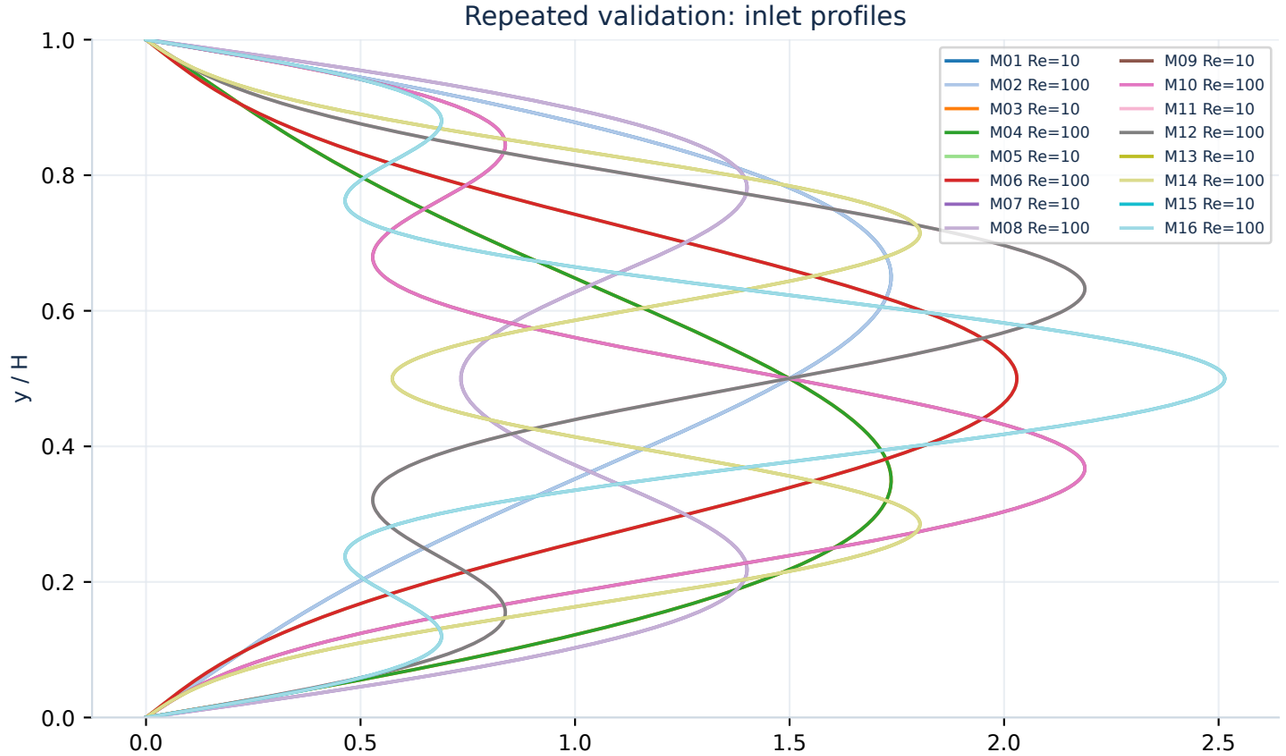
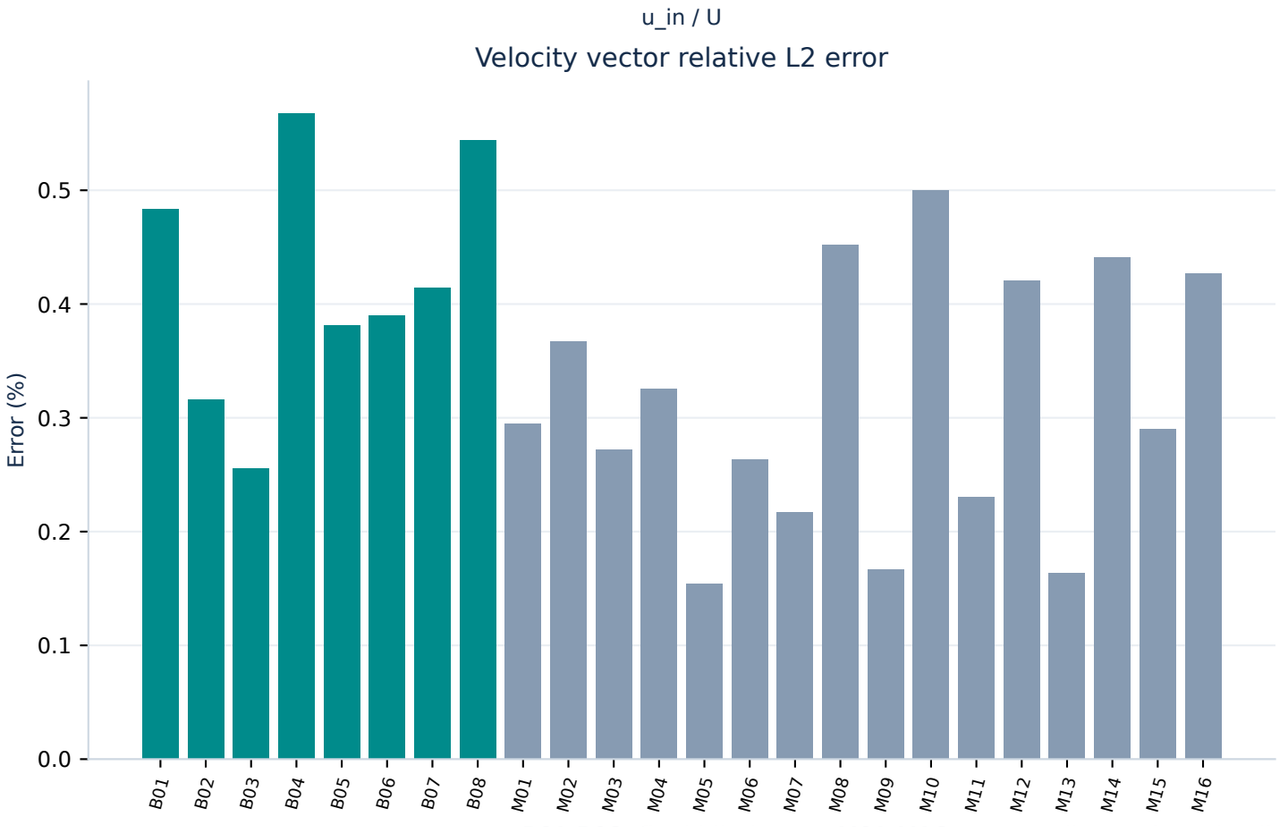
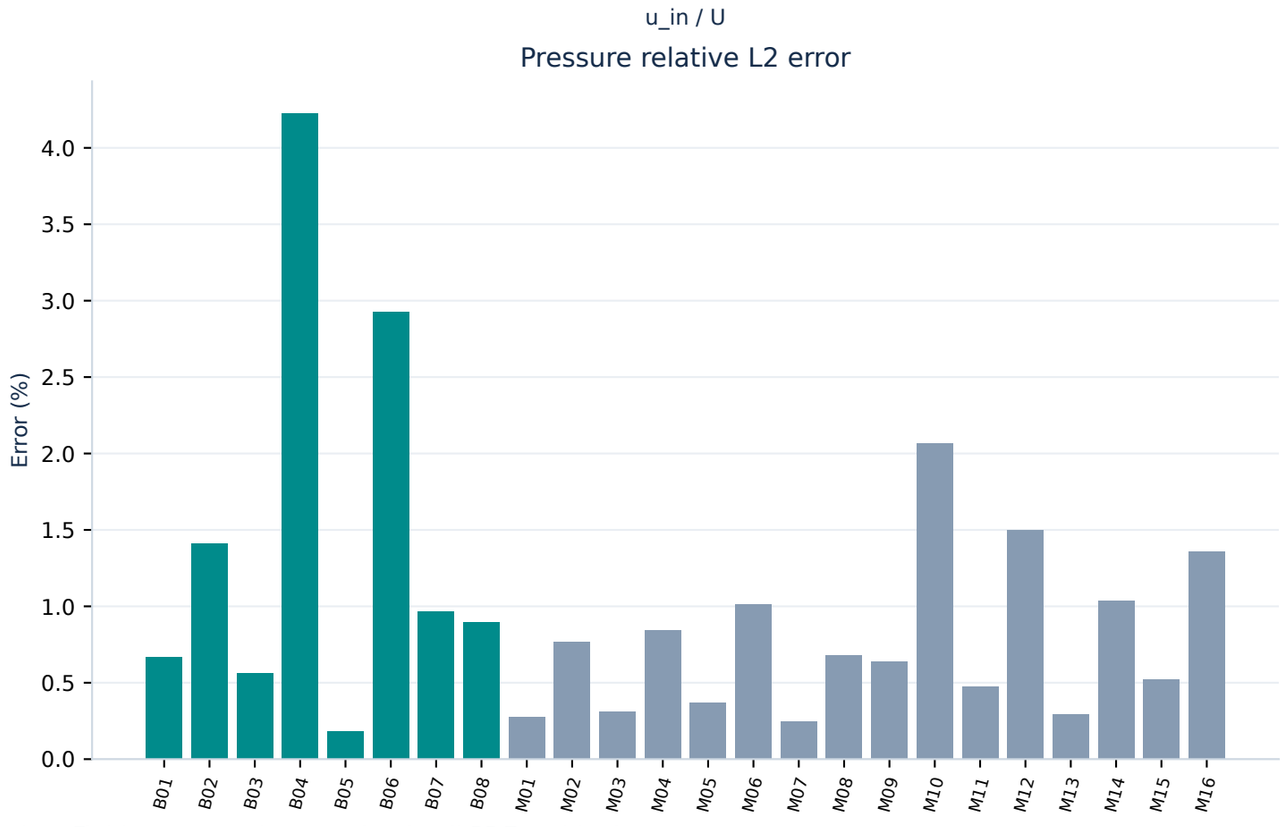
CHANNEL FLOW / FIXED OPERATOR vs FEM
24 held-out conditions | 70,000 training updates | no per-case training



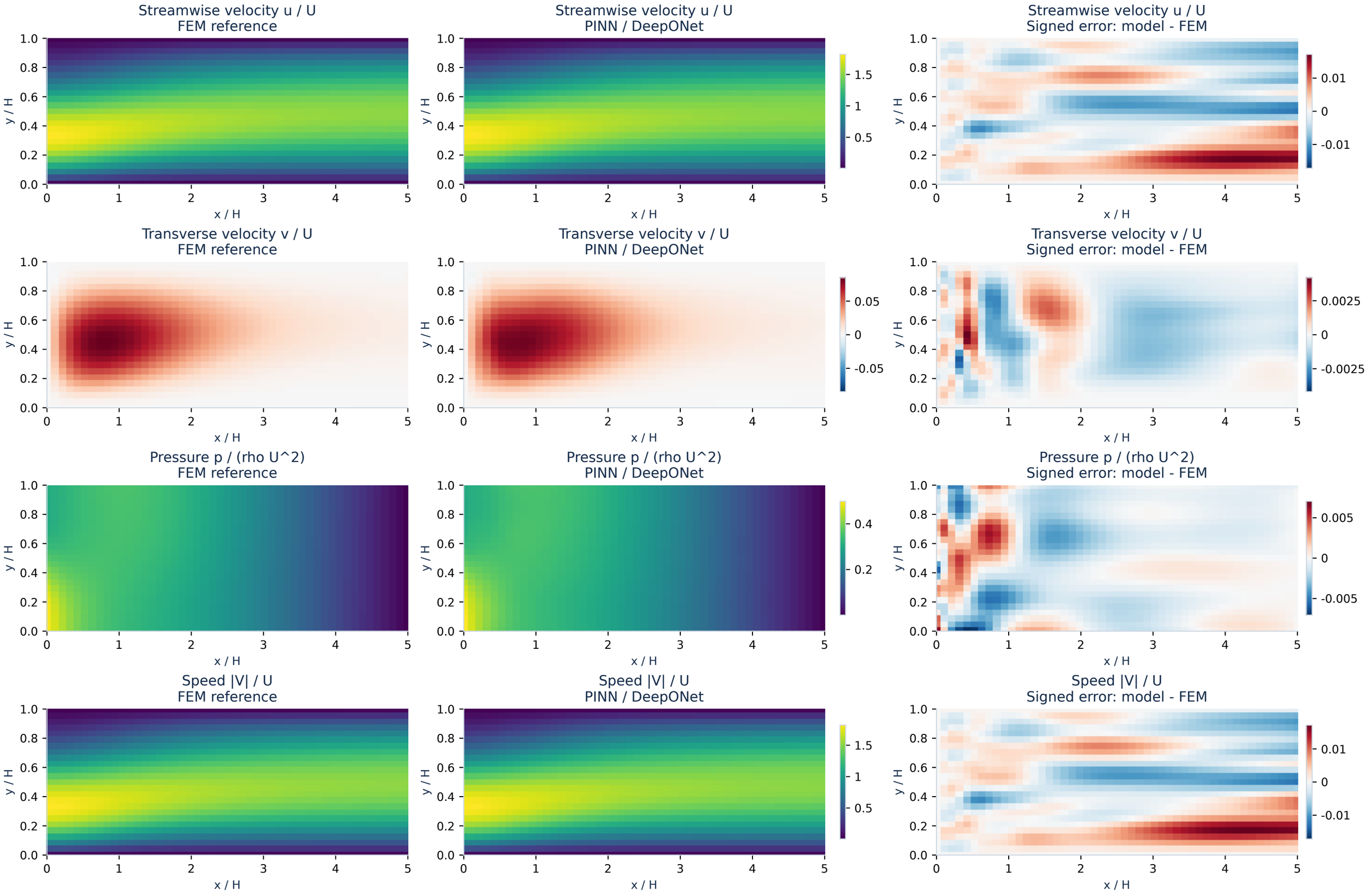






B01-B08: final blind tests. M01-M16: repeatedly monitored validation. Field accuracy does not imply all PDE residual gates pass.

B01 | Re = 100 | a = [+0.730, +0.000, +0.000, +0.000]
Velocity L2: 0.483% Pressure L2: 0.670% v RMSE: 0.00098 U

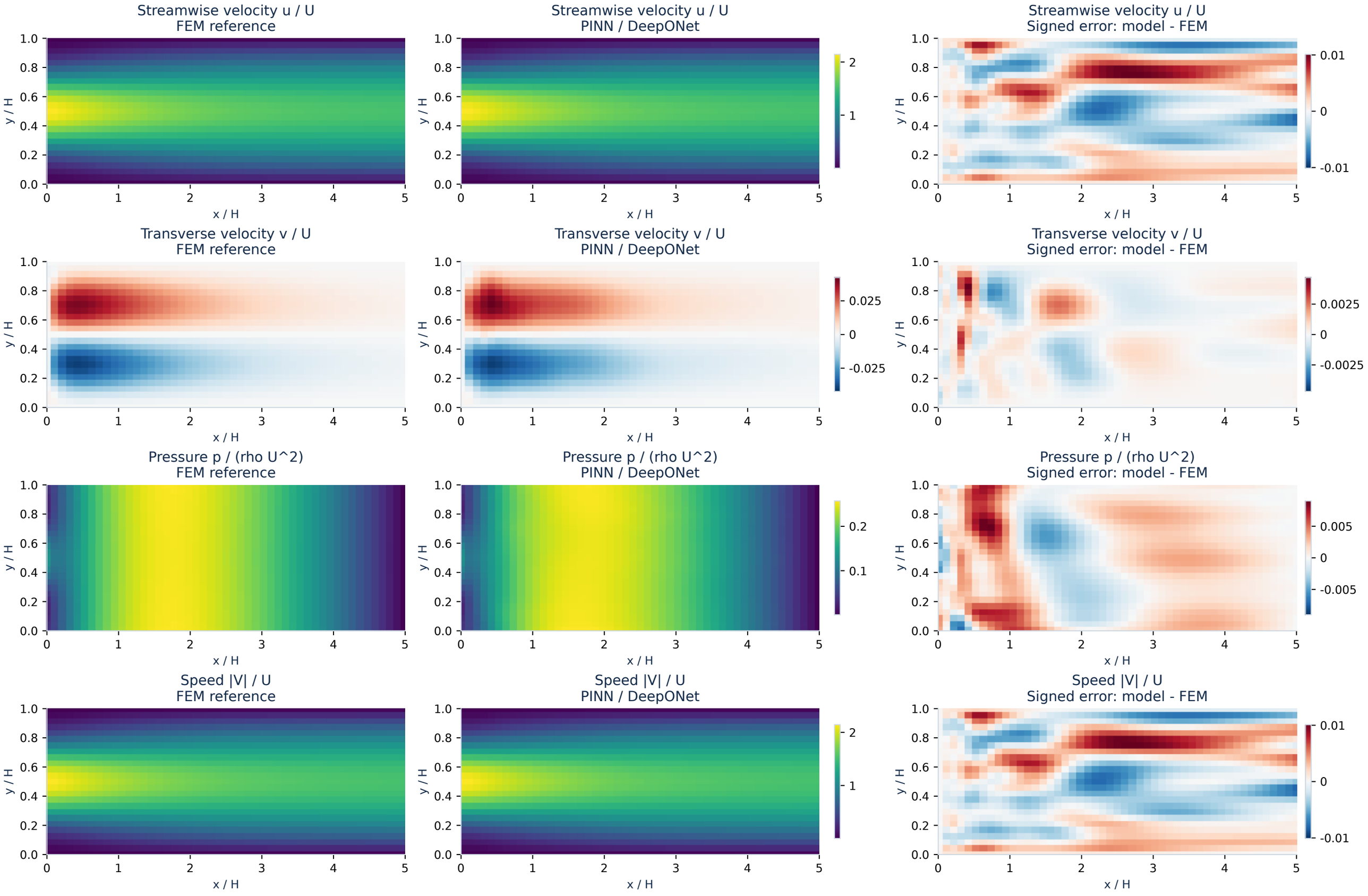


Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

B01 | Re = 100 | a = [+0.730, +0.000, +0.000, +0.000]
Velocity L2: 0.483% Pressure L2: 0.670% v RMSE: 0.00098 U

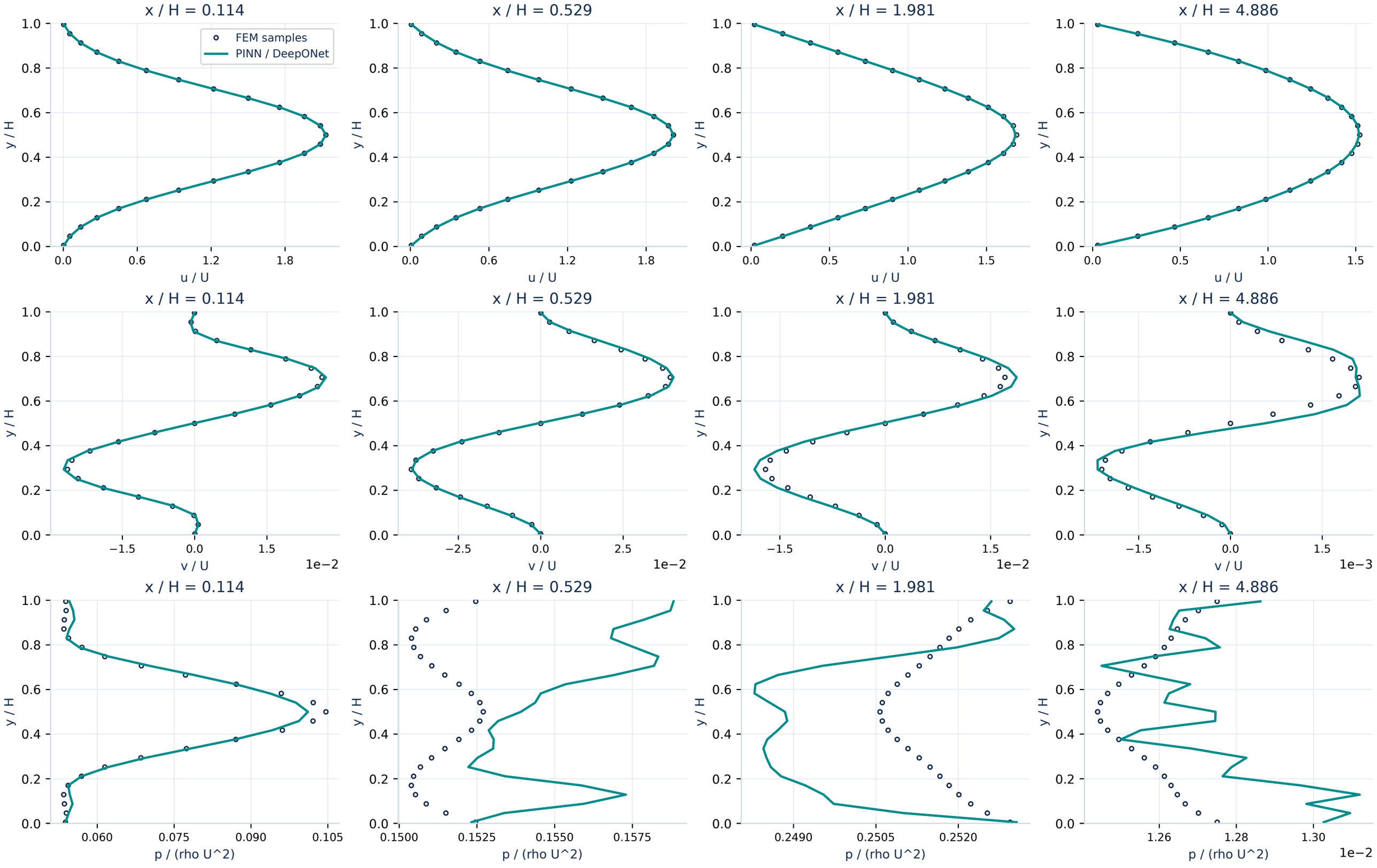
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

B02 | Re = 100 | a = [+0.000, -0.760, +0.000, +0.000]
Velocity L2: 0.316% Pressure L2: 1.412% v RMSE: 0.00072 U



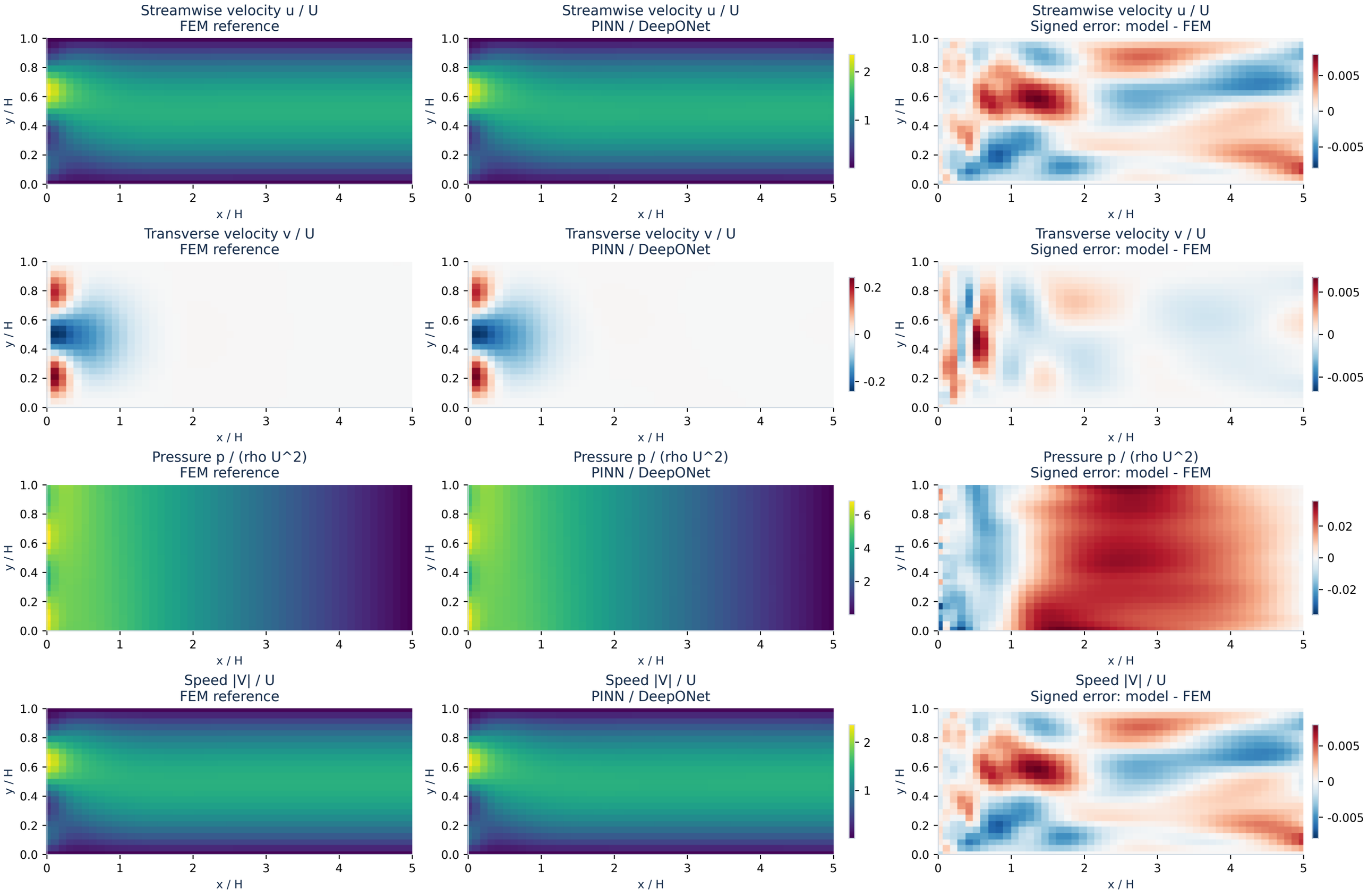
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

B02 | Re = 100 | a = [+0.000, -0.760, +0.000, +0.000]
Velocity L2: 0.316% Pressure L2: 1.412% v RMSE: 0.00072 U



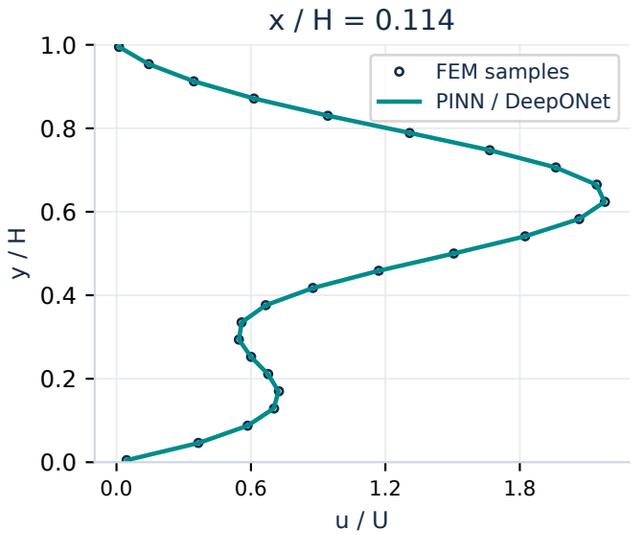
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

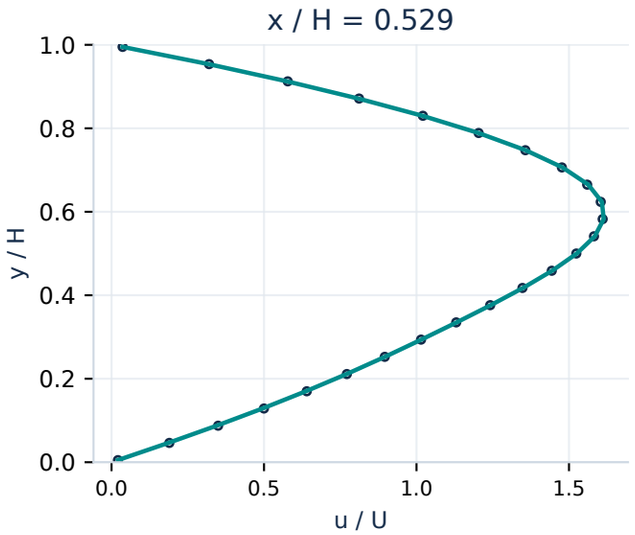
B03 | Re = 10 | a = [+0.000, +0.000, +0.740, +0.000]
Velocity L2: 0.256% Pressure L2: 0.562% v RMSE: 0.00108 U

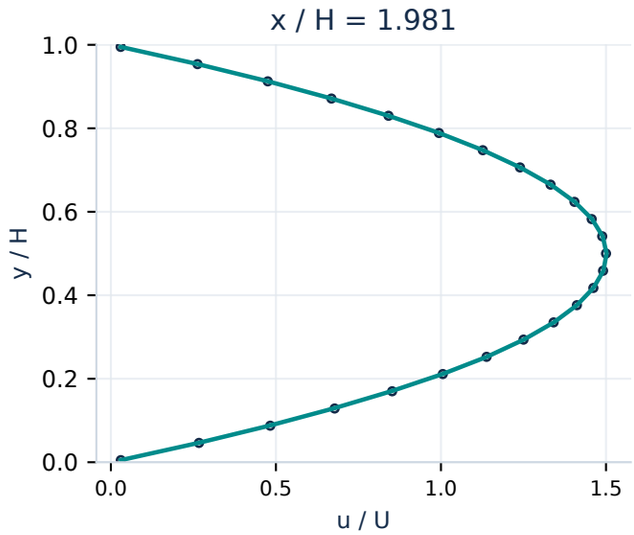


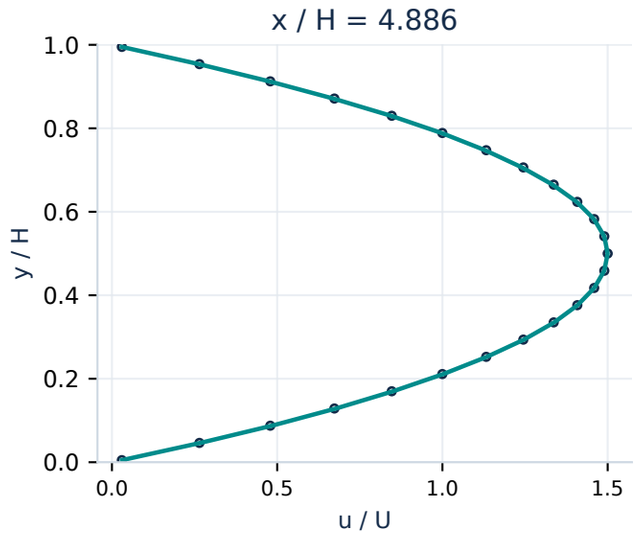
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

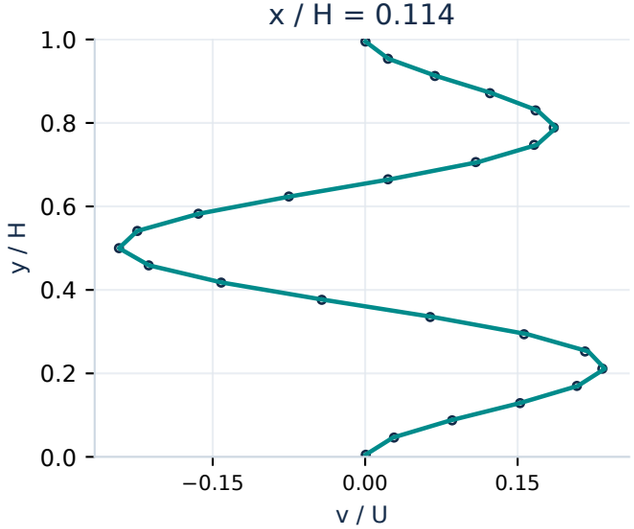
B03 | Re = 10 | a = [+0.000, +0.000, +0.740, +0.000]
Velocity L2: 0.256% Pressure L2: 0.562% v RMSE: 0.00108 U

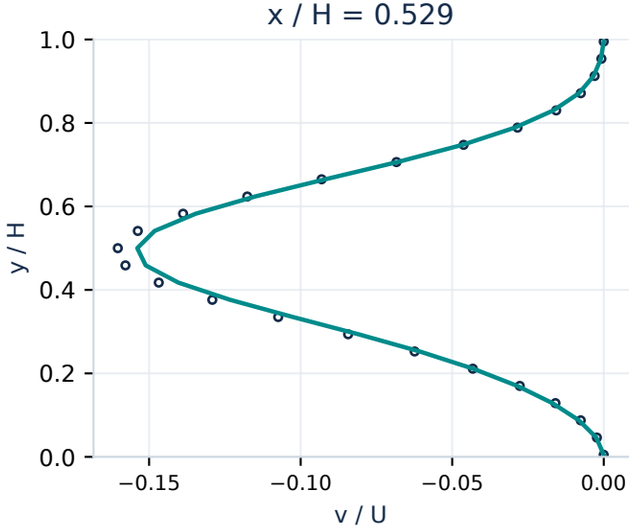


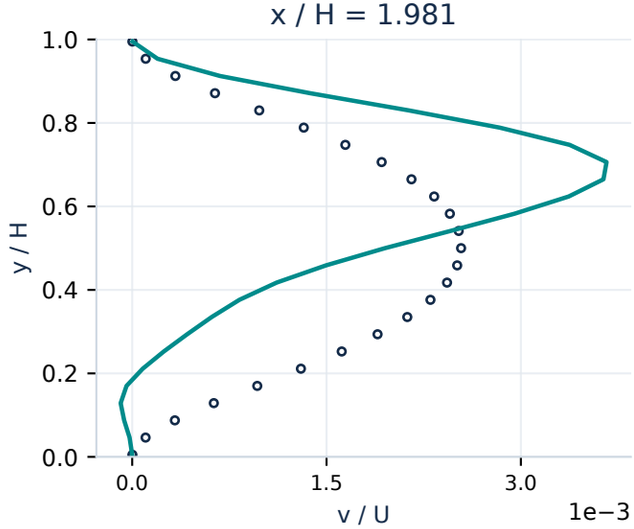


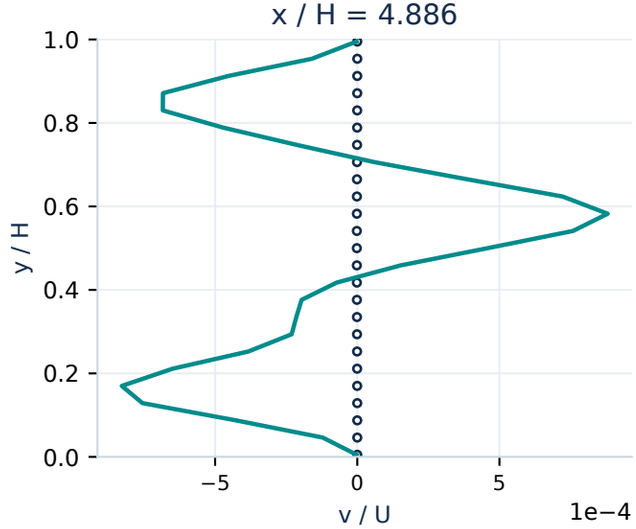


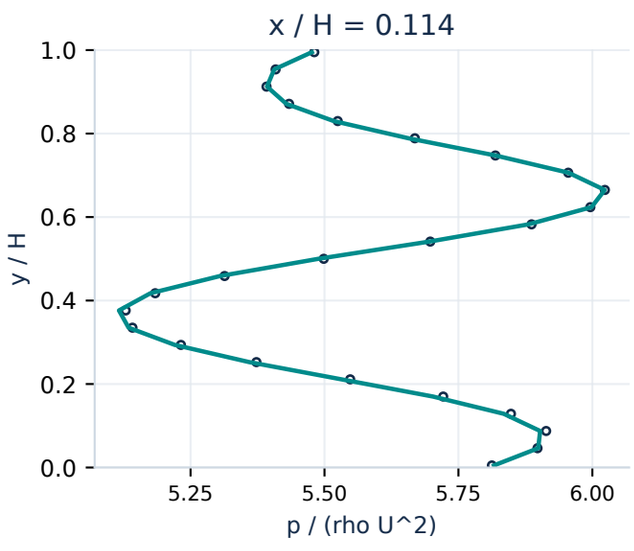


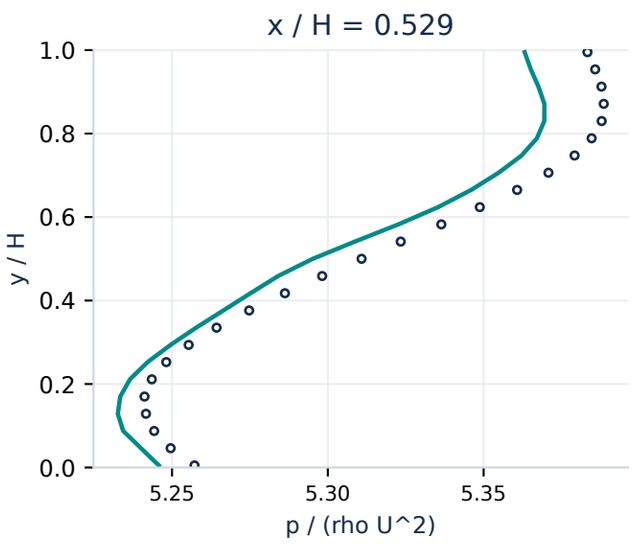


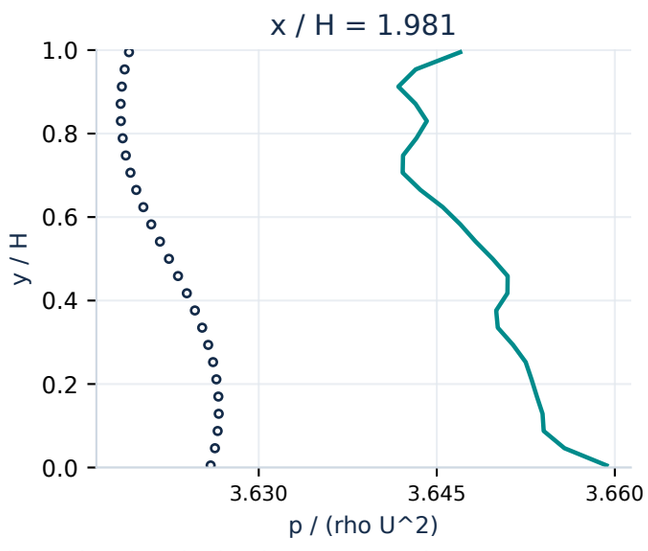


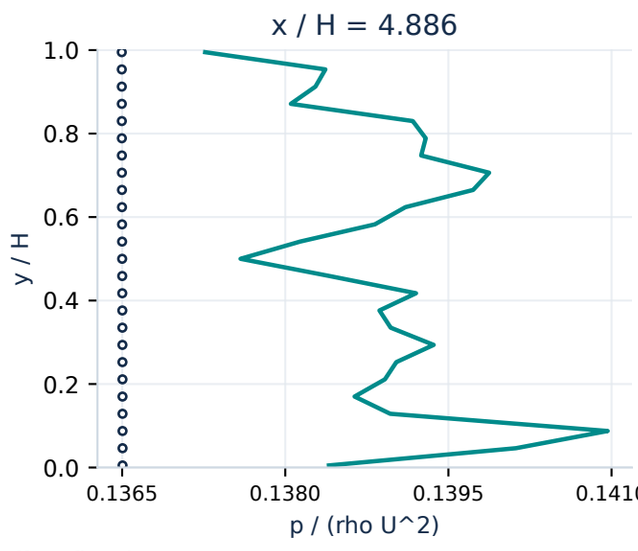






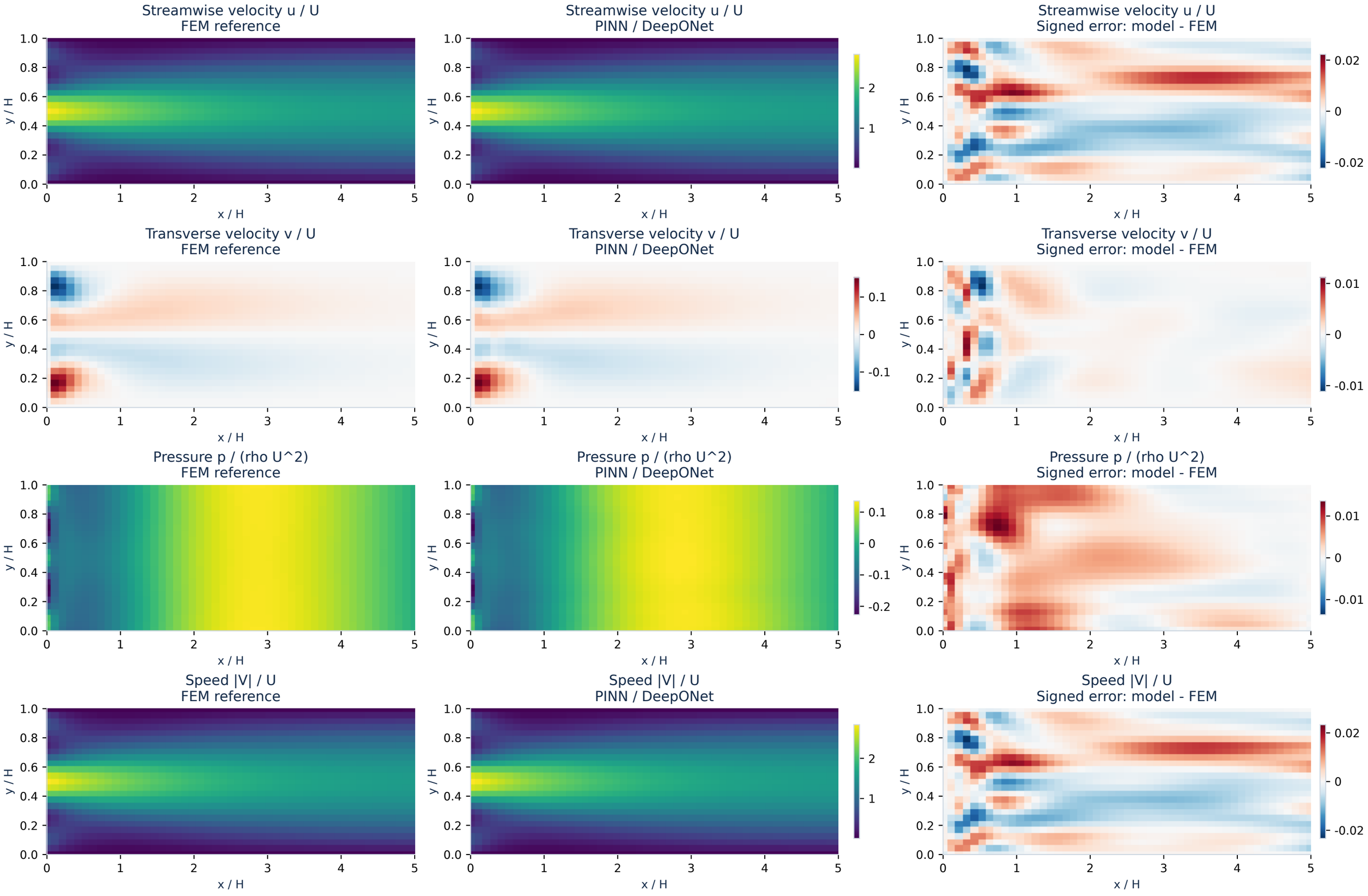






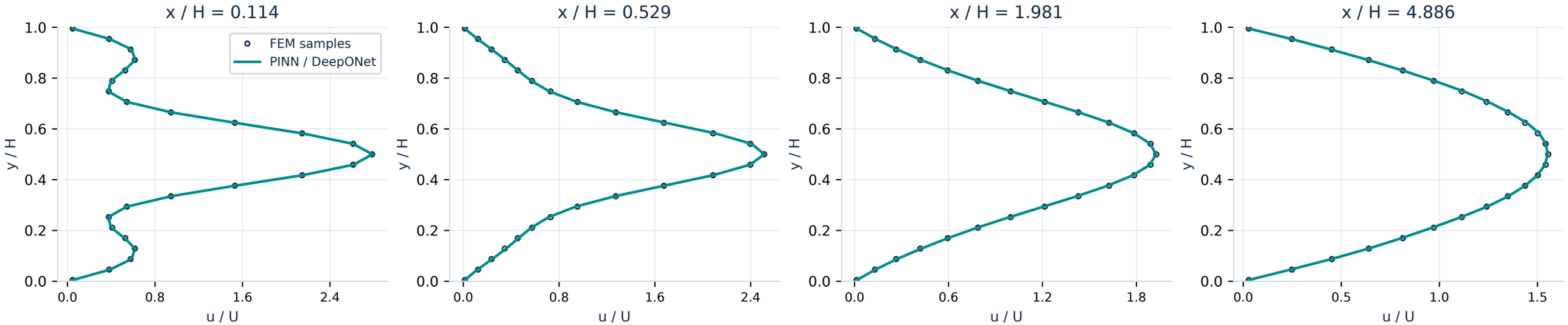
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

B04 | Re = 100 | a = [+0.000, +0.000, +0.000, +0.780]
Velocity L2: 0.567% Pressure L2: 4.224% v RMSE: 0.00159 U



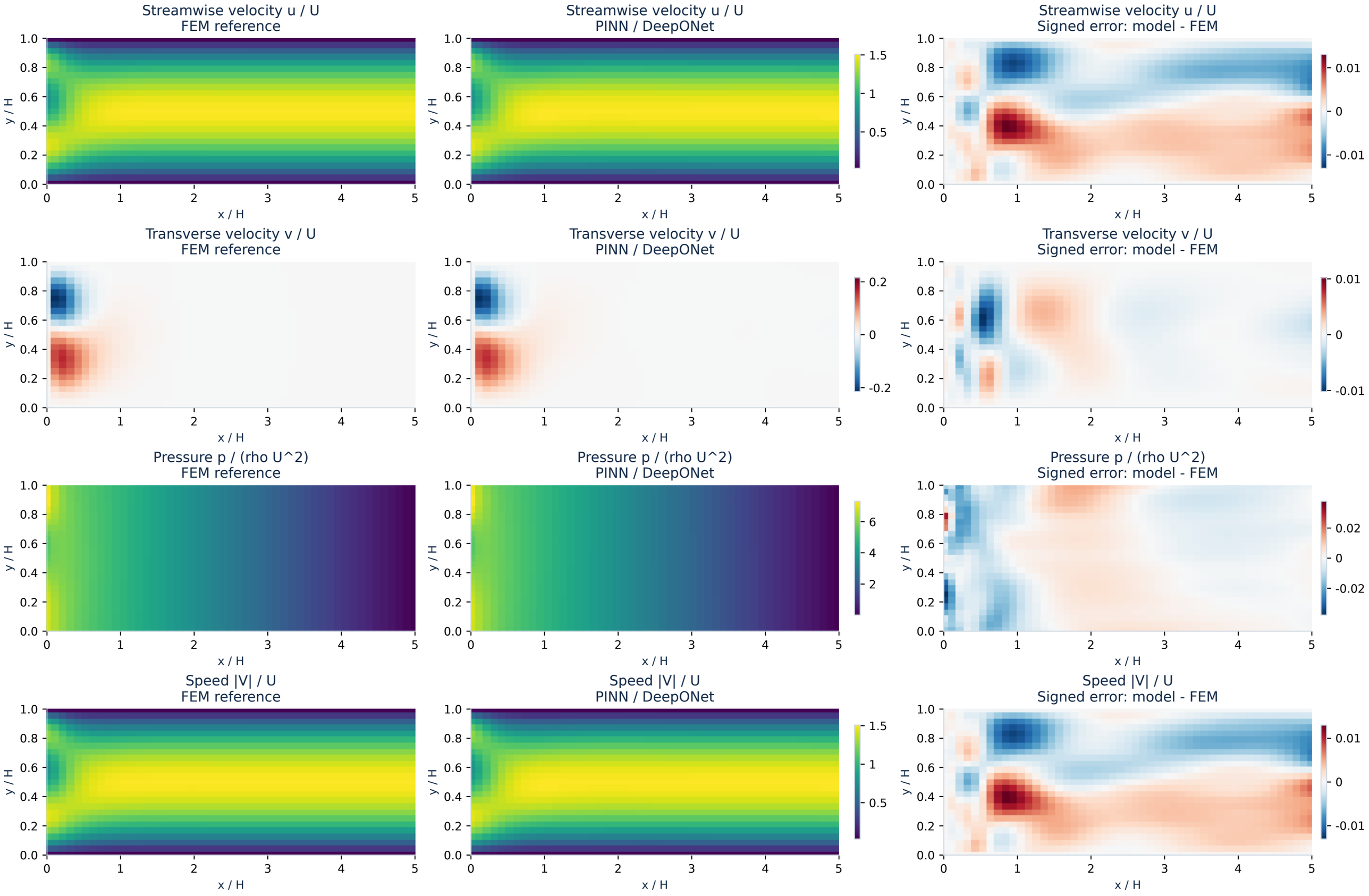
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

B04 | Re = 100 | a = [+0.000, +0.000, +0.000, +0.780]
Velocity L2: 0.567% Pressure L2: 4.224% v RMSE: 0.00159 U



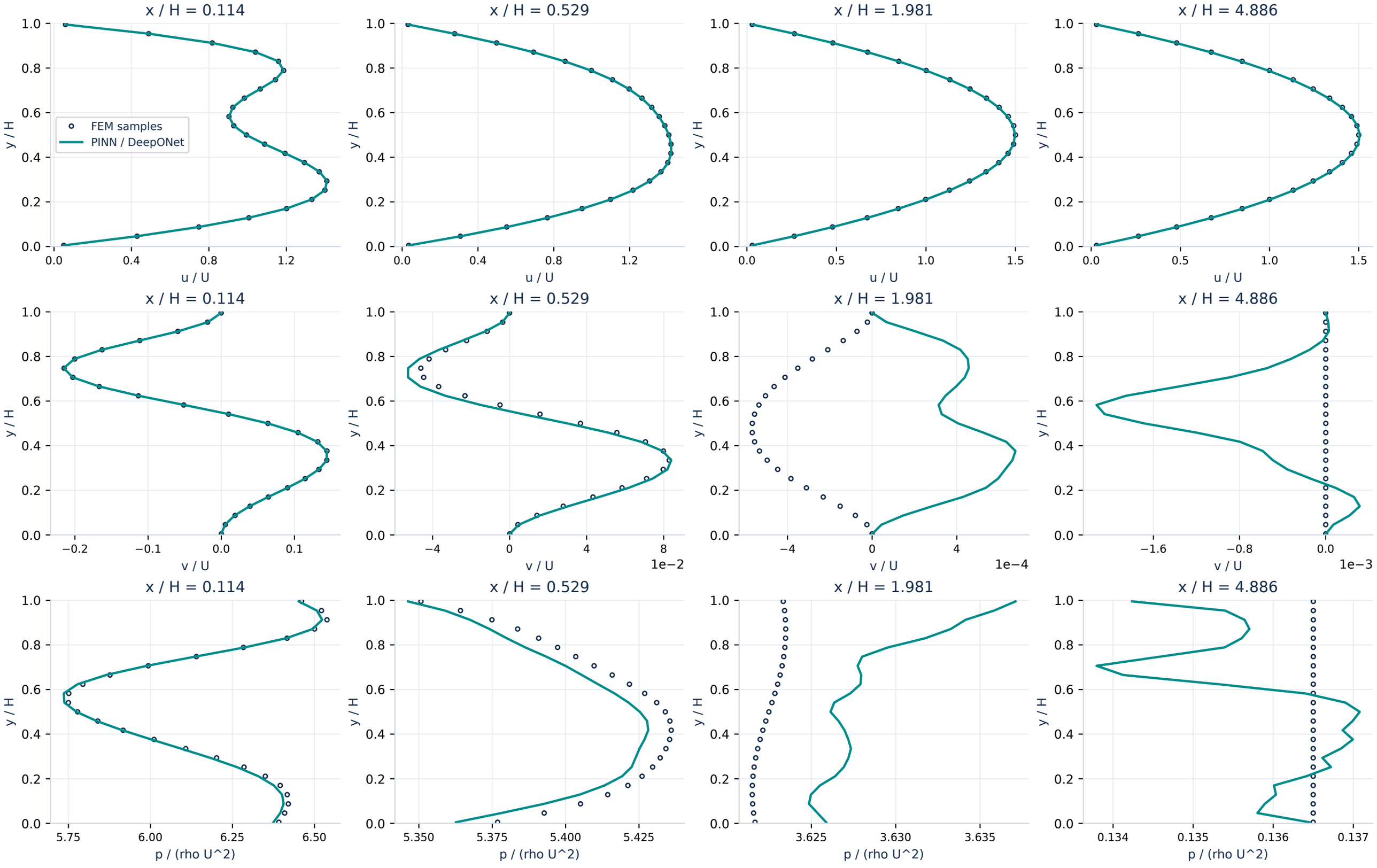
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

B05 | Re = 10 | a = [-0.008, +0.528, -0.158, +0.038]
Velocity L2: 0.382% Pressure L2: 0.185% v RMSE: 0.00145 U



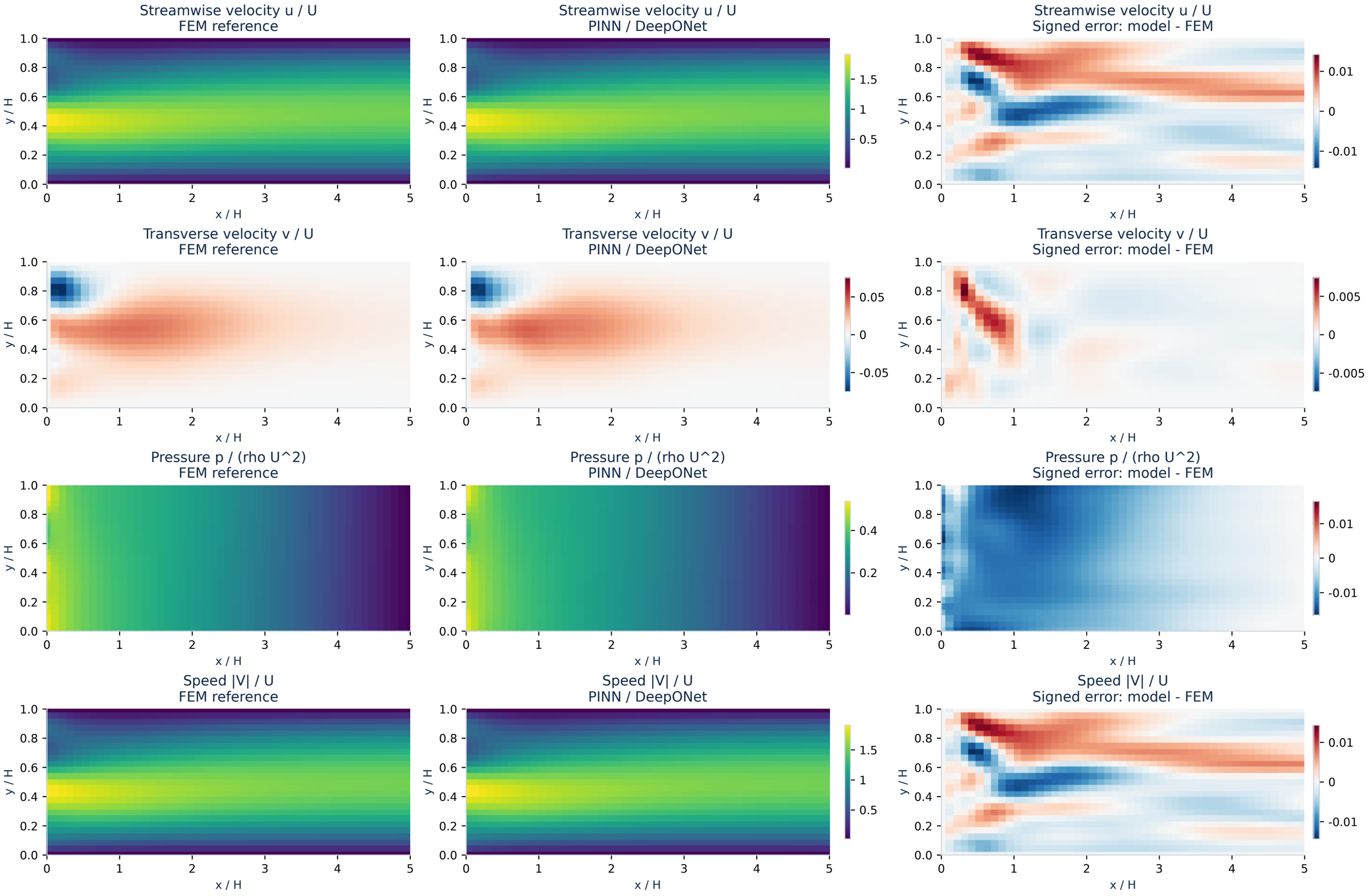
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

B05 | Re = 10 | a = [-0.008, +0.528, -0.158, +0.038]
Velocity L2: 0.382% Pressure L2: 0.185% v RMSE: 0.00145 U



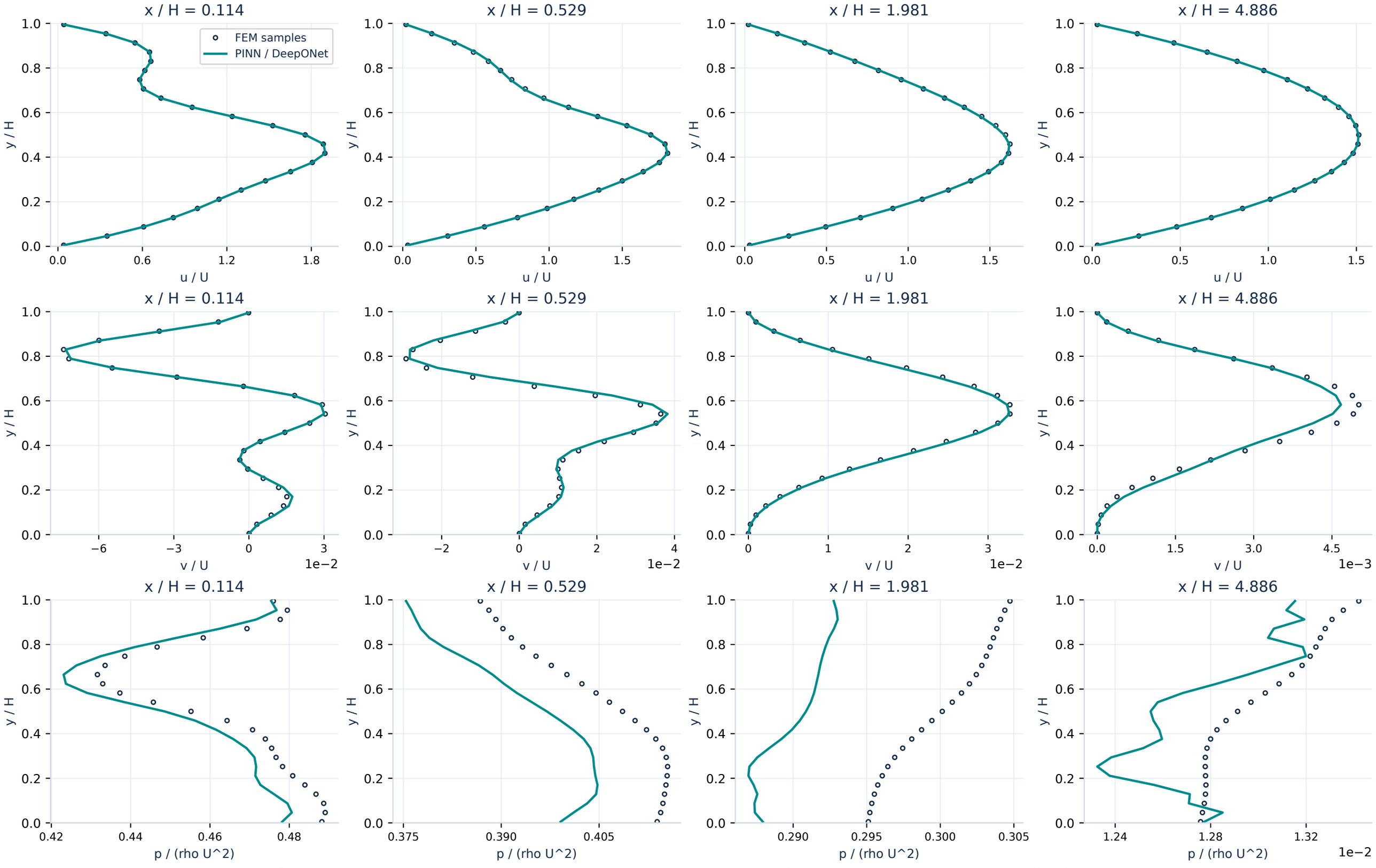
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

B06 | Re = 100 | a = [+0.197, +0.114, -0.243, +0.234]
Velocity L2: 0.390% Pressure L2: 2.929% v RMSE: 0.00098 U



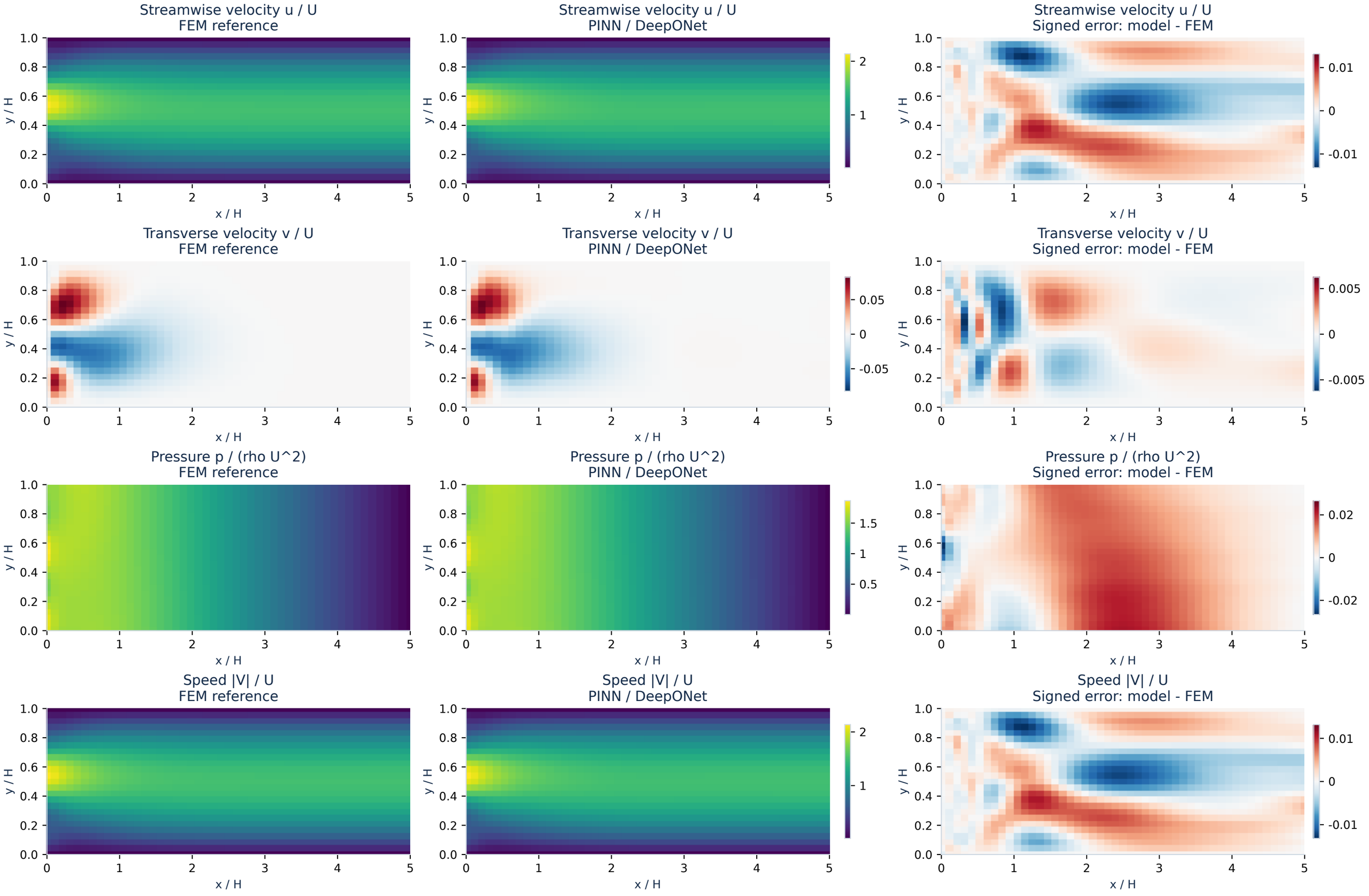
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

B06 | Re = 100 | a = [+0.197, +0.114, -0.243, +0.234]
Velocity L2: 0.390% Pressure L2: 2.929% v RMSE: 0.00098 U



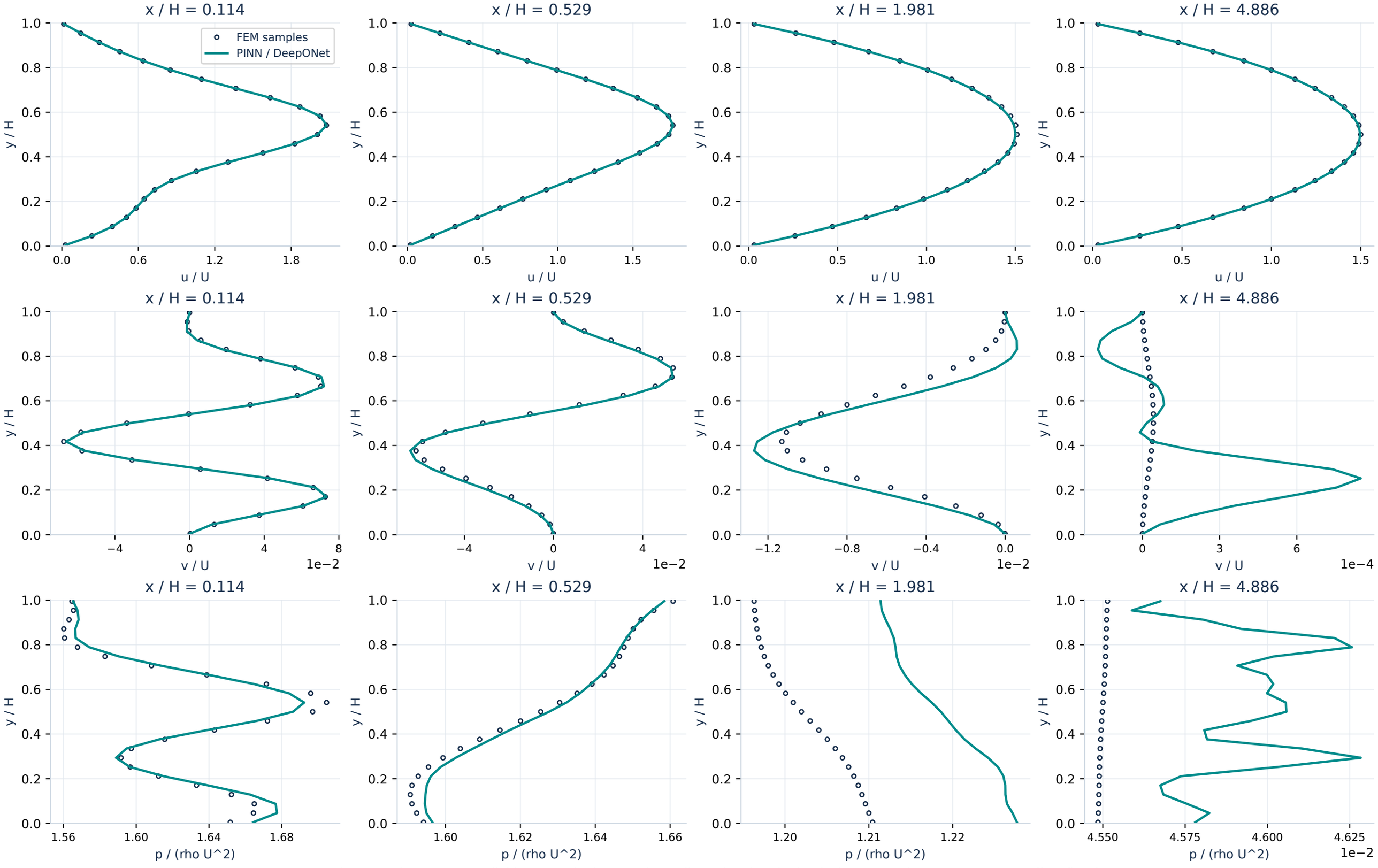
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

B07 | Re = 30 | a = [-0.001, -0.290, +0.262, +0.186]
Velocity L2: 0.415% Pressure L2: 0.968% v RMSE: 0.00133 U



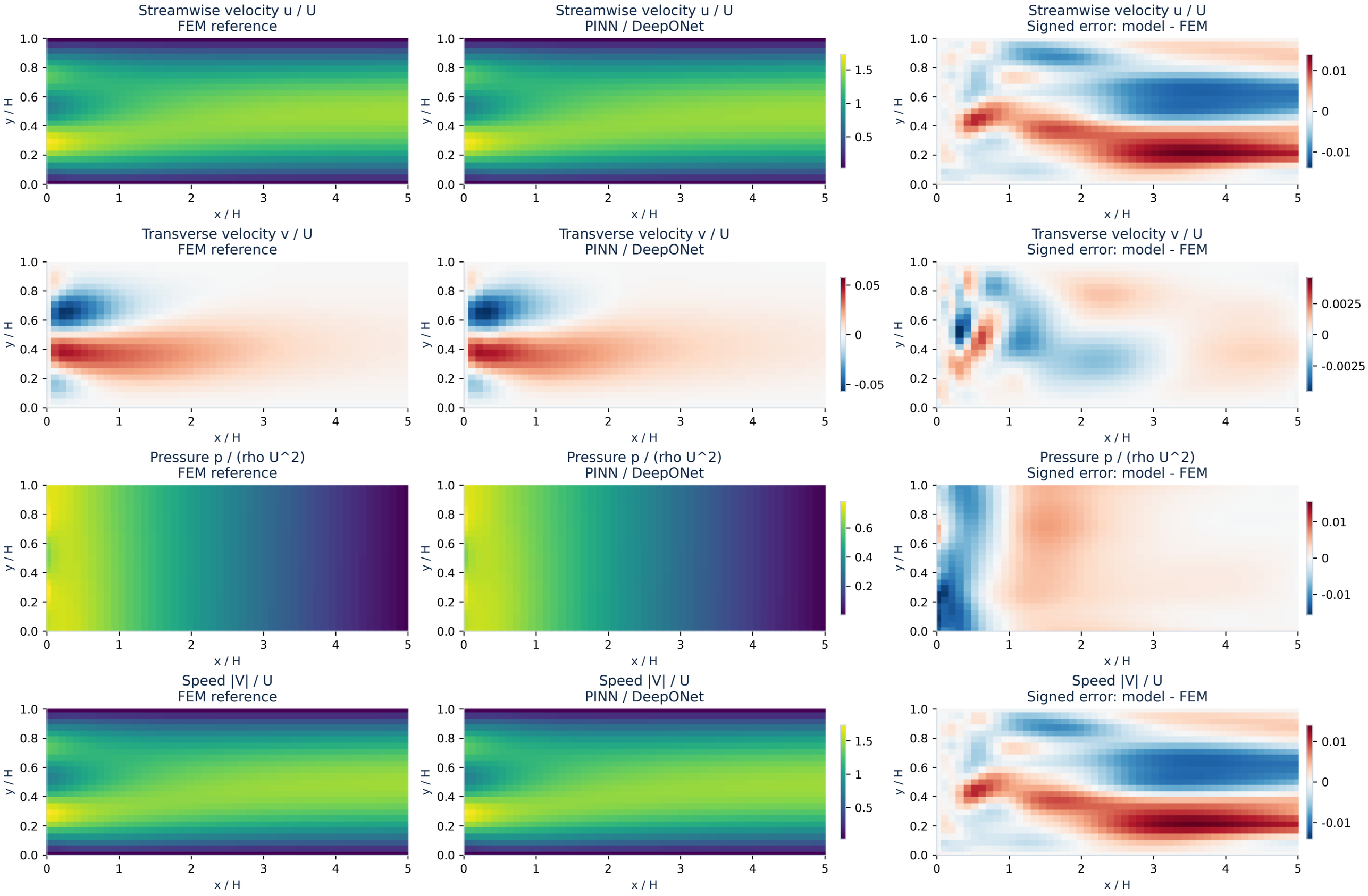
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

B07 | Re = 30 | a = [-0.001, -0.290, +0.262, +0.186]
Velocity L2: 0.415% Pressure L2: 0.968% v RMSE: 0.00133 U



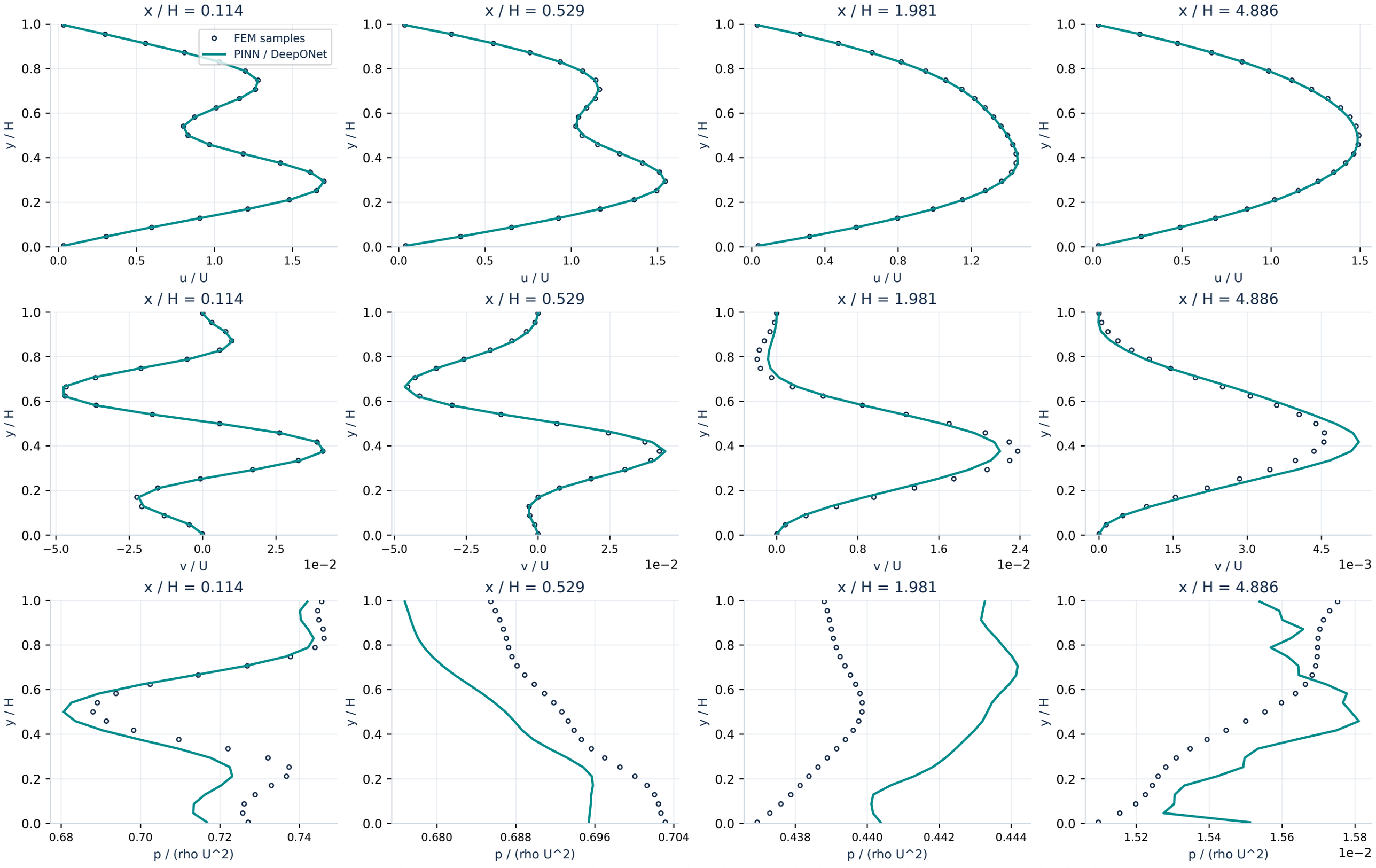
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

B08 | Re = 90 | a = [+0.109, +0.256, -0.110, -0.249]
Velocity L2: 0.544% Pressure L2: 0.897% v RMSE: 0.00089 U



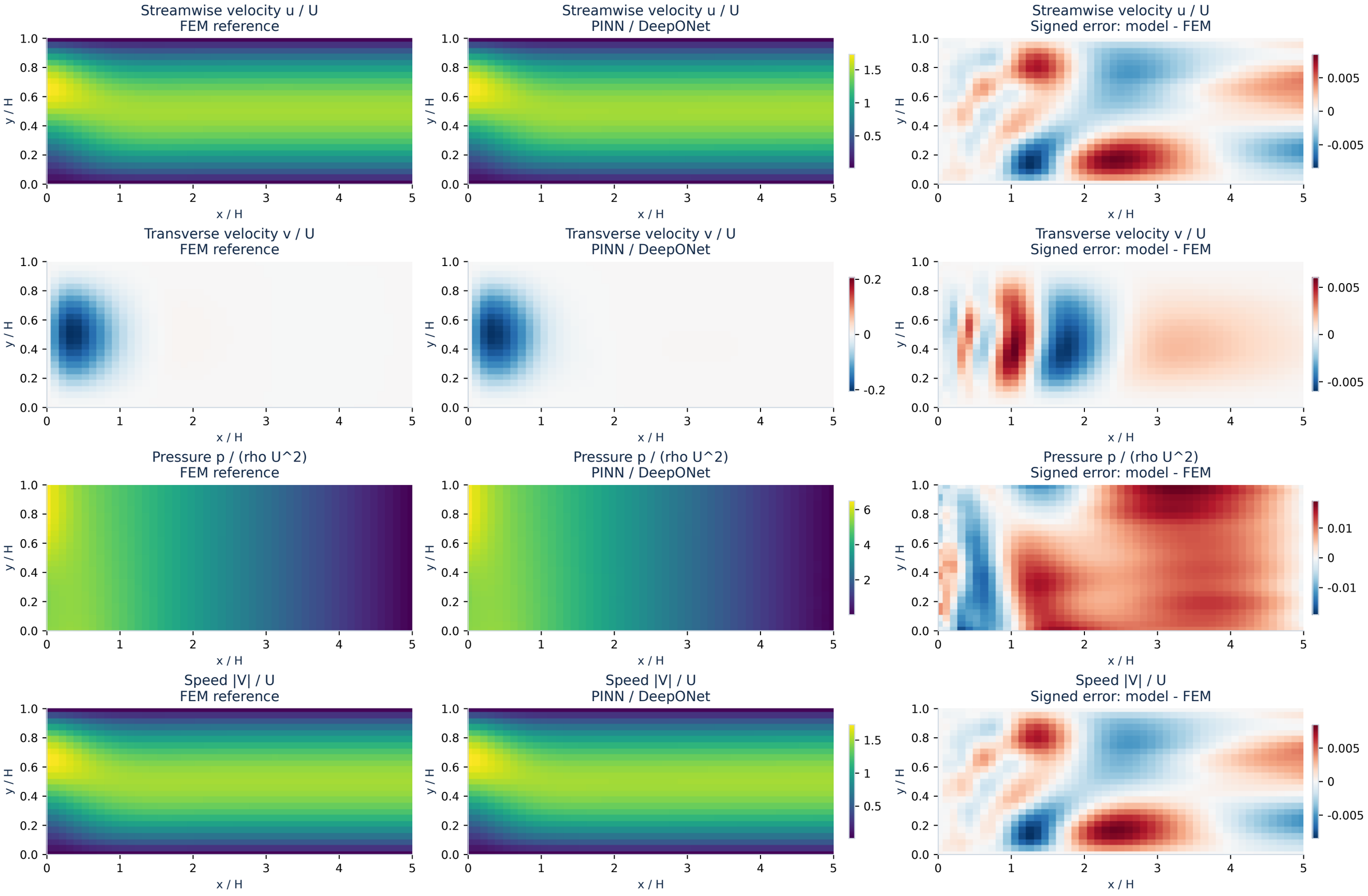
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

B08 | Re = 90 | a = [+0.109, +0.256, -0.110, -0.249]
Velocity L2: 0.544% Pressure L2: 0.897% v RMSE: 0.00089 U



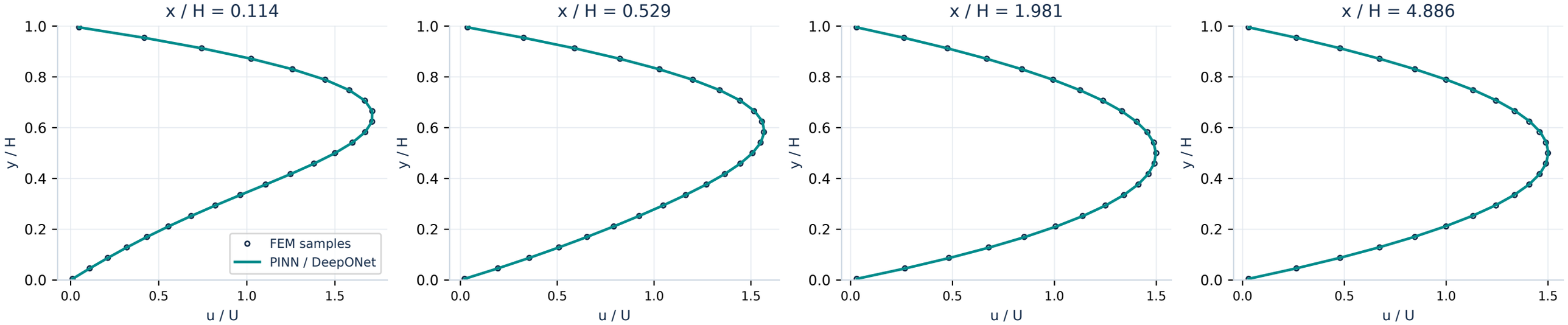
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

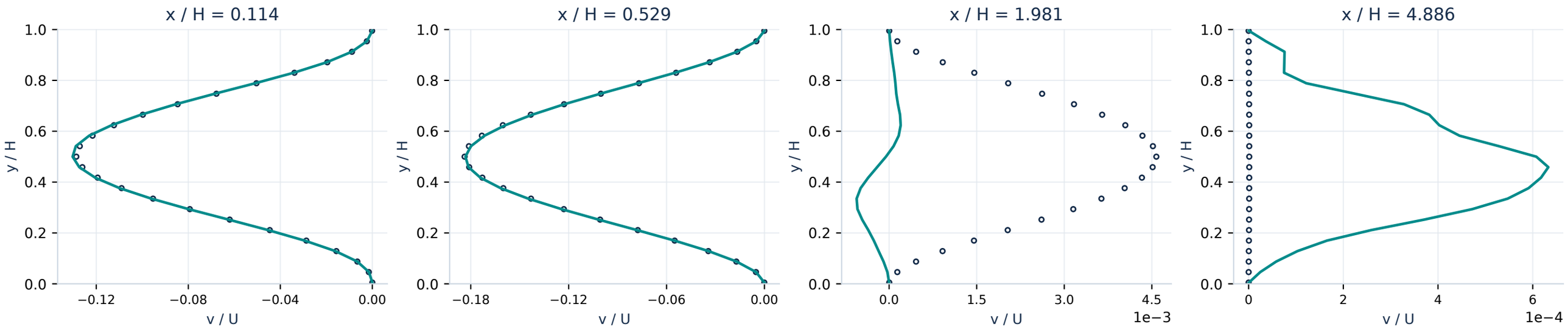
M01 | Re = 10 | a = [-0.600, +0.000, +0.000, +0.000]
Velocity L2: 0.295% Pressure L2: 0.273% v RMSE: 0.00172 U

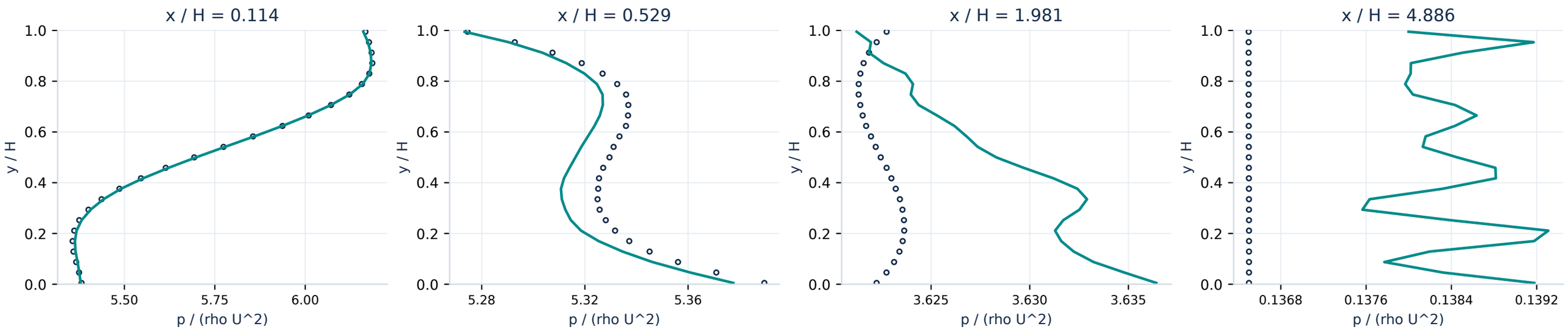


Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M01 | Re = 10 | a = [-0.600, +0.000, +0.000, +0.000]
Velocity L2: 0.295% Pressure L2: 0.273% v RMSE: 0.00172 U

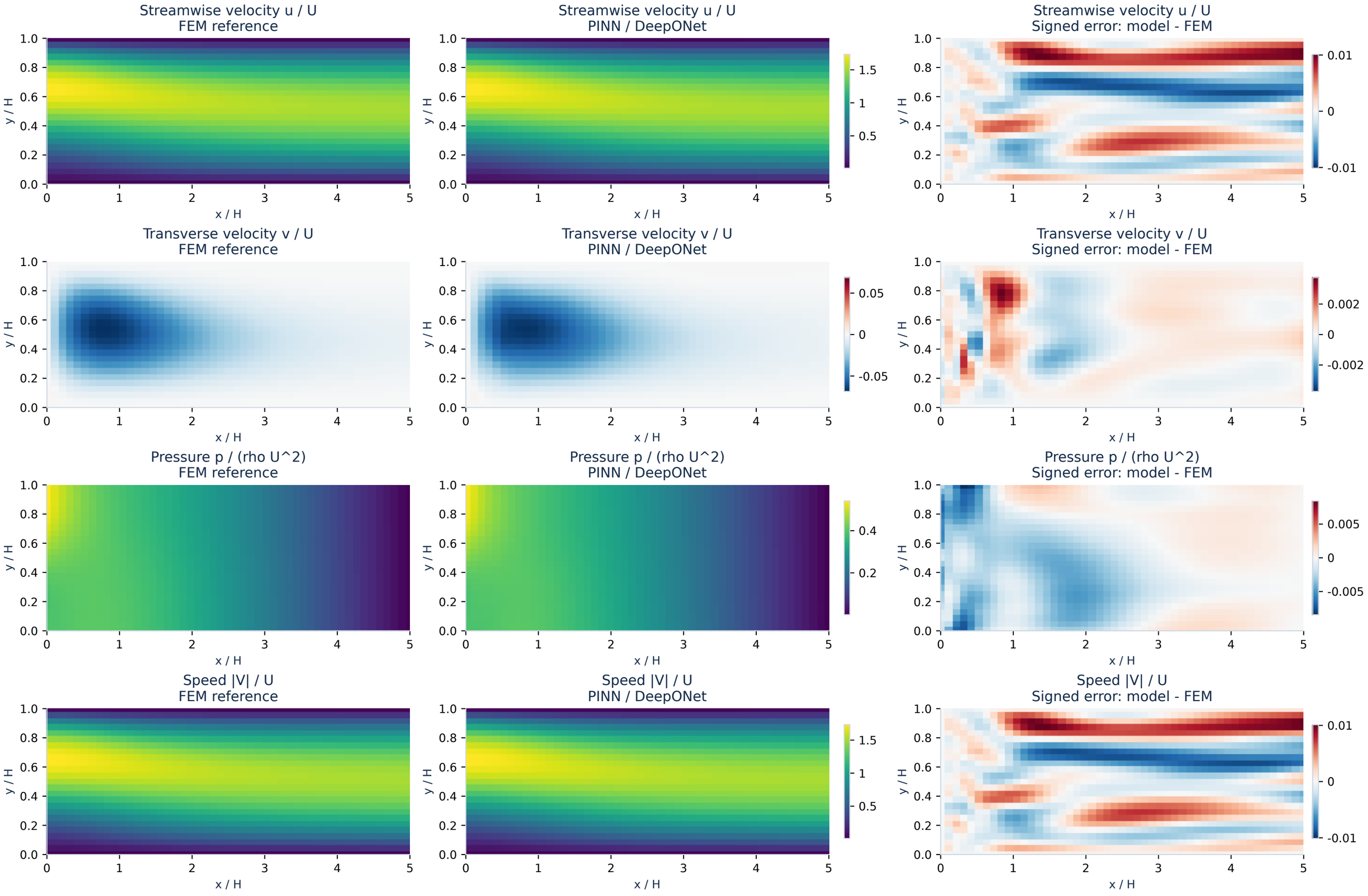






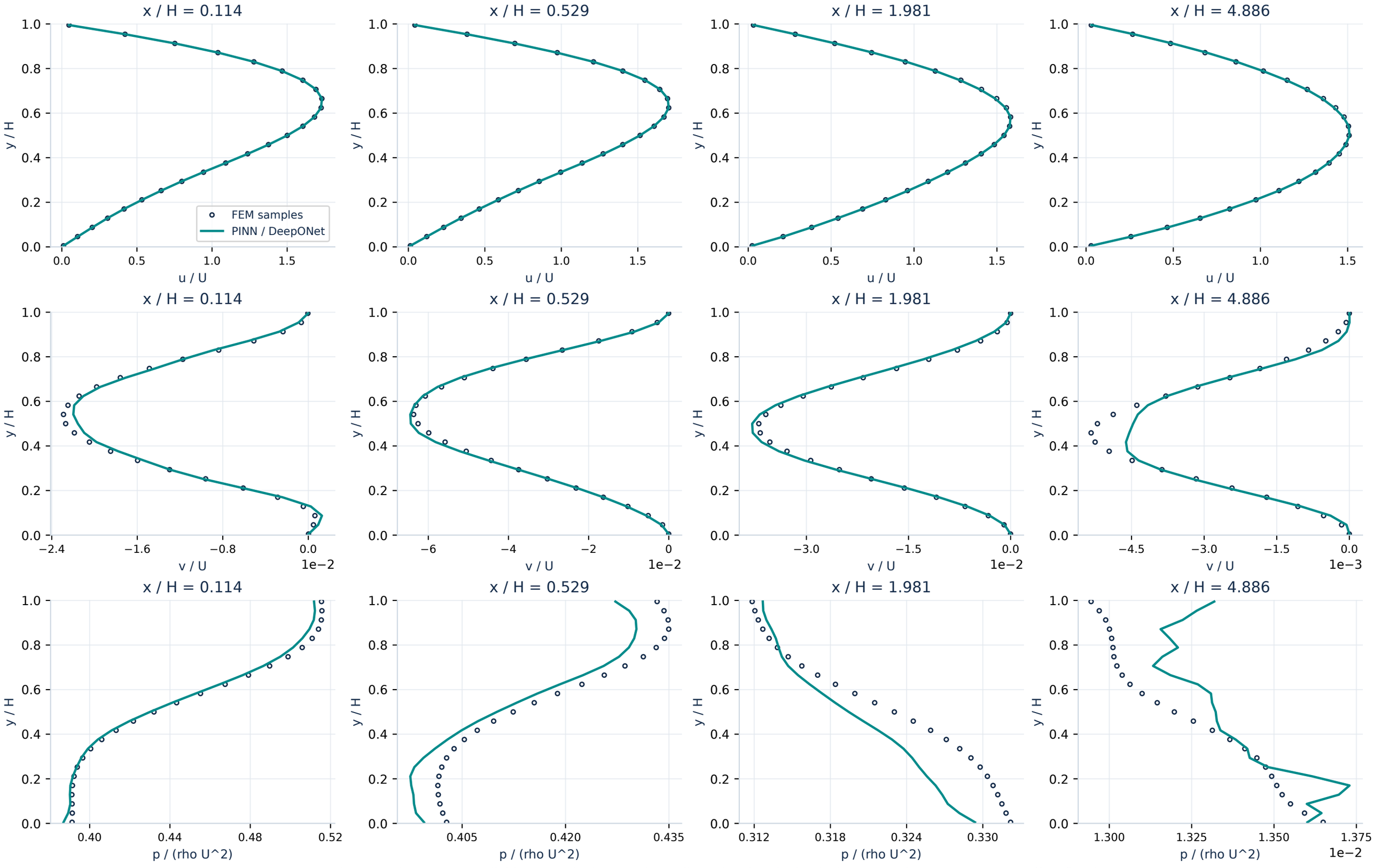
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

M02 | Re = 100 | a = [-0.600, +0.000, +0.000, +0.000]
Velocity L2: 0.367% Pressure L2: 0.767% v RMSE: 0.00067 U



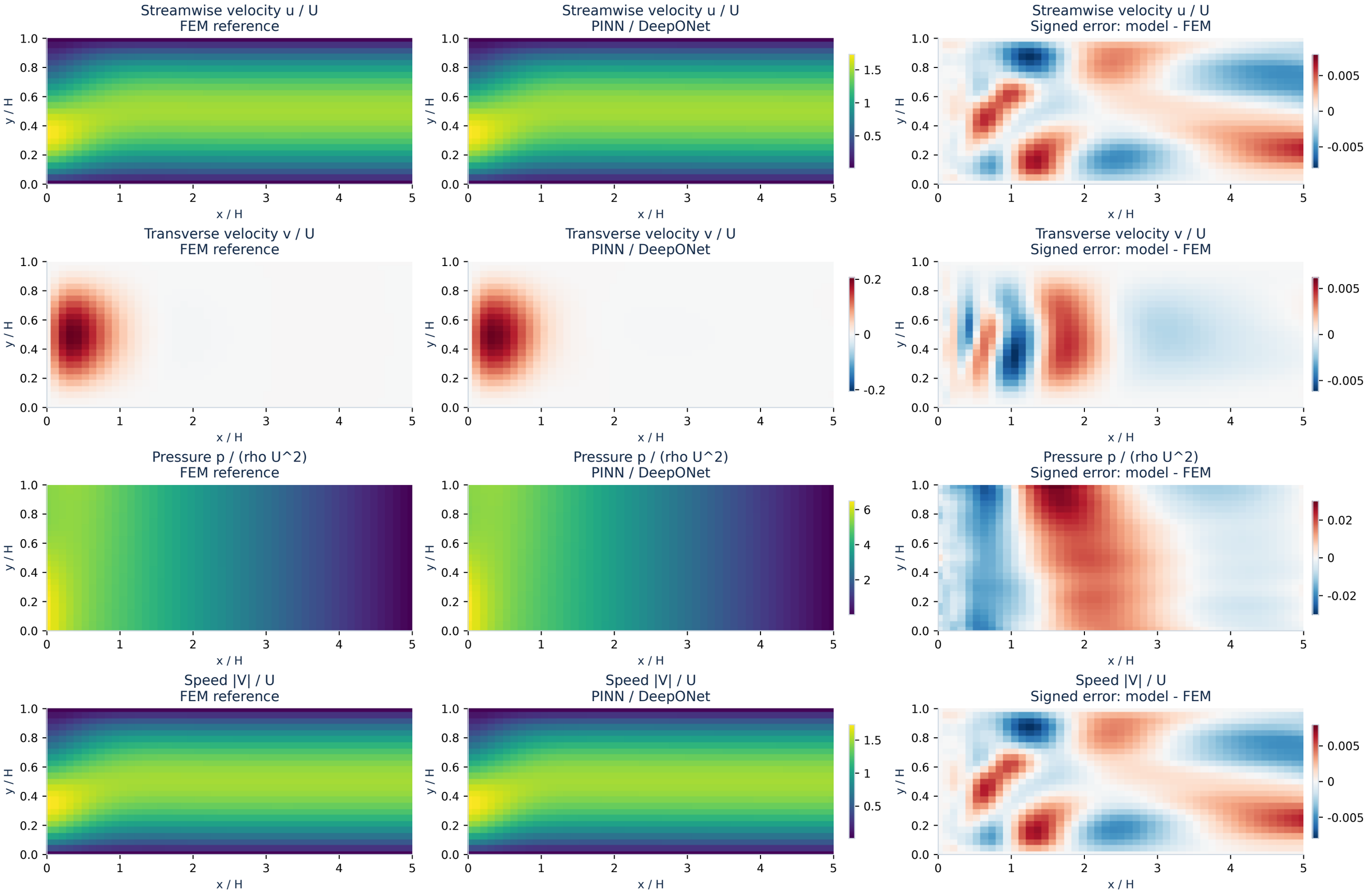
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M02 | Re = 100 | a = [-0.600, +0.000, +0.000, +0.000]
Velocity L2: 0.367% Pressure L2: 0.767% v RMSE: 0.00067 U



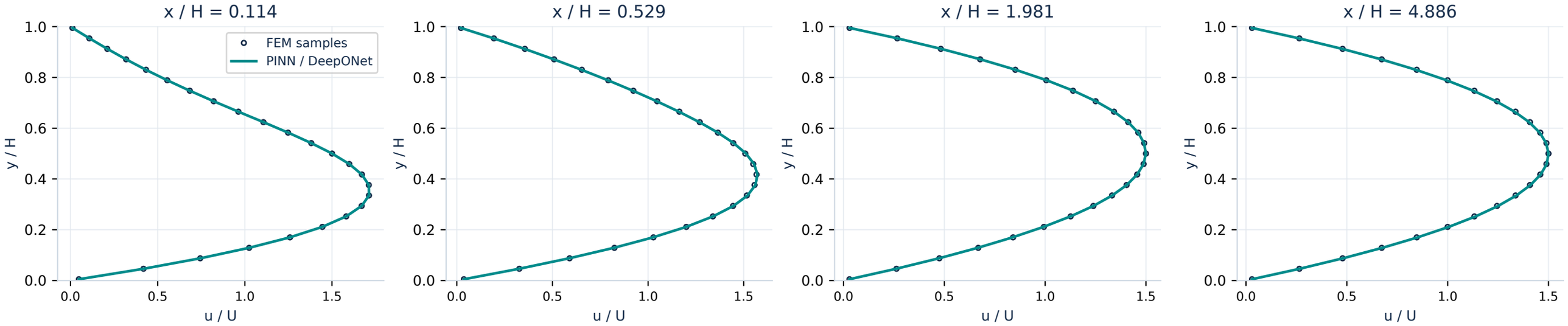
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

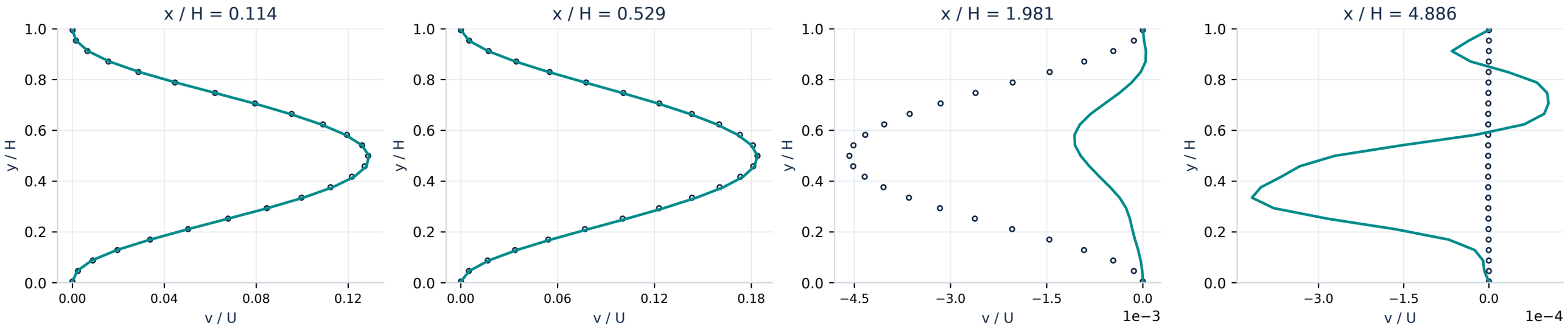
M03 | Re = 10 | a = [+0.600, +0.000, +0.000, +0.000]
Velocity L2: 0.272% Pressure L2: 0.314% v RMSE: 0.00155 U

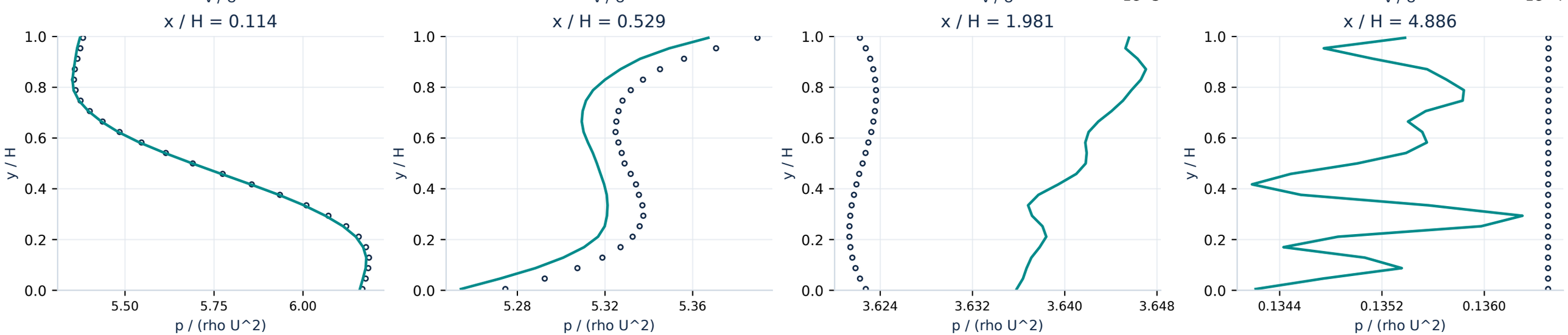


Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M03 | Re = 10 | a = [+0.600, +0.000, +0.000, +0.000]
Velocity L2: 0.272% Pressure L2: 0.314% v RMSE: 0.00155 U

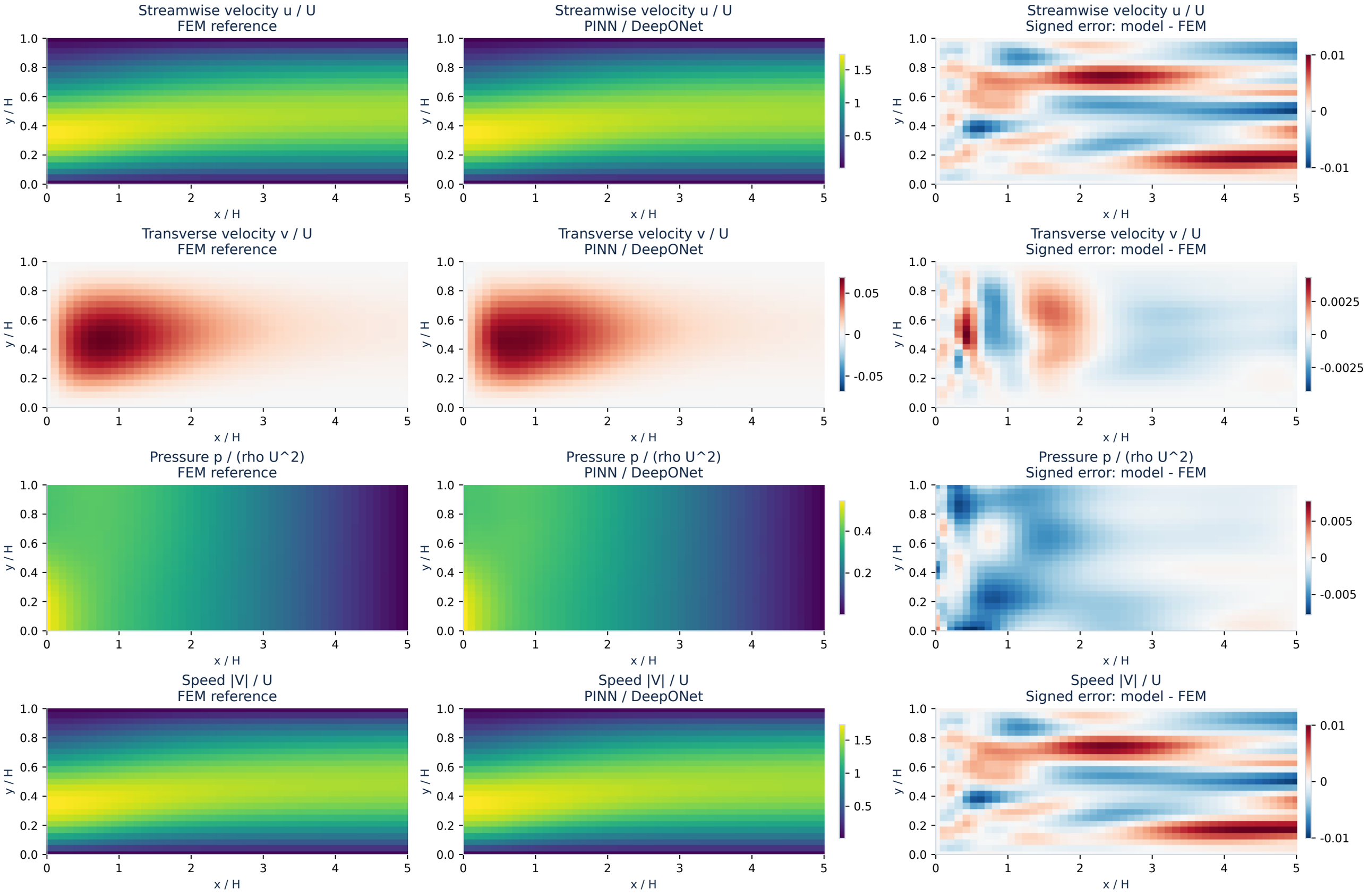






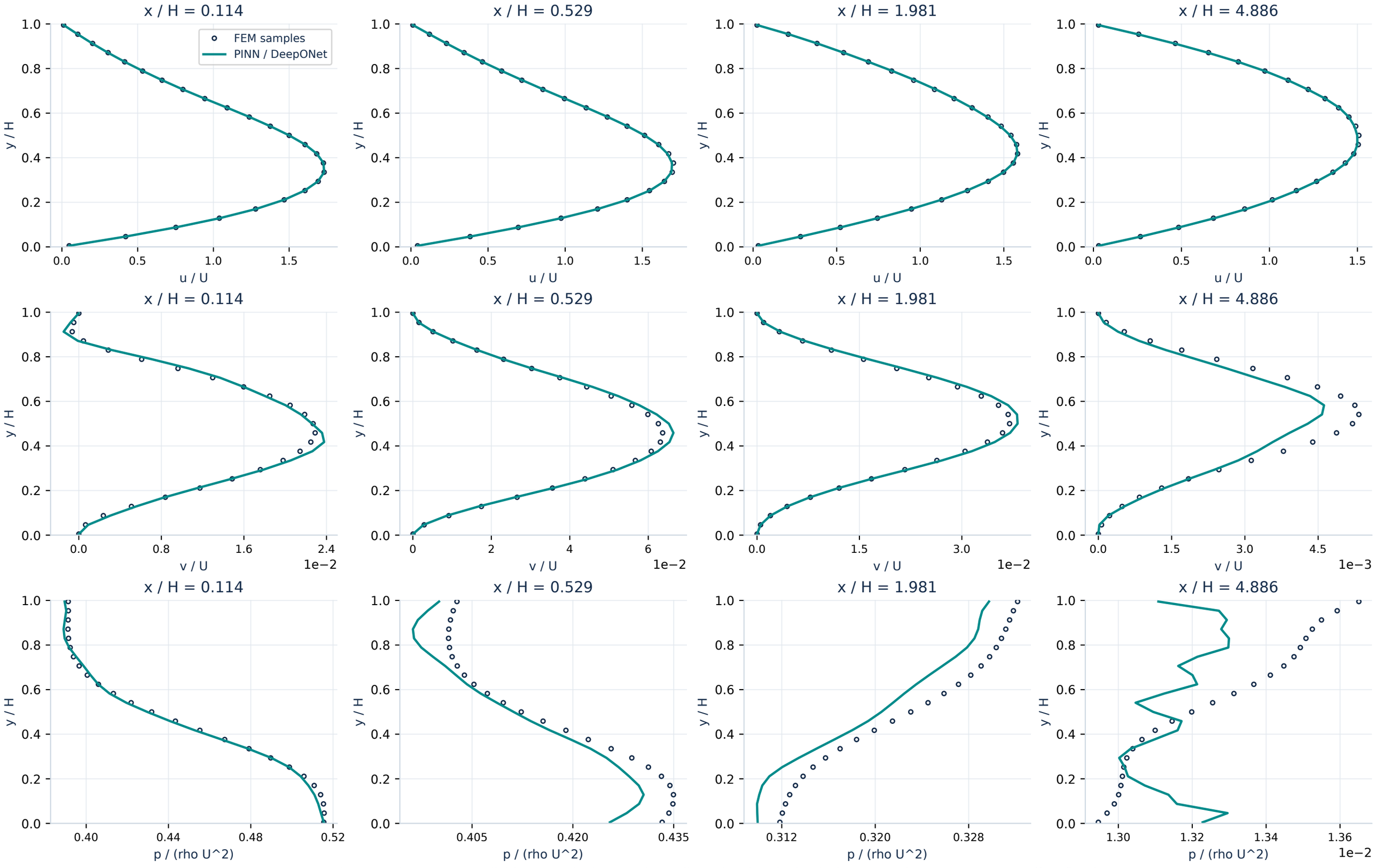
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

M04 | Re = 100 | a = [+0.600, +0.000, +0.000, +0.000]
Velocity L2: 0.325% Pressure L2: 0.844% v RMSE: 0.00087 U



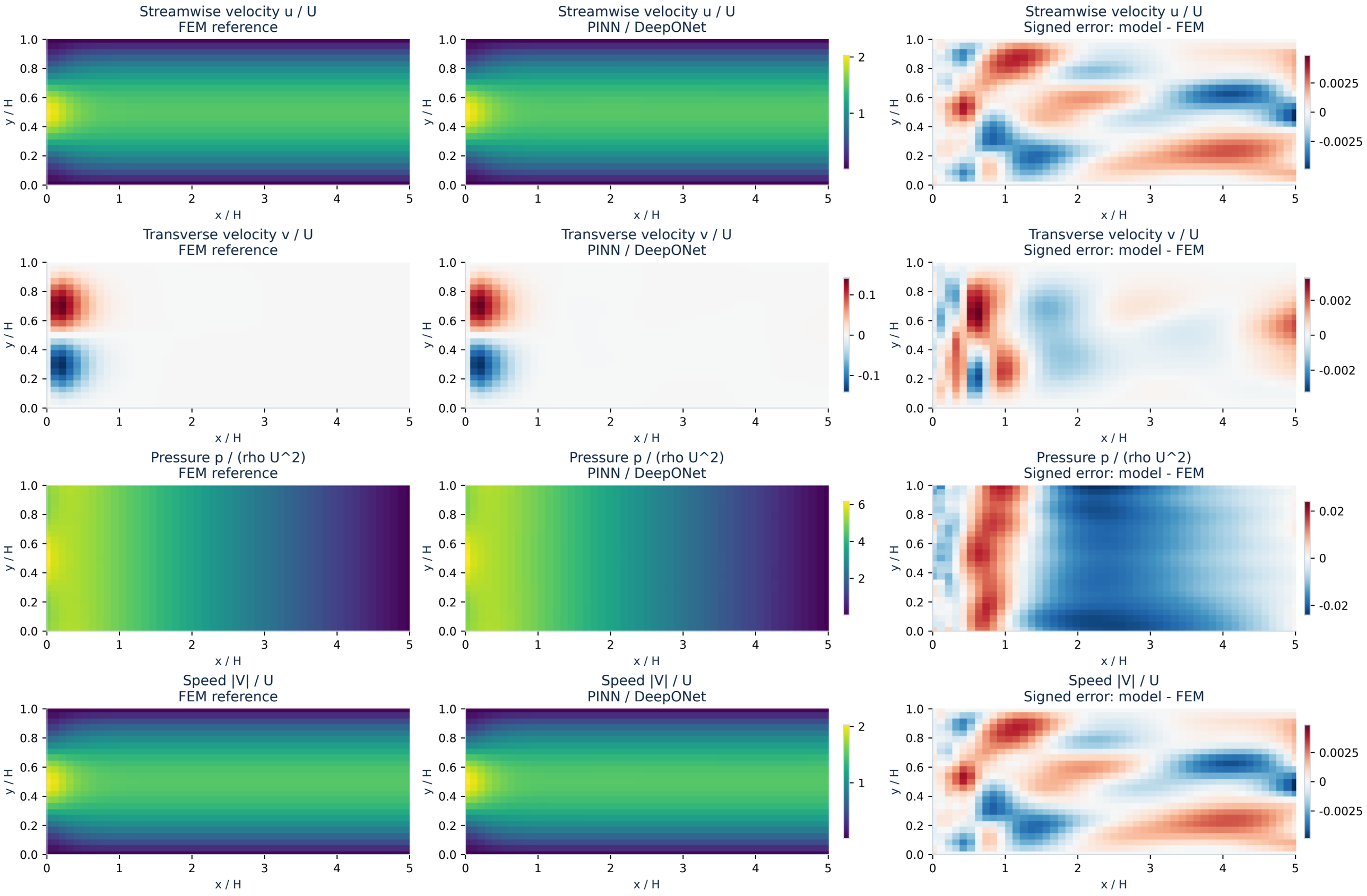
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M04 | Re = 100 | a = [+0.600, +0.000, +0.000, +0.000]
Velocity L2: 0.325% Pressure L2: 0.844% v RMSE: 0.00087 U



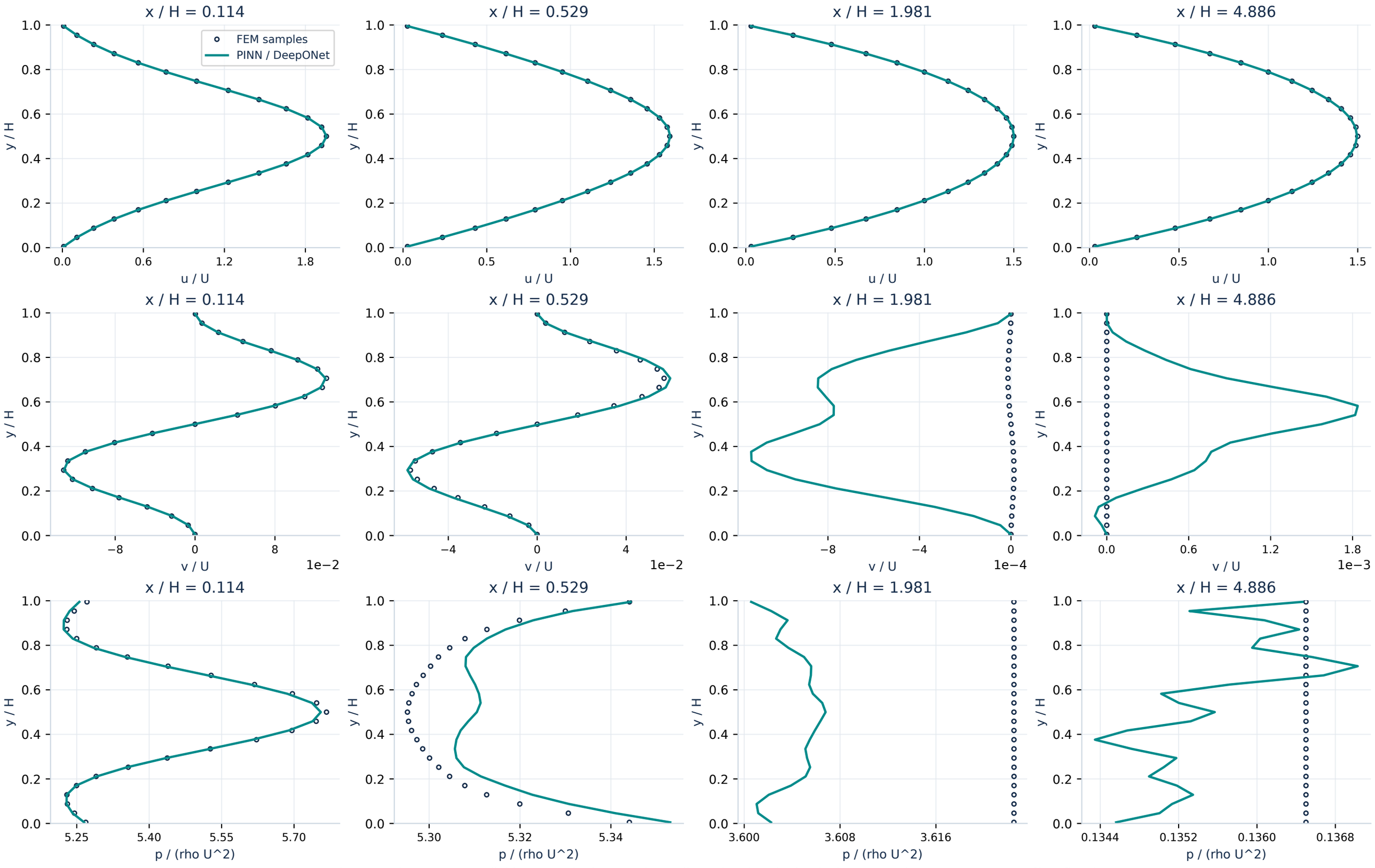
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

M05 | Re = 10 | a = [+0.000, -0.600, +0.000, +0.000]
Velocity L2: 0.155% Pressure L2: 0.369% v RMSE: 0.00073 U



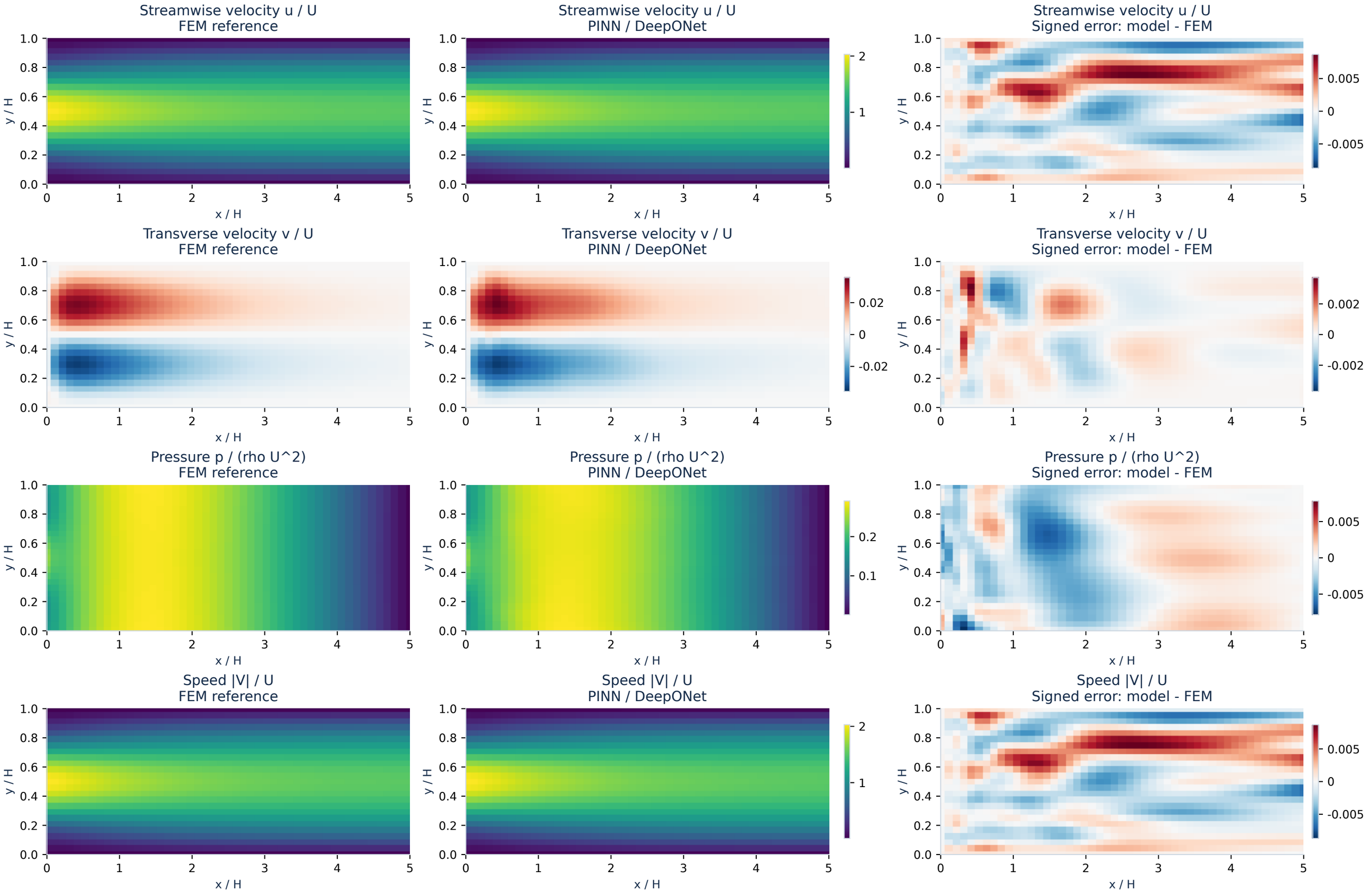
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M05 | Re = 10 | a = [+0.000, -0.600, +0.000, +0.000]
Velocity L2: 0.155% Pressure L2: 0.369% v RMSE: 0.00073 U



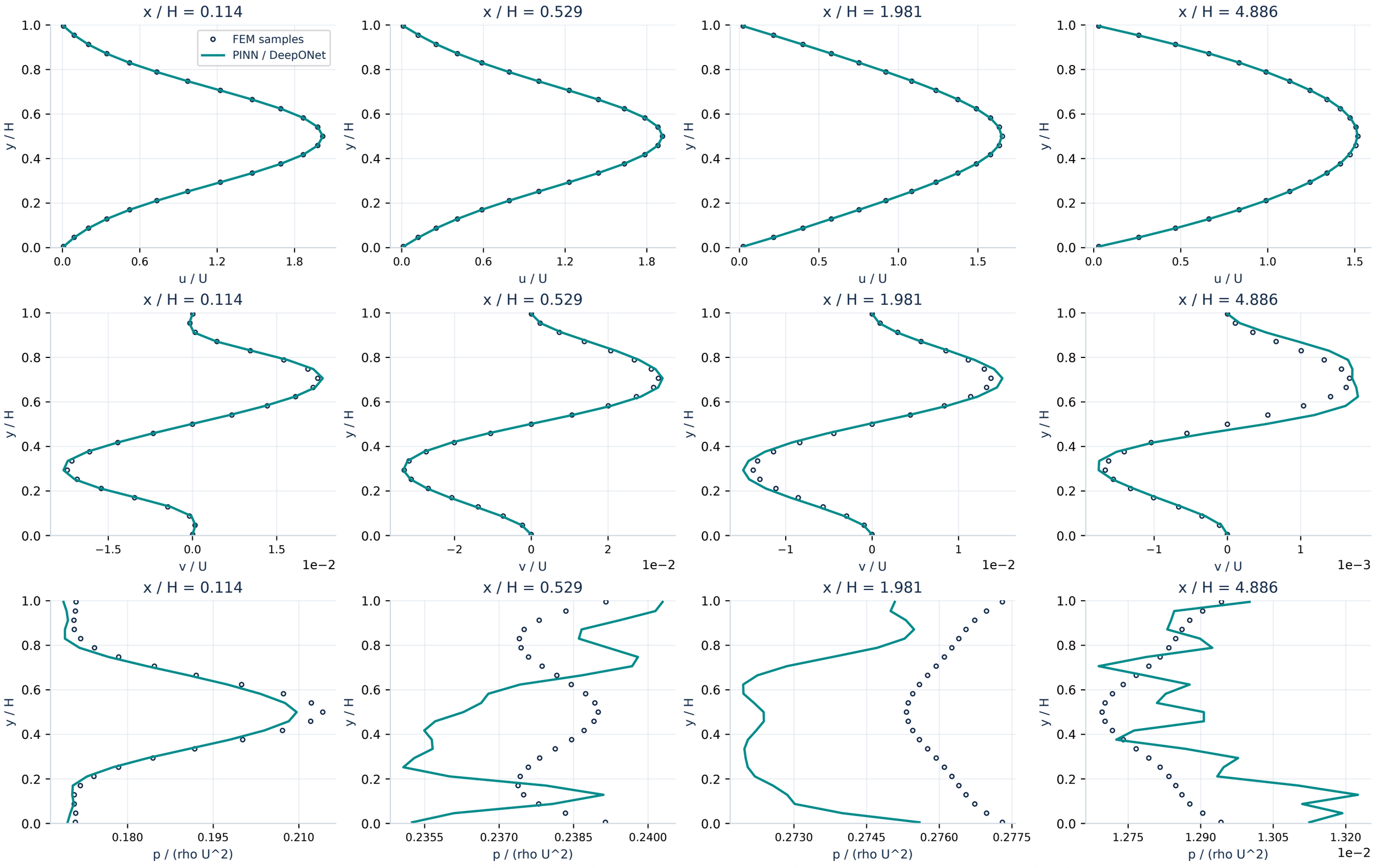
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

M06 | Re = 100 | a = [+0.000, -0.600, +0.000, +0.000]
Velocity L2: 0.263% Pressure L2: 1.014% v RMSE: 0.00059 U



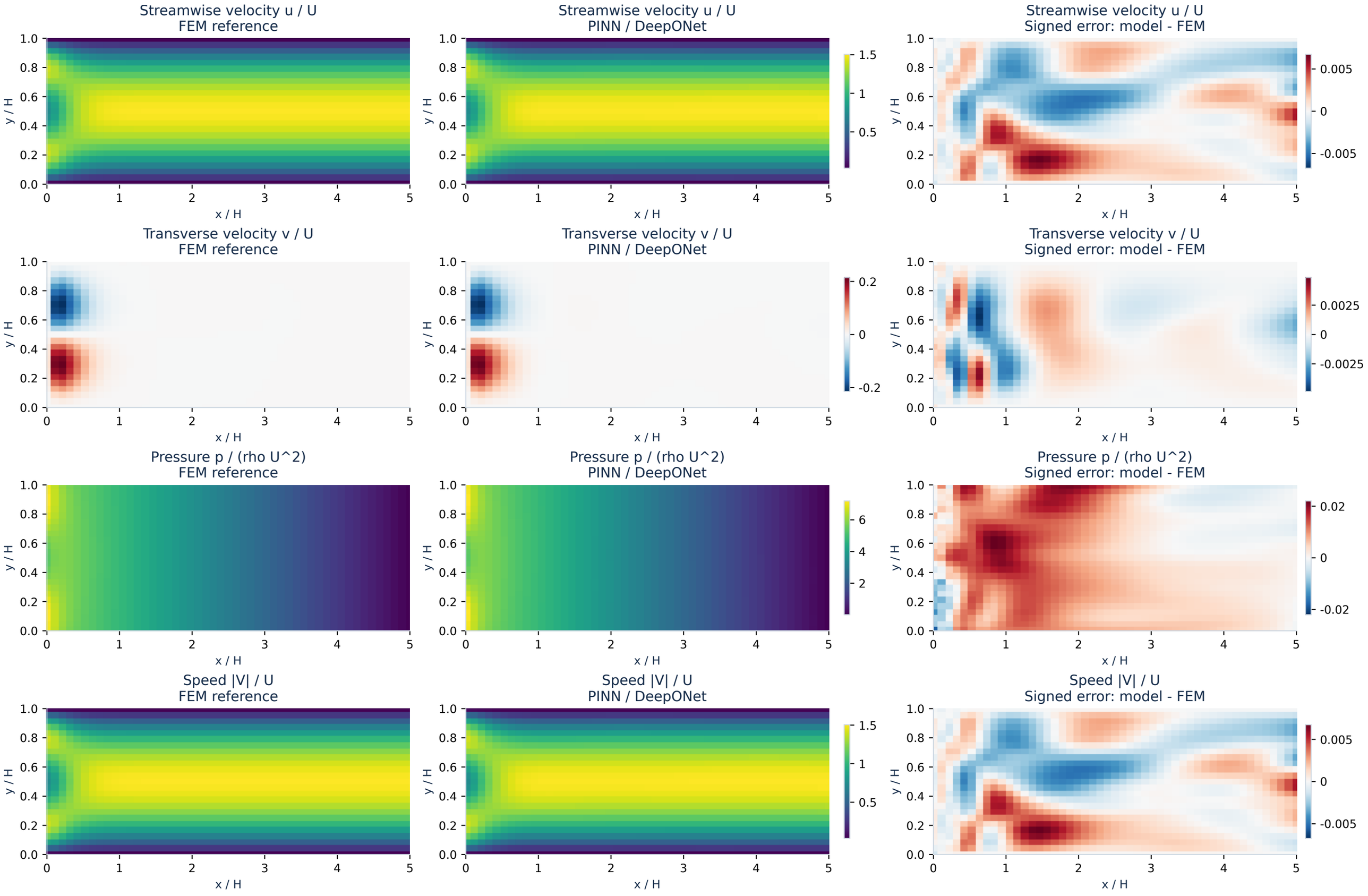
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M06 | Re = 100 | a = [+0.000, -0.600, +0.000, +0.000]
Velocity L2: 0.263% Pressure L2: 1.014% v RMSE: 0.00059 U



Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

M07 | Re = 10 | a = [+0.000, +0.600, +0.000, +0.000]
Velocity L2: 0.217% Pressure L2: 0.247% v RMSE: 0.00105 U

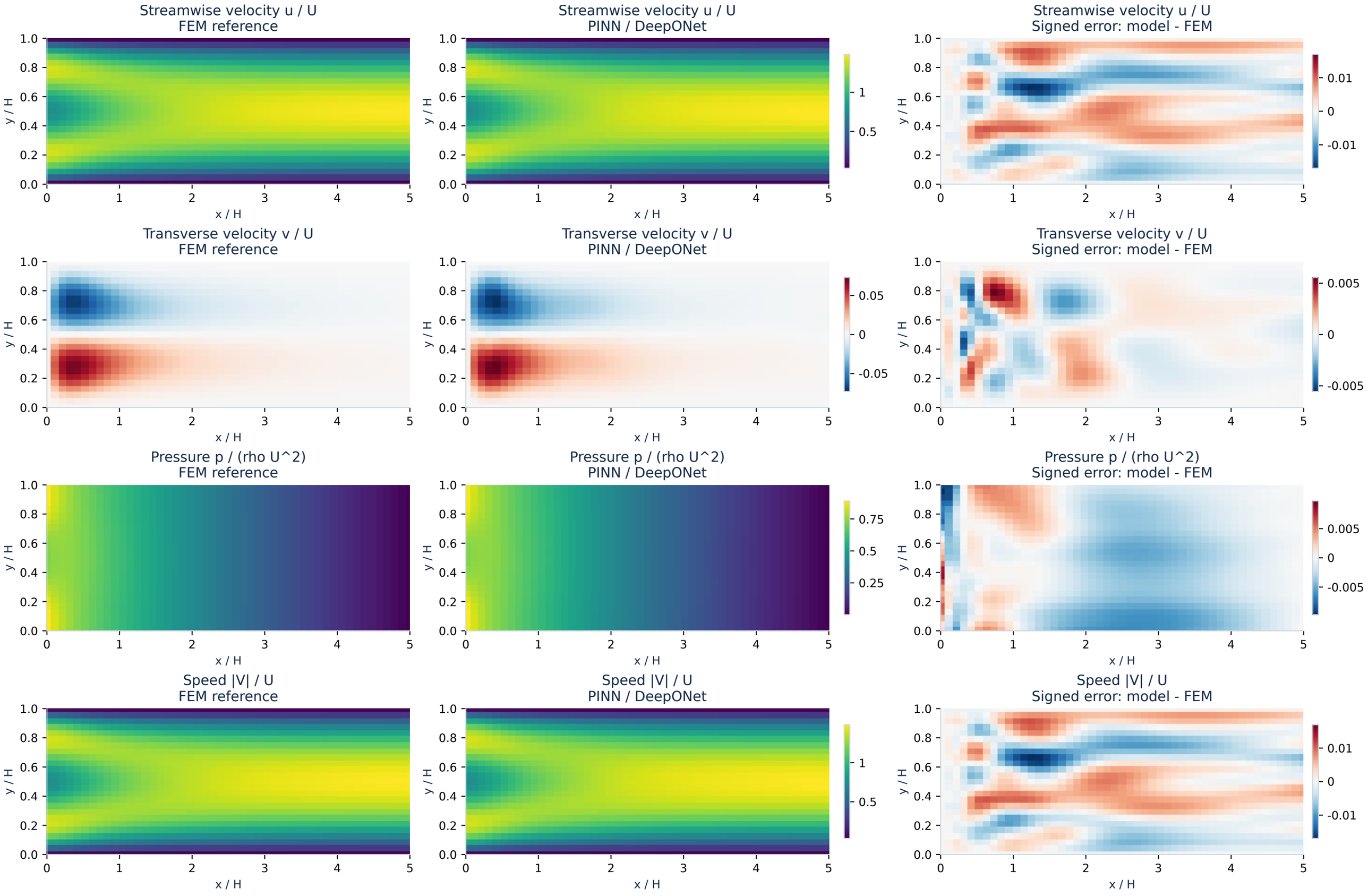


Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M07 | Re = 10 | a = [+0.000, +0.600, +0.000, +0.000]
Velocity L2: 0.217% Pressure L2: 0.247% v RMSE: 0.00105 U

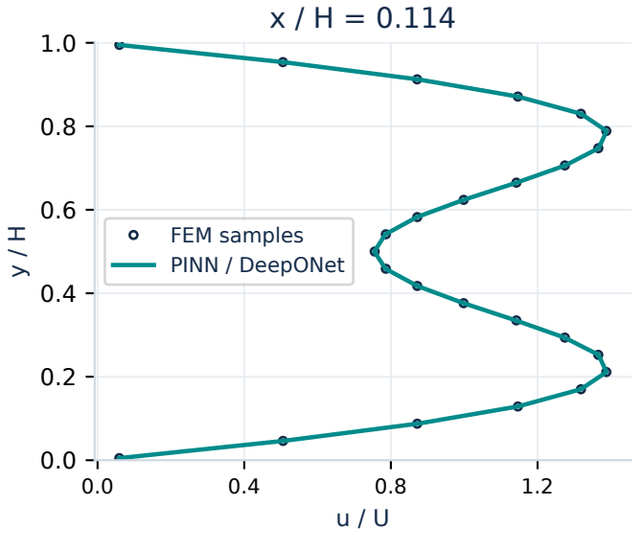
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

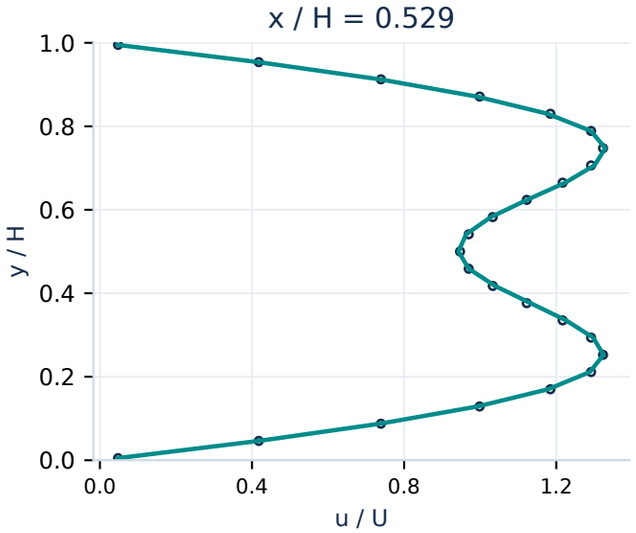
M08 | Re = 100 | a = [+0.000, +0.600, +0.000, +0.000]
Velocity L2: 0.452% Pressure L2: 0.683% v RMSE: 0.00107 U

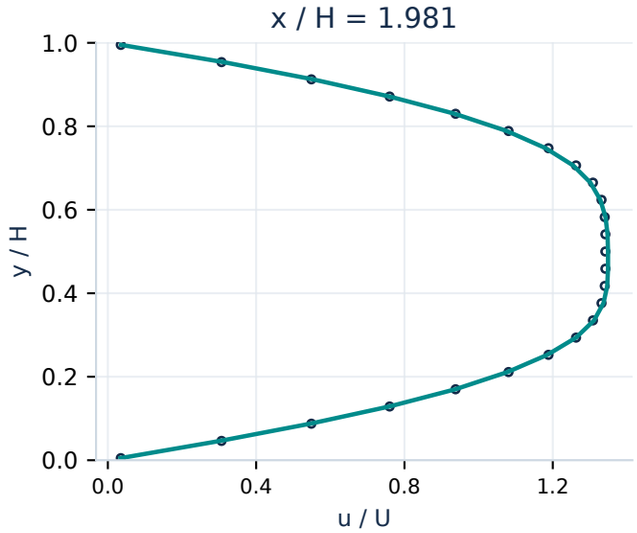


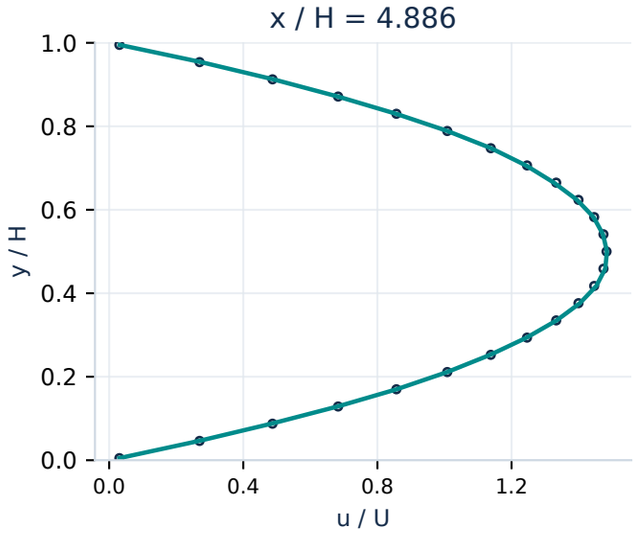
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

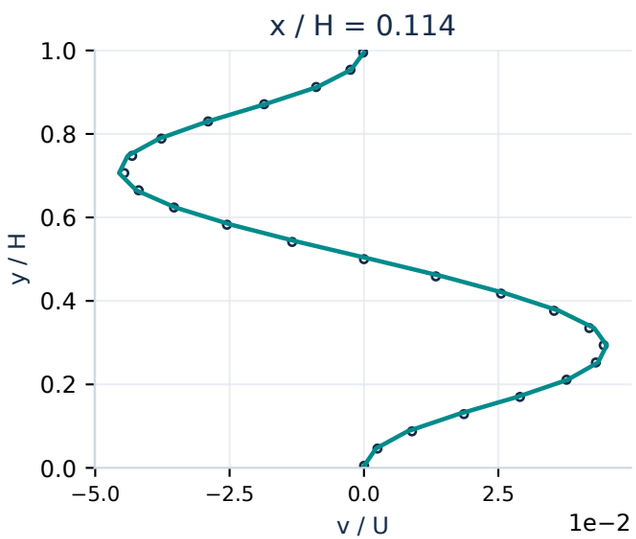
M08 | Re = 100 | a = [+0.000, +0.600, +0.000, +0.000]
Velocity L2: 0.452% Pressure L2: 0.683% v RMSE: 0.00107 U

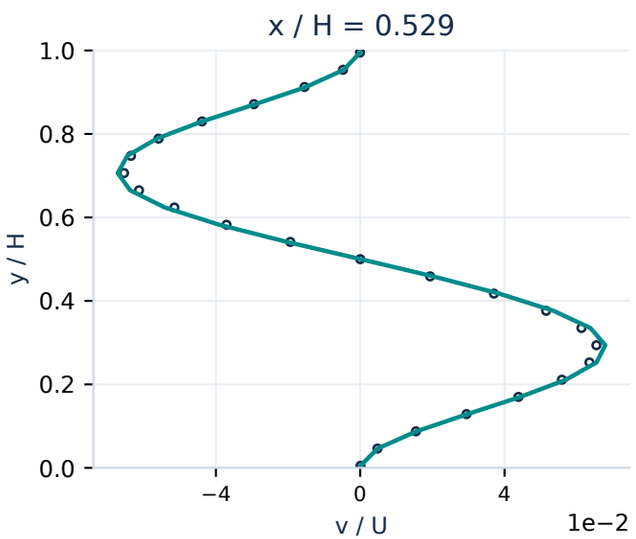


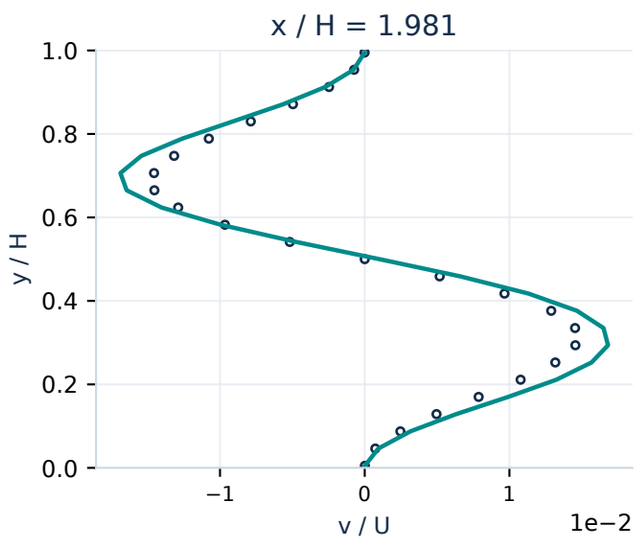


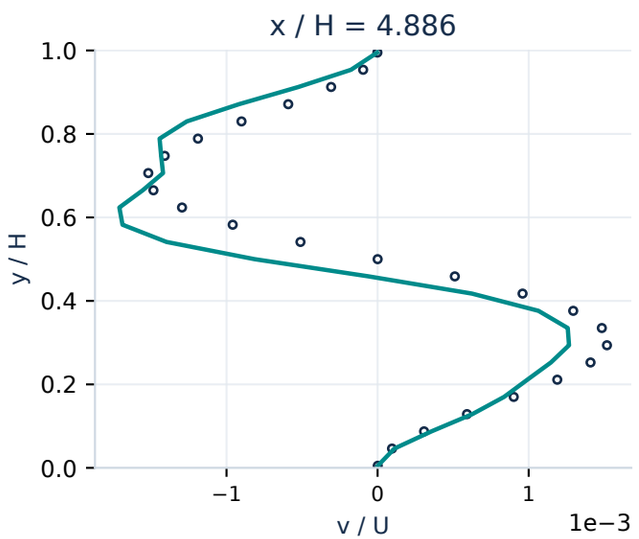


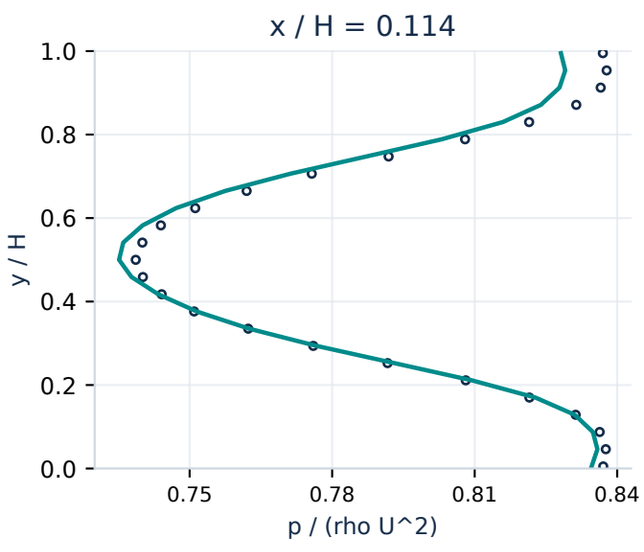


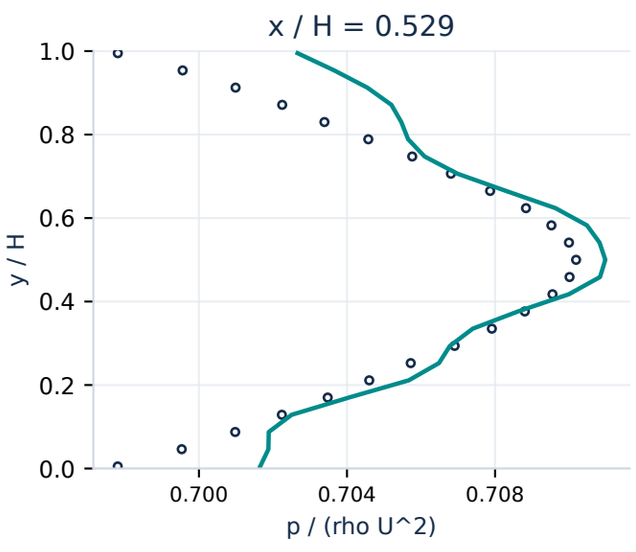


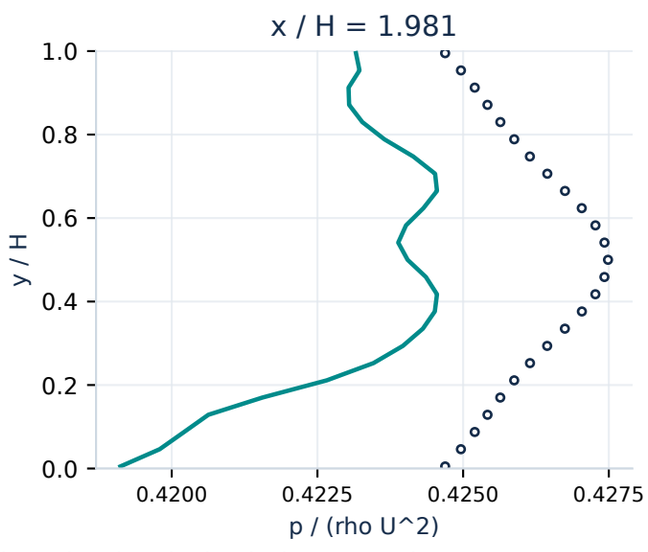


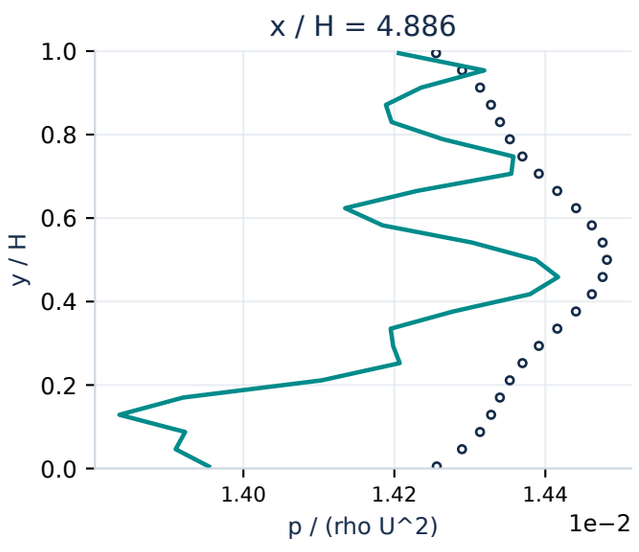






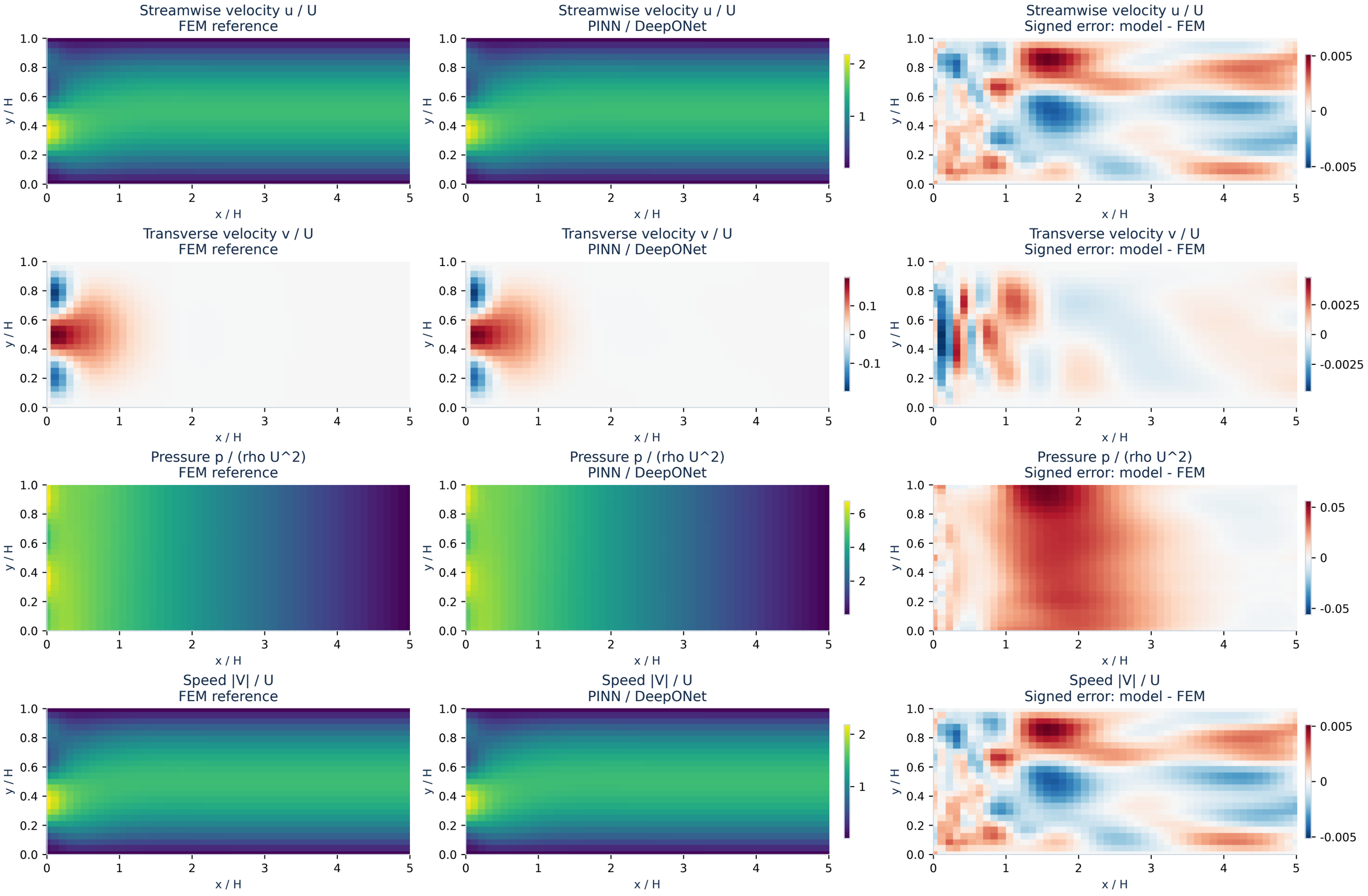






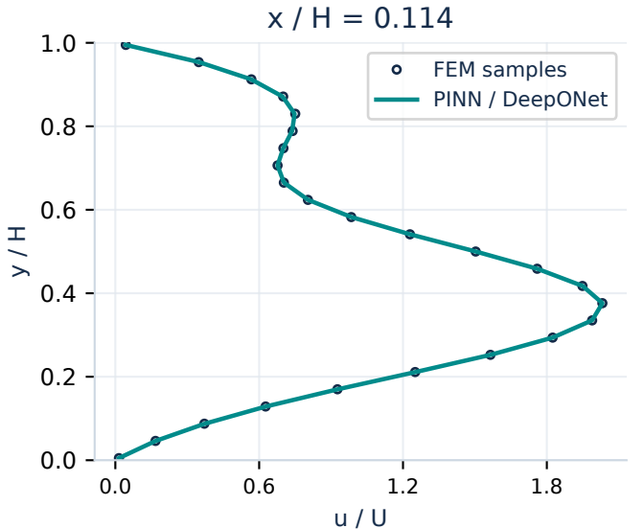
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

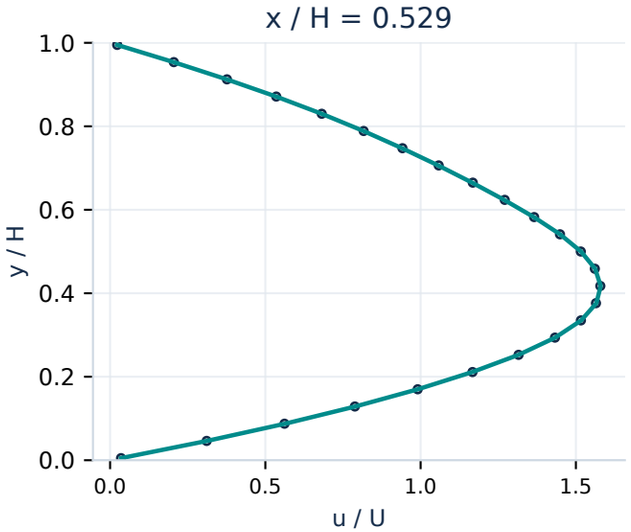
M09 | Re = 10 | a = [+0.000, +0.000, -0.600, +0.000]
Velocity L2: 0.167% Pressure L2: 0.638% v RMSE: 0.00090 U

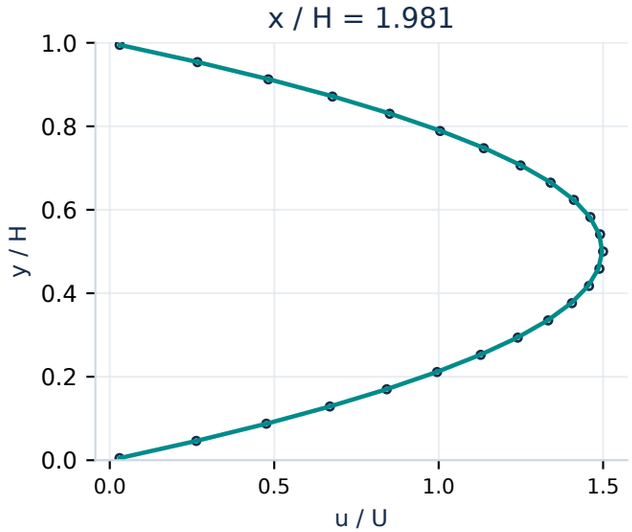


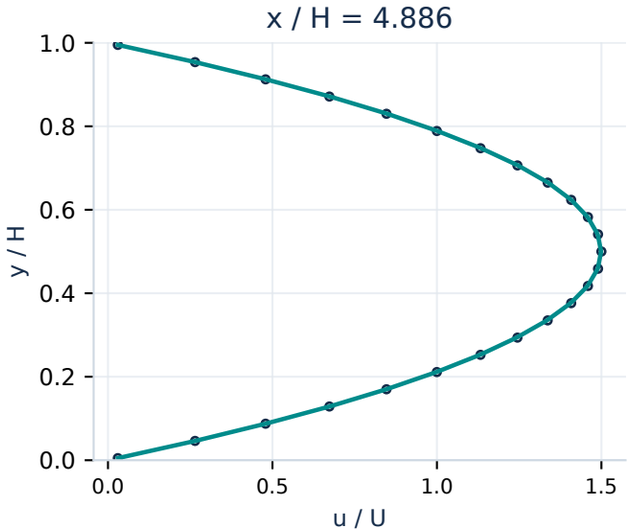
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

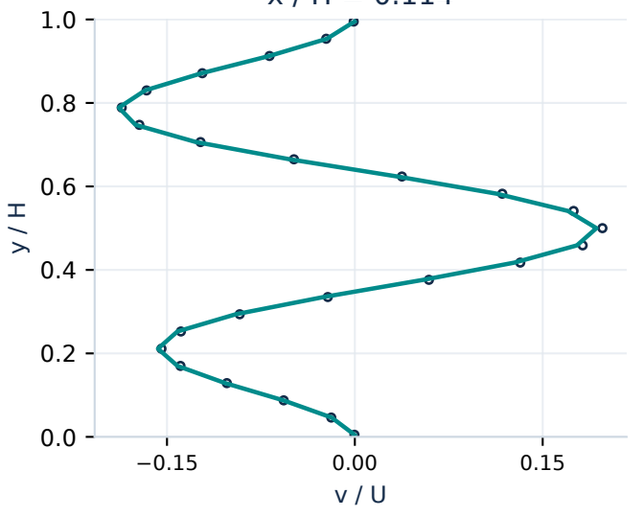
M09 | Re = 10 | a = [+0.000, +0.000, -0.600, +0.000]
Velocity L2: 0.167% Pressure L2: 0.638% v RMSE: 0.00090 U

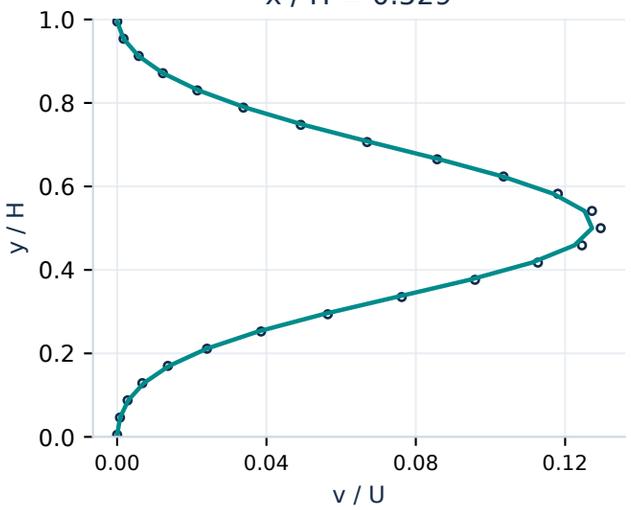


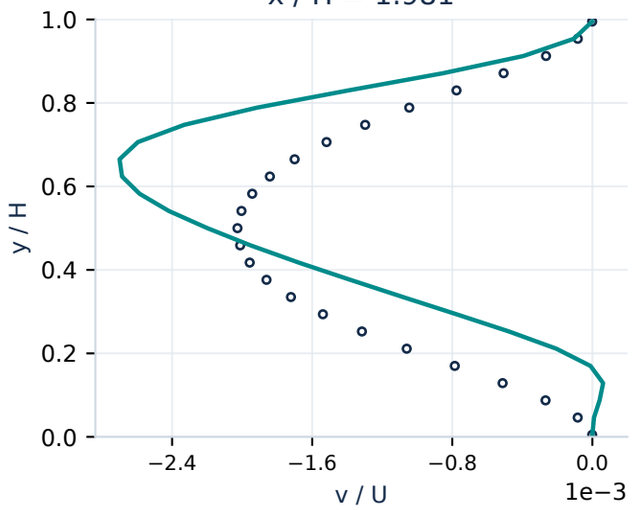


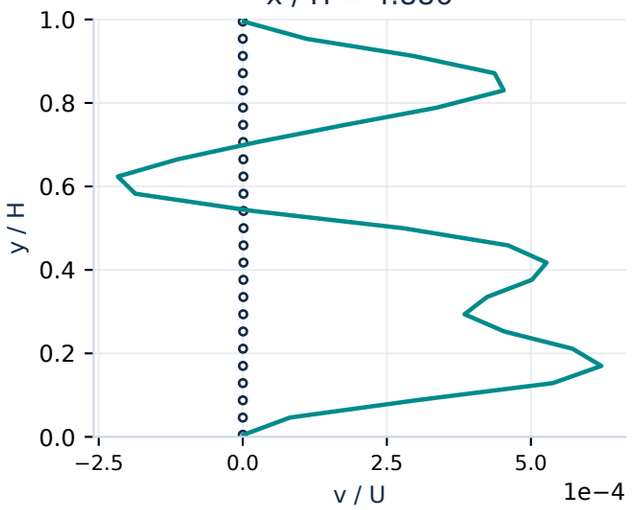


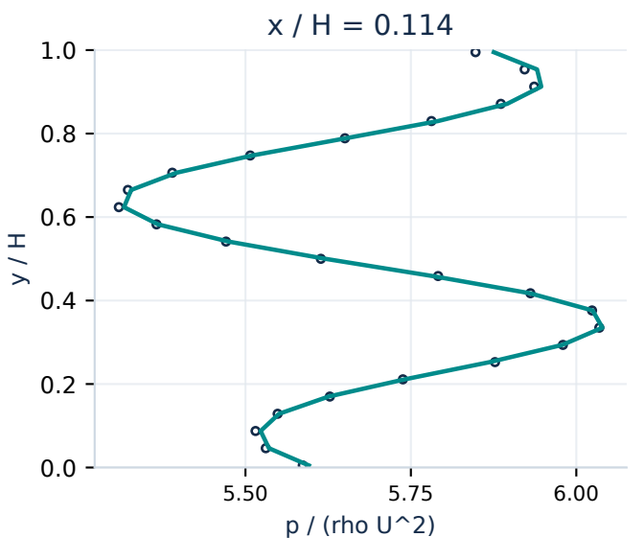


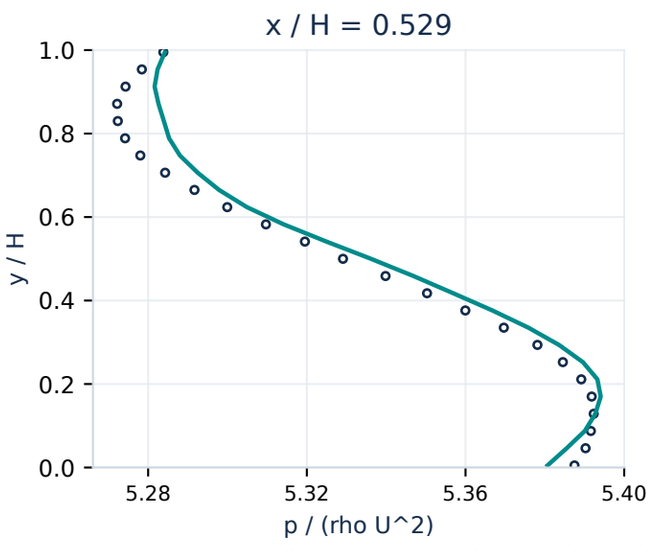


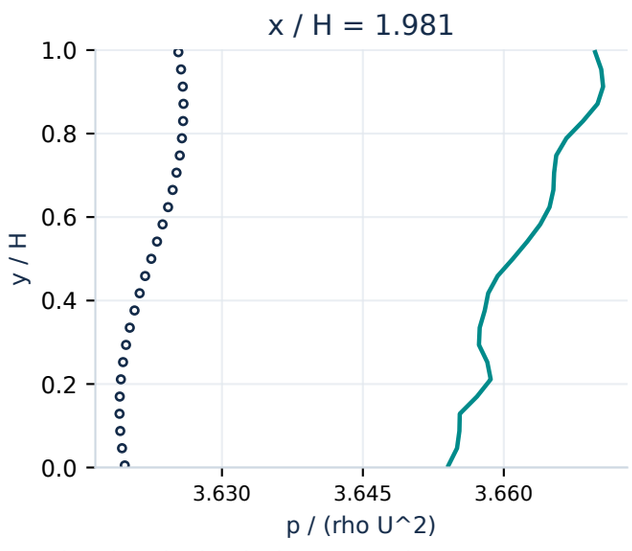


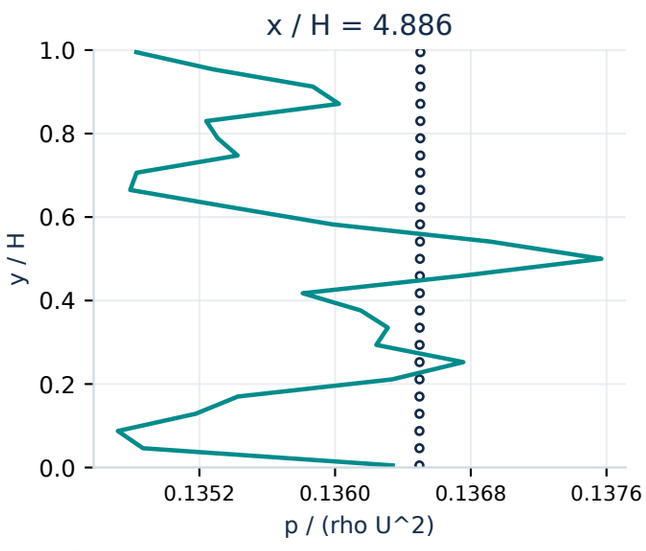






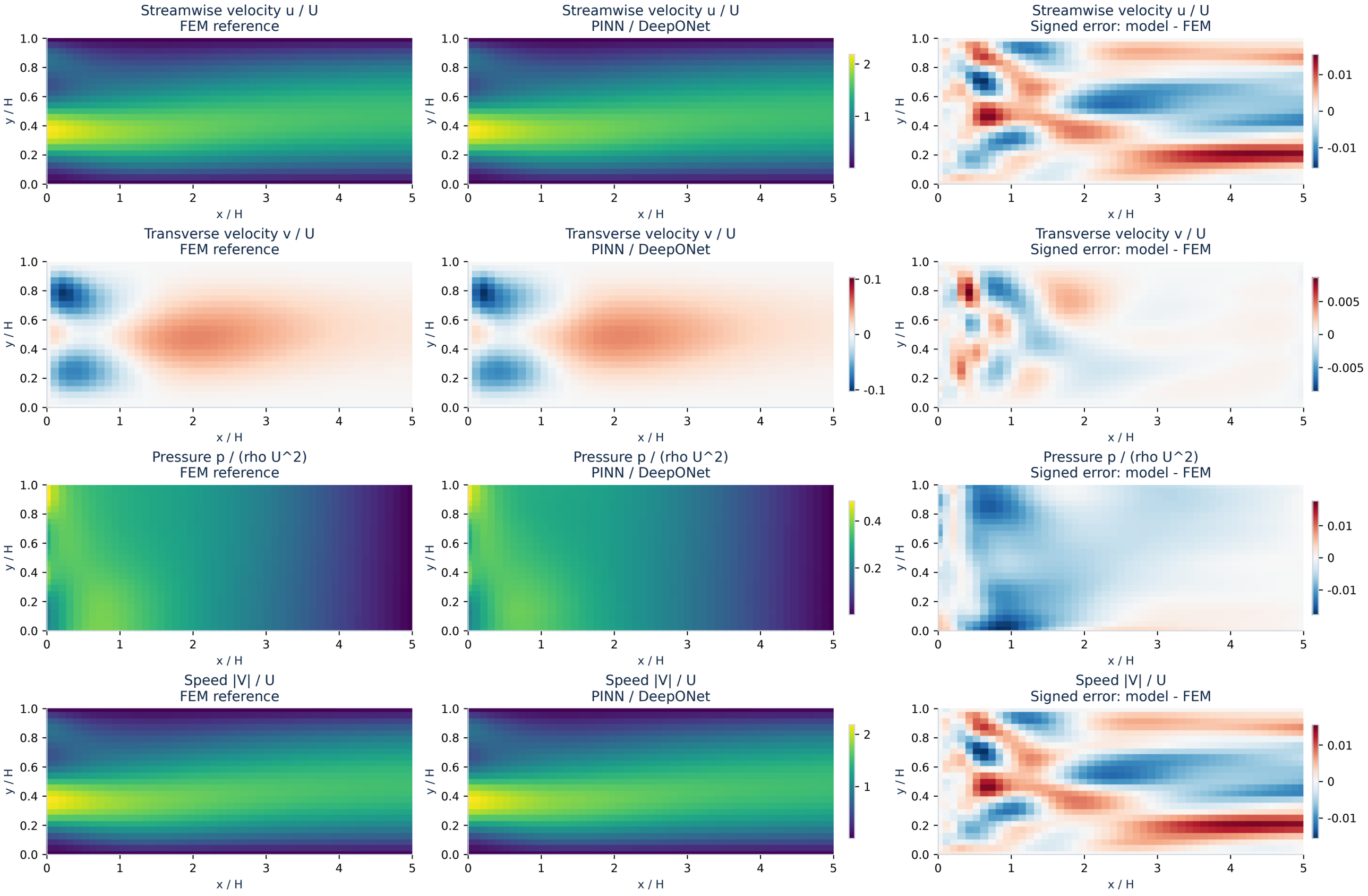






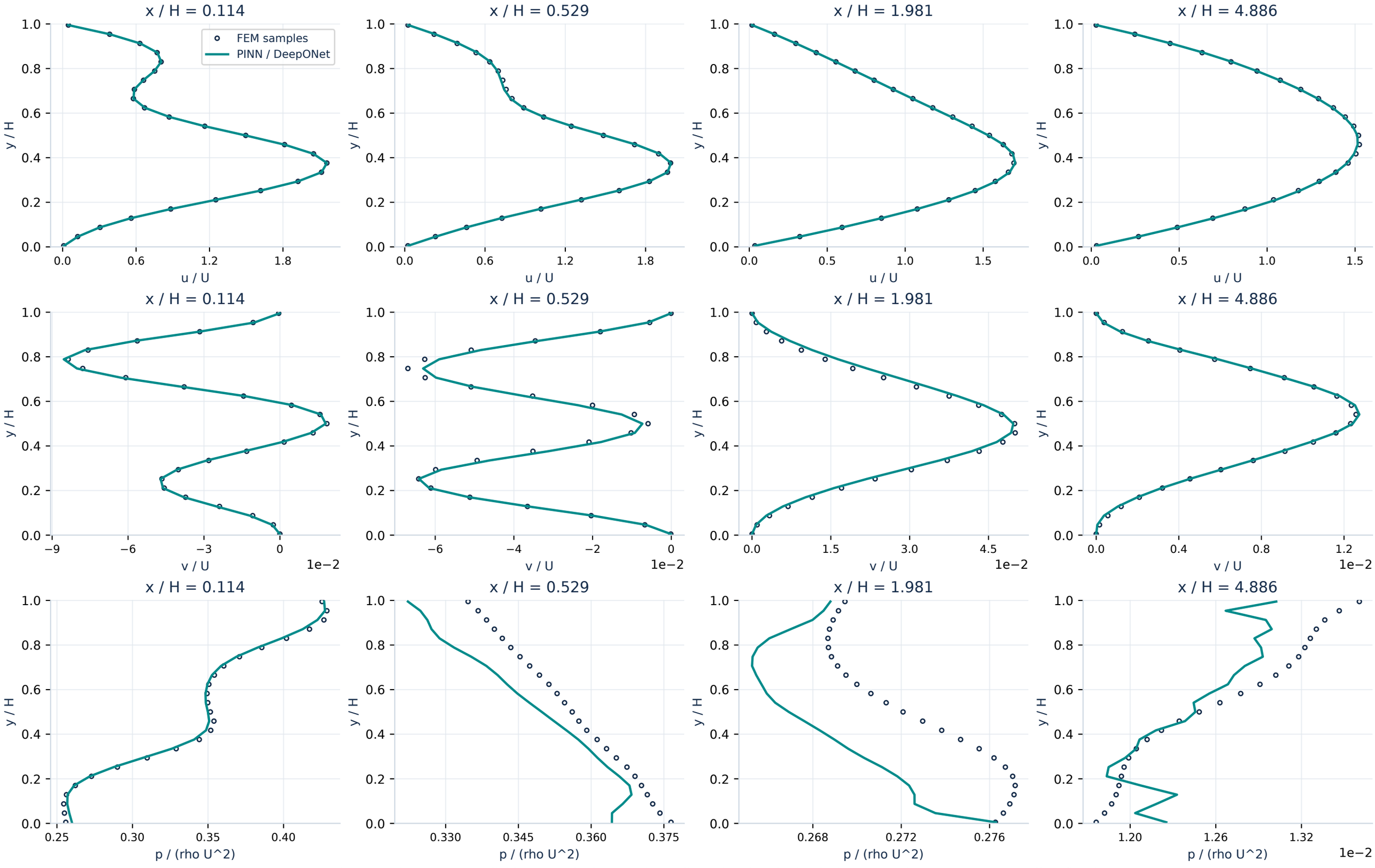
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

M10 | Re = 100 | a = [+0.000, +0.000, -0.600, +0.000]
Velocity L2: 0.500% Pressure L2: 2.065% v RMSE: 0.00131 U



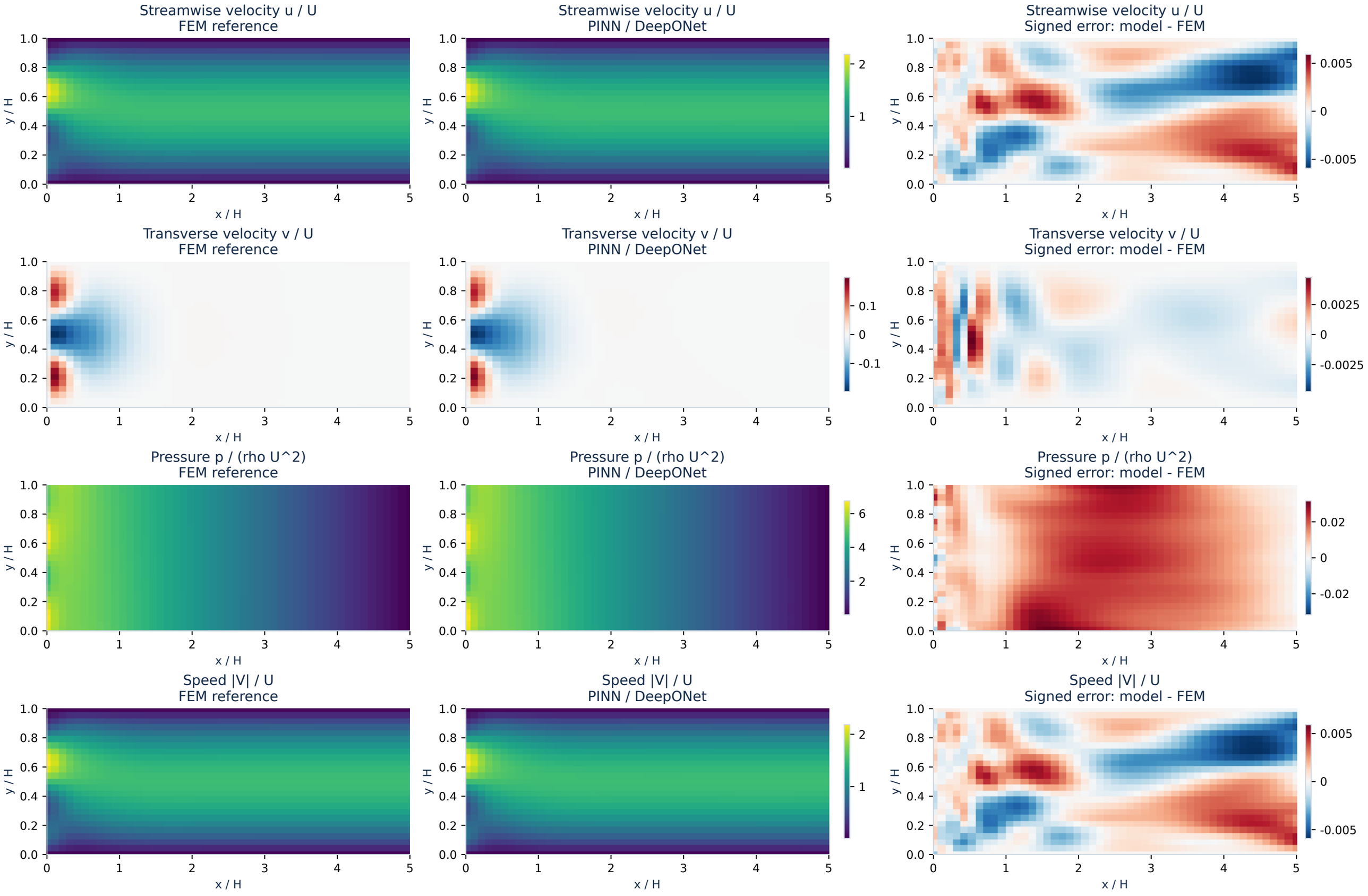
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M10 | Re = 100 | a = [+0.000, +0.000, -0.600, +0.000]
Velocity L2: 0.500% Pressure L2: 2.065% v RMSE: 0.00131 U



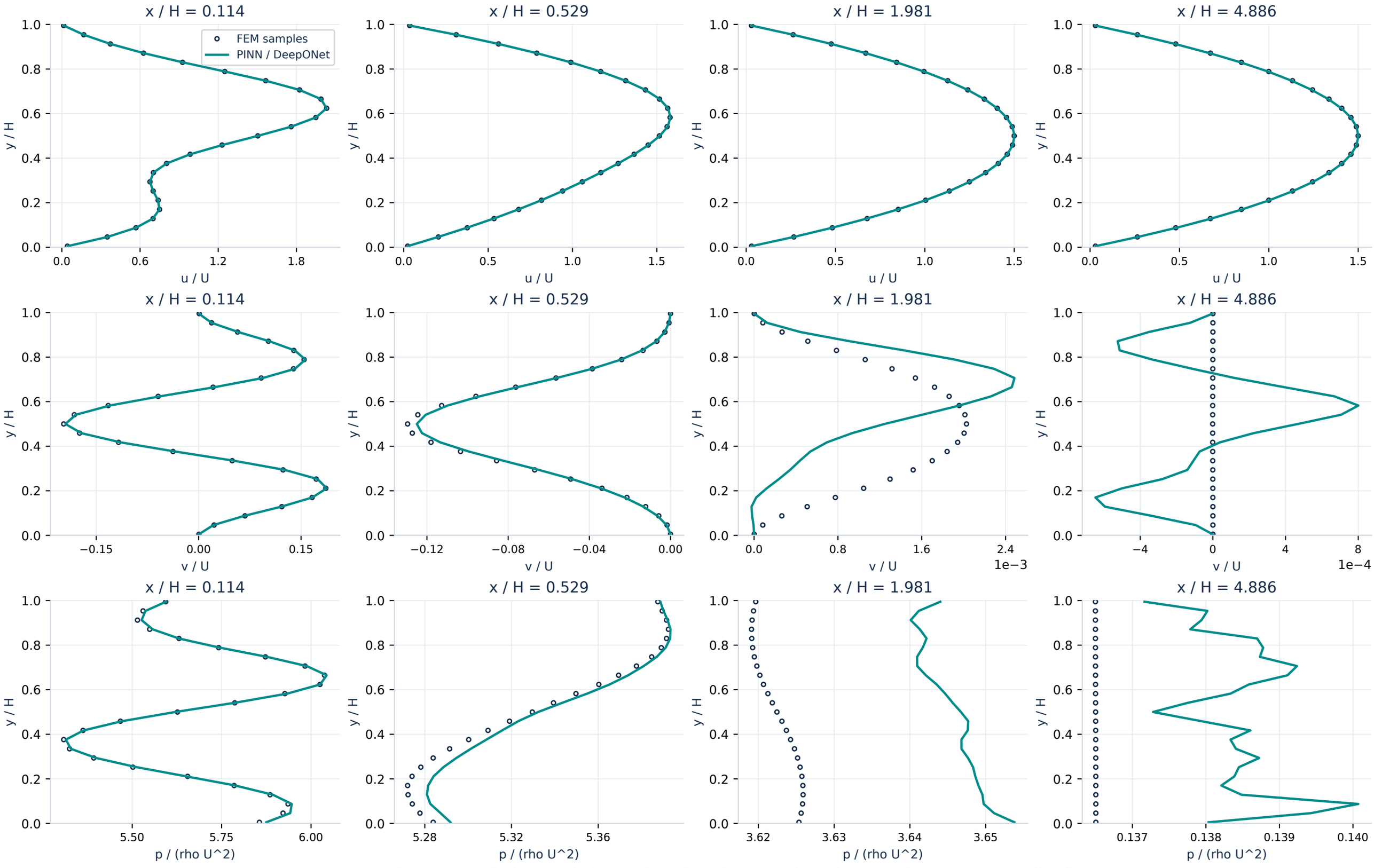
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

M11 | Re = 10 | a = [+0.000, +0.000, +0.600, +0.000]
Velocity L2: 0.230% Pressure L2: 0.478% v RMSE: 0.00087 U



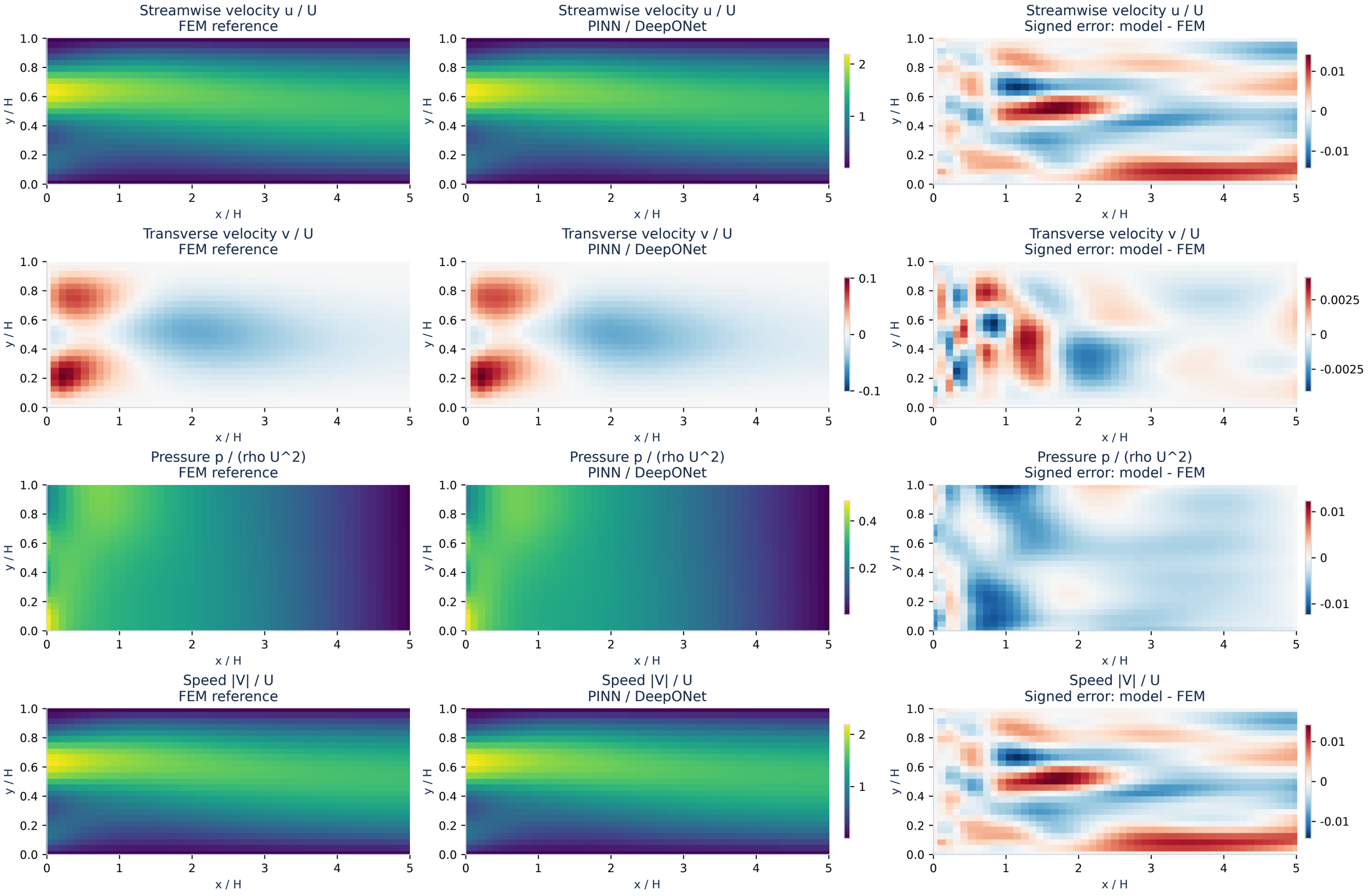
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M11 | Re = 10 | a = [+0.000, +0.000, +0.600, +0.000]
Velocity L2: 0.230% Pressure L2: 0.478% v RMSE: 0.00087 U



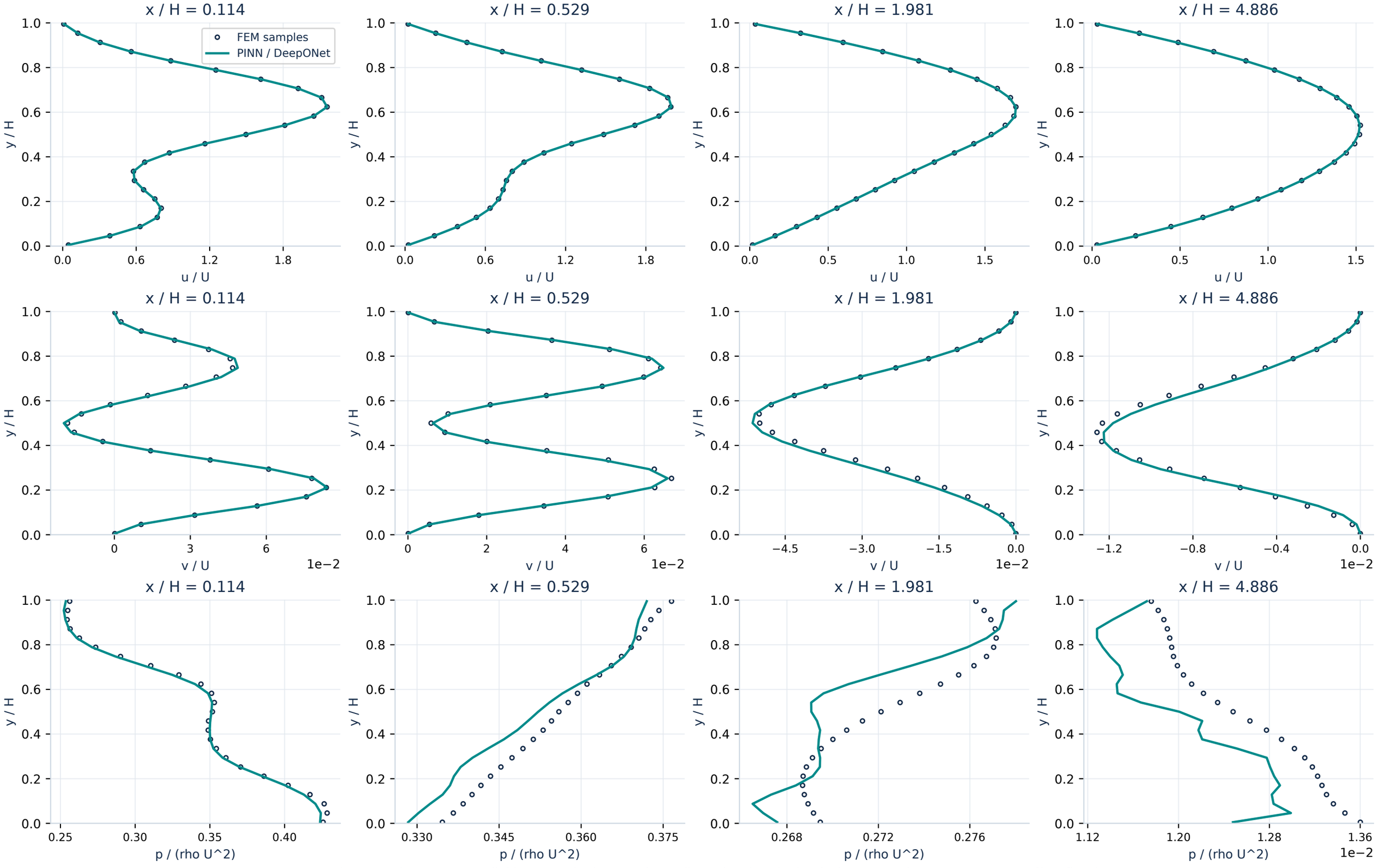
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

M12 | Re = 100 | a = [+0.000, +0.000, +0.600, +0.000]
Velocity L2: 0.420% Pressure L2: 1.501% v RMSE: 0.00100 U



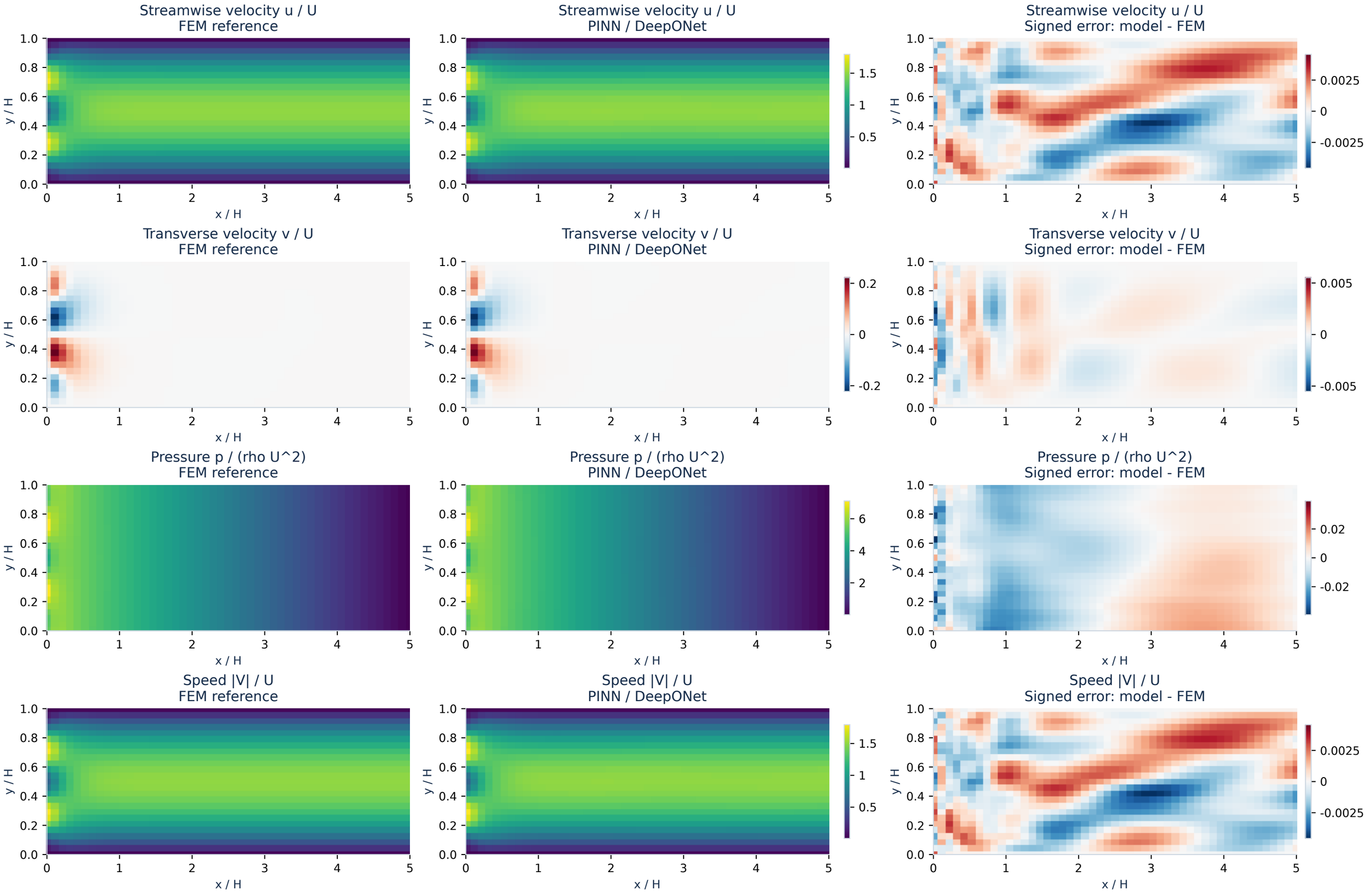
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M12 | Re = 100 | a = [+0.000, +0.000, +0.600, +0.000]
Velocity L2: 0.420% Pressure L2: 1.501% v RMSE: 0.00100 U



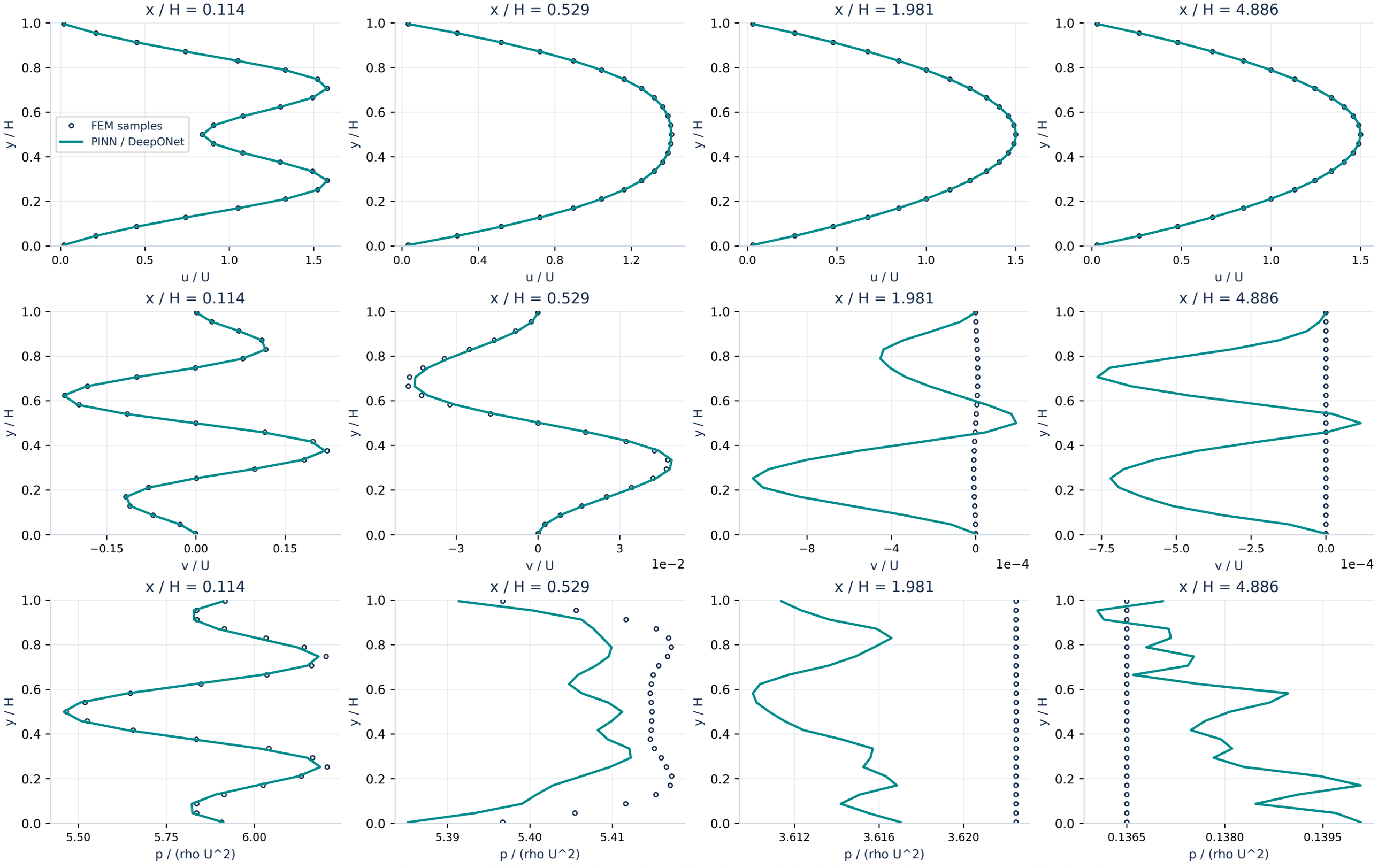
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

M13 | Re = 10 | a = [+0.000, +0.000, +0.000, -0.600]
Velocity L2: 0.164% Pressure L2: 0.291% v RMSE: 0.00073 U



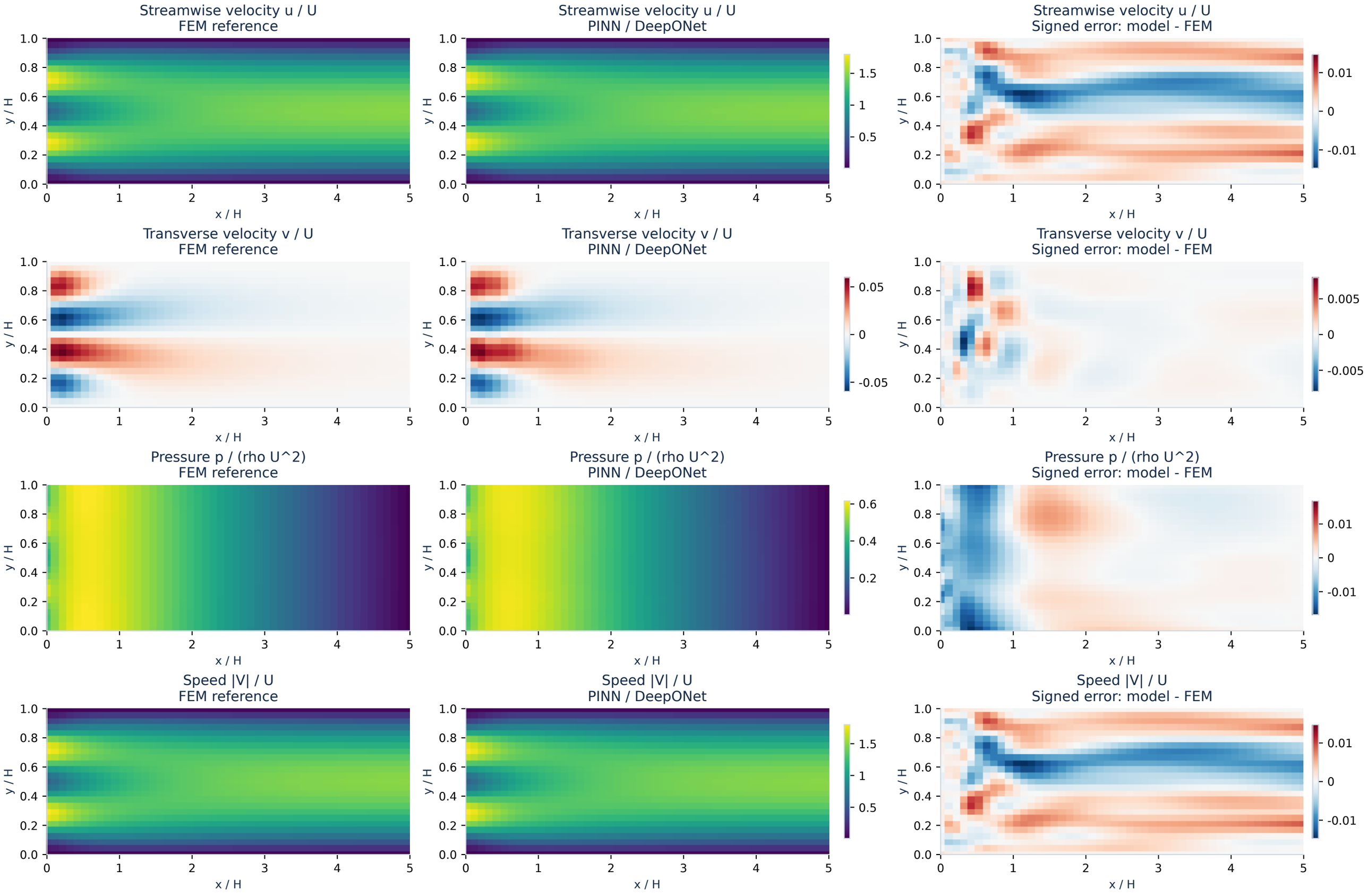
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M13 | Re = 10 | a = [+0.000, +0.000, +0.000, -0.600]
Velocity L2: 0.164% Pressure L2: 0.291% v RMSE: 0.00073 U



Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

M14 | Re = 100 | a = [+0.000, +0.000, +0.000, -0.600]
Velocity L2: 0.441% Pressure L2: 1.037% v RMSE: 0.00096 U

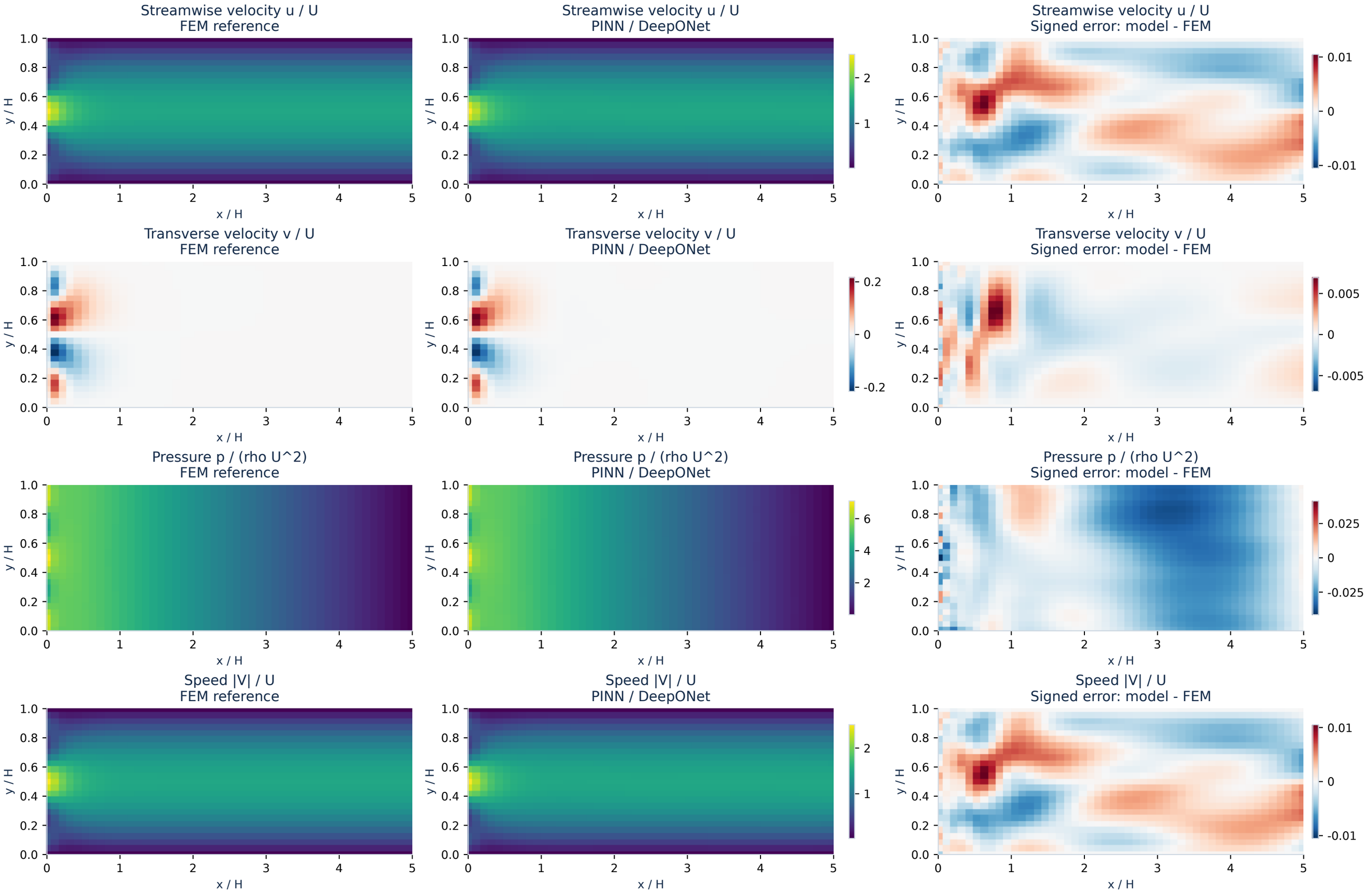


Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M14 | Re = 100 | a = [+0.000, +0.000, +0.000, -0.600]
Velocity L2: 0.441% Pressure L2: 1.037% v RMSE: 0.00096 U

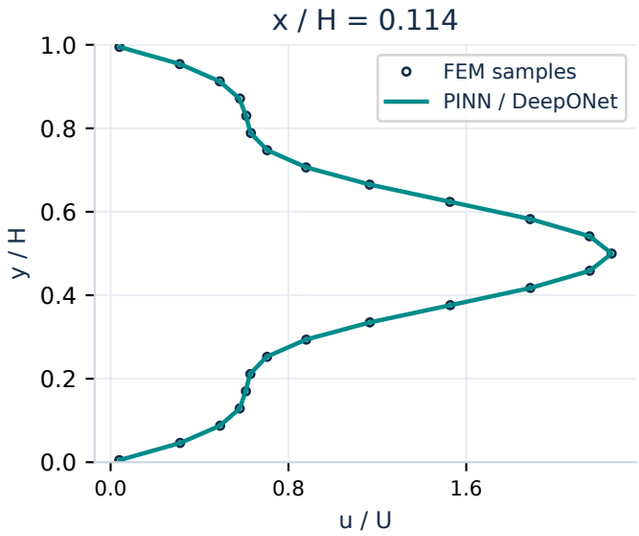
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

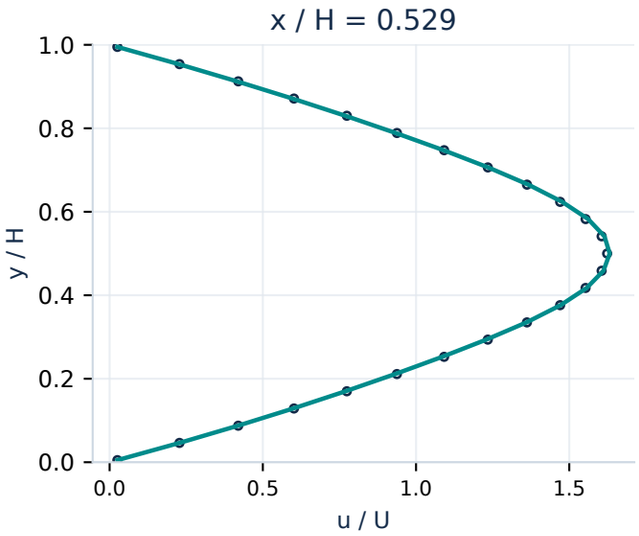
M15 | Re = 10 | a = [+0.000, +0.000, +0.000, +0.600]
Velocity L2: 0.291% Pressure L2: 0.525% v RMSE: 0.00120 U

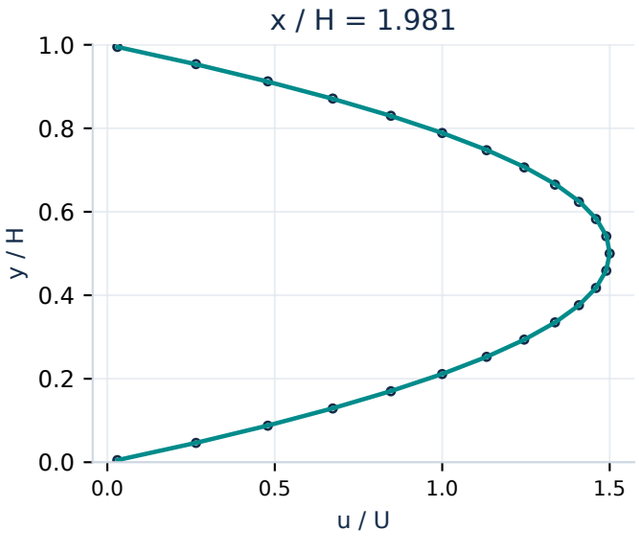


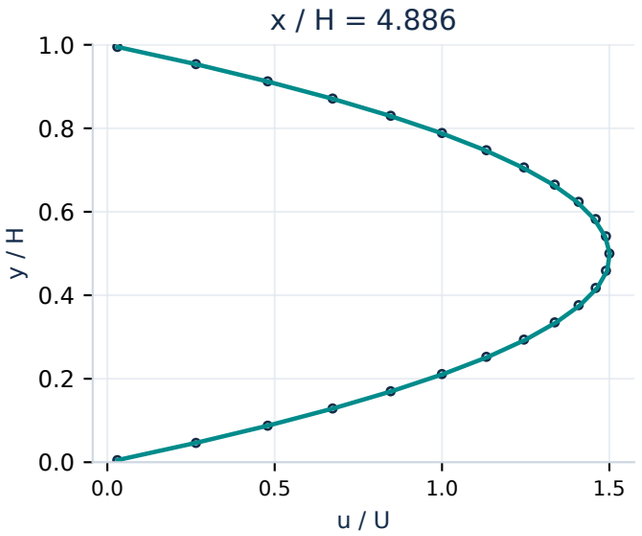
Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

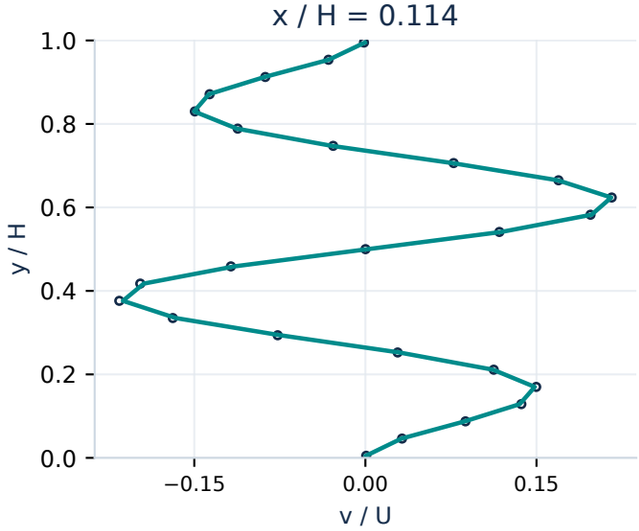
M15 | Re = 10 | a = [+0.000, +0.000, +0.000, +0.600]
Velocity L2: 0.291% Pressure L2: 0.525% v RMSE: 0.00120 U

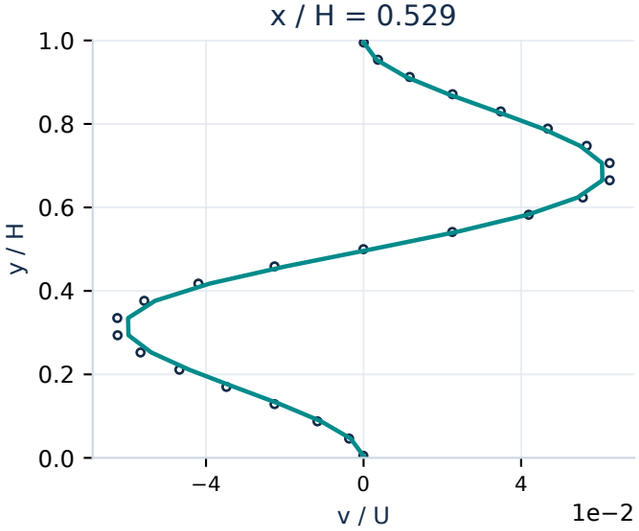


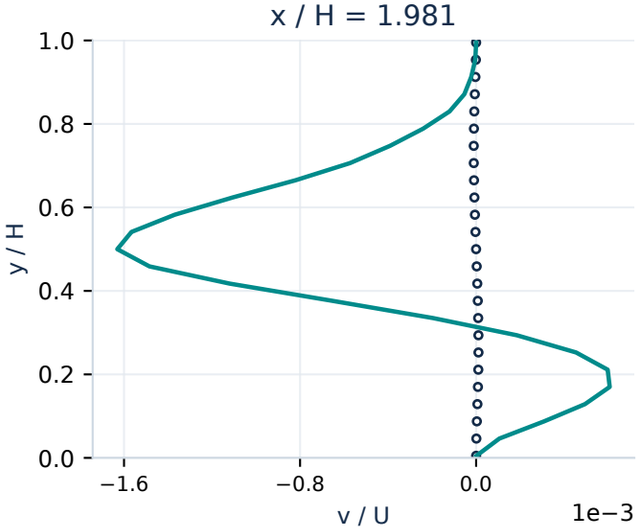


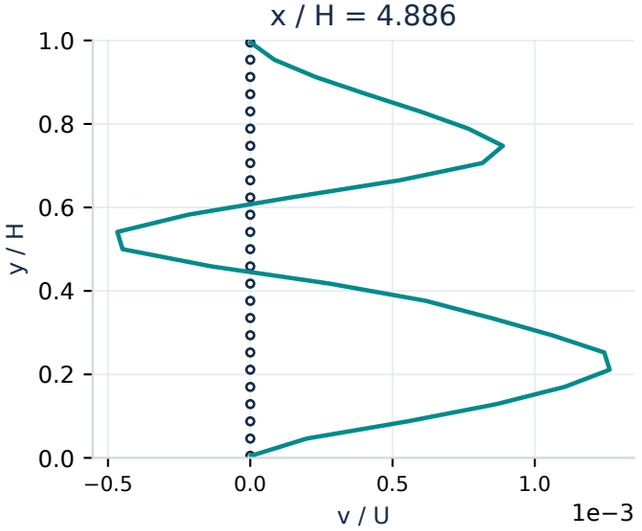


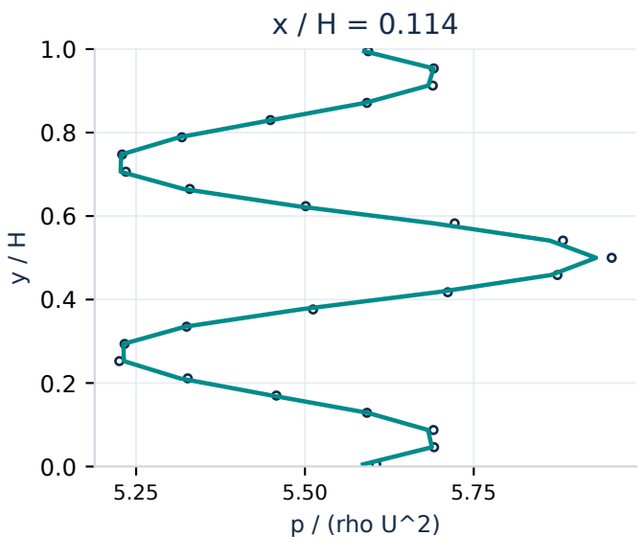


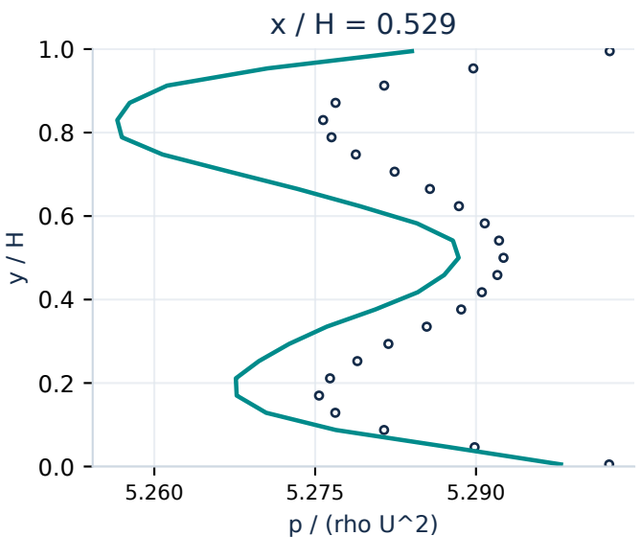


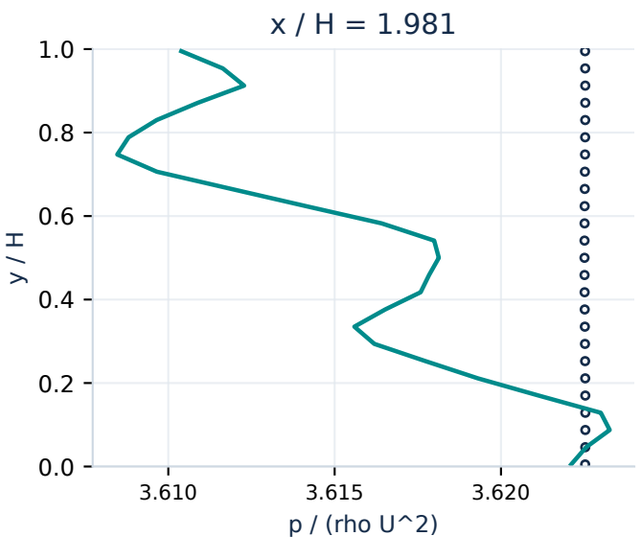


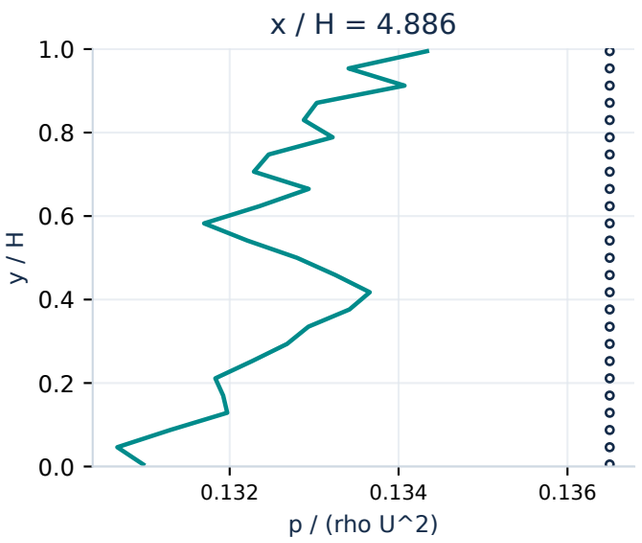






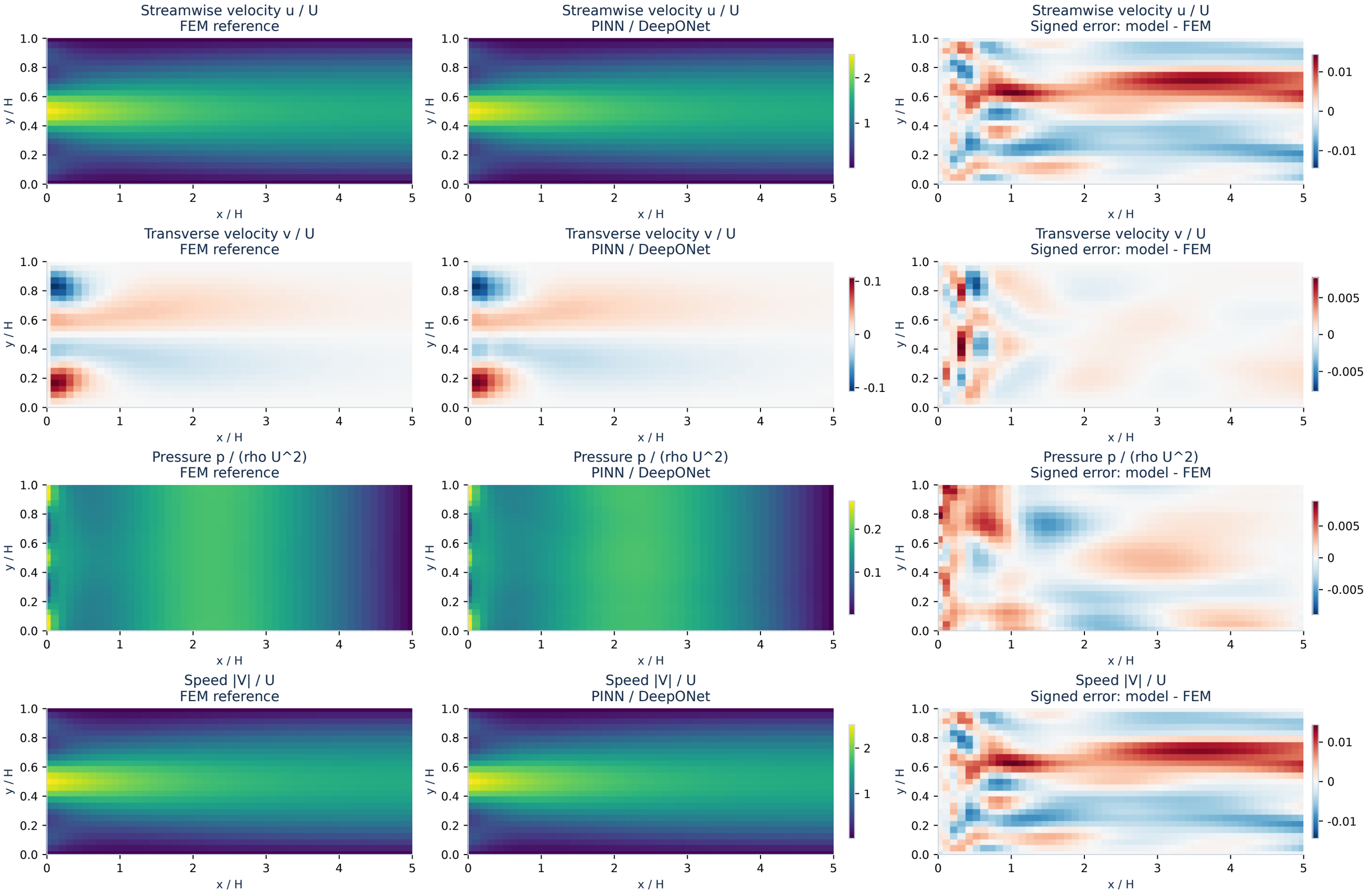






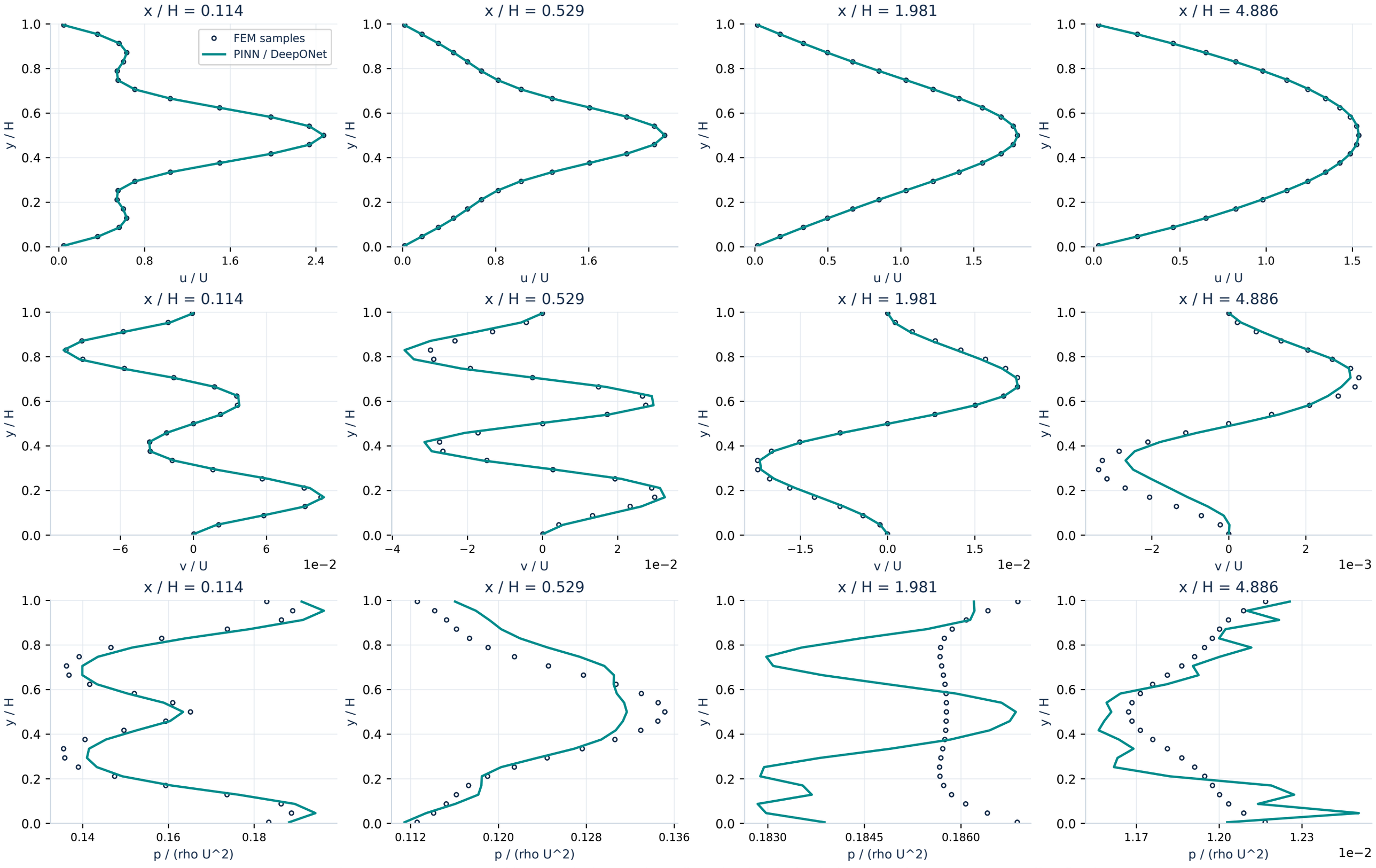
Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.

M16 | Re = 100 | a = [+0.000, +0.000, +0.000, +0.600]
Velocity L2: 0.427% Pressure L2: 1.361% v RMSE: 0.00101 U



Shared color scale for FEM and model in each row; error scale is separate. Native comparison samples: 49 x 25. All fields dimensionless.

M16 | Re = 100 | a = [+0.000, +0.000, +0.000, +0.600]
Velocity L2: 0.427% Pressure L2: 1.361% v RMSE: 0.00101 U



Cross-sections use the nearest available x sample; its actual coordinate is printed. Identical pressure datum; no pressure offset fitted.